

Chapter 5 :

BJT Biasing Circuits

5.1 Introduction

Transistors are used for many applications like: as a switch, as a comparator, as a current source, as an amplifier etc. The most common application of a transistor is as amplifier. Transistor used in an amplifier circuit must be biased in the active region with suitable levels of base, collector and emitter currents and constant terminal voltages. The levels of these currents and voltages set a particular operating point of the transistor which is called the **DC operating point** or **quiescent point**. In Chapter 4, we have seen that, to use a transistor in active region, the $B-E$ junction must be forward biased and the $B-C$ junction must be reverse biased. We have ensured this biasing conditions using two set of batteries which is not practical. We have to design circuits that can provide both the forward bias and reverse bias voltages from a single battery or power source. This type of circuits are called **biasing circuits**. Ideally the DC currents and voltage levels (without any AC signal) in a biasing circuit should remain absolutely constant. In practical circuits, these quantities may change due to the variations of temperature and the transistor current gain (β or h_{FE}). The aim of designing good bias circuits is to provide stable DC current and voltage levels irrespective of the temperature and h_{FE} . The simplest bias circuit consists of the smallest number of components and hence, they are the least costly. But they are not able to provide stable biasing. On the other hand, a complicated bias circuit is costly, but can provide better biasing for the transistor. In this chapter, we will study different biasing circuits and their performance.

5.2 Biasing From a Single Source

In Chapter 4, we have seen how to bias a transistor using two sets of batteries. Fig.5-1(a) shows such a circuit to bias a transistor in CE configuration. Here, resistors R_B and R_C are used to limit the base current and the collector current to our desired levels. The battery V_{BB} , used to forward bias the $B-E$ junction, has small value (say 3 V) and V_{CC} , used to bias the $B-C$ junction, has higher value

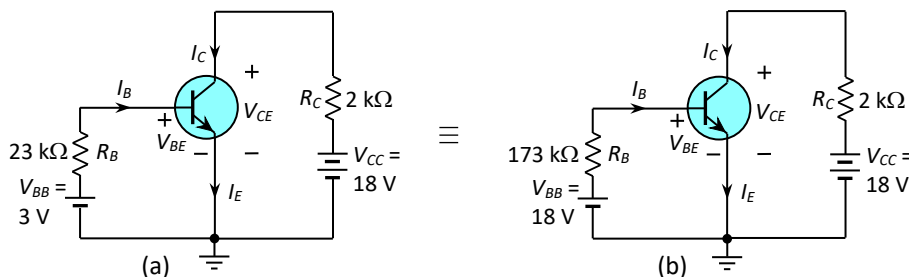


Fig.5-1: Biasing circuit of a transistor (a) $V_{BB} = 3\text{ V}$ and $R_B = 23\text{ k}\Omega$, (b) $V_{BB} = 18\text{ V}$ and $R_B = 173\text{ k}\Omega$

(say 18 V). But using suitable resistance values we can also use higher voltage for the $B-E$ junction. Fig.5-1(b) shows the biasing circuit using same values for V_{BB} and V_{CC} . We can redraw the circuit as shown in Fig.5-1(c). Now, connecting the terminals of same potential in a single point we get the circuit of Fig.5-1(d). Thus, we have redrawn the first bias circuit [Fig.5-1(a)] with a single voltage source (supply voltage) in Fig.5-1(d).

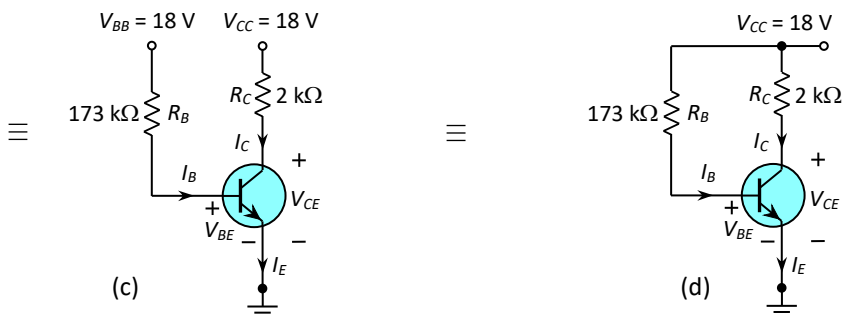


Fig.5-1: Biasing circuit of a transistor (c) and (d) alternate representations of circuit (b)

Example 5-1

For the biasing circuits of Fig.5-1(a) and (b), calculate the values of I_B , I_C and V_{CE} assuming $\beta = 50$.

Solution:

For circuit (a):

Considering a Si transistor $V_{BE} = 0.7$ V.

The value of I_B can be calculated as (the equations are derived in Section 5.7),

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{3 \text{ V} - 0.7 \text{ V}}{23 \text{ k}\Omega} = 100 \text{ }\mu\text{A} \text{ [Ans.]}$$

Therefore, the value of I_C will be, $I_C = \beta I_B = 50 \times 100 \text{ }\mu\text{A} = 5 \text{ mA}$ [Ans.]

Now, the value of V_{CE} can be calculated as,

$$V_{CE} = V_{CC} - I_C R_C = 18 \text{ V} - (5 \text{ mA})(2 \text{ k}\Omega) = 8 \text{ V} \text{ [Ans.]}$$

For circuit (b):

Considering a Si transistor $V_{BE} = 0.7$ V.

The value of I_B can be calculated as, $I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{18 \text{ V} - 0.7 \text{ V}}{173 \text{ k}\Omega} = 100 \text{ }\mu\text{A} \text{ [Ans.]}$

Therefore, the value of I_C will be, $I_C = \beta I_B = 50 \times 100 \text{ }\mu\text{A} = 5 \text{ mA}$ [Ans.]

Now, the value of V_{CE} can be calculated as,

$$V_{CE} = V_{CC} - I_C R_C = 18 \text{ V} - (5 \text{ mA})(2 \text{ k}\Omega) = 8 \text{ V} \text{ [Ans.]}$$

As the circuits of Fig.5-1(c) and (d) use the same voltages and resistances, same results will be found for them.

Comments: Same collector current and collector-emitter voltage is found using two batteries and a single battery using different base resistance R_B .

5.3 DC Load Line and Operating Point

As we are interested with the output of a transistor and the power of a transistor depends on the output voltage and output current, the operating point of a transistor is represented in terms of them. For example, in case of CB configuration the values of output current (I_C) and output voltage (V_{CE}) represent a point on the output characteristic curves which is called the **operating point**. Similarly, for CE configuration, the operating point is represented by the values of I_C and V_{CE} . That is, the point that gives the values of output voltage and output current of a transistor is called the **operating point**. Depending on the bias voltage and the circuit components, the operating point can be at any location on the output characteristics. But, for a fixed value of bias voltage and output resistance (or load resistance) operating point can be at any location within a line on the output characteristics. This line is called the **load line**. Thus, the line, drawn on the transistor output characteristics, that gives all possible positions of the operating point is called the load line.

To draw a load line we will consider a common-emitter (CE) circuit as shown in Fig.5-2. For CE configuration, the load-line is a graph of collector current (I_C) versus collector-emitter voltage (V_{CE}) for a given value of collector resistance R_C and given supply voltage V_{CC} . That is, the load line shows all levels of I_C and V_{CE} that can exist in a particular circuit.

To draw the load line, we need an expression that relates I_C and V_{CE} . This can easily be found from the C-E circuit of Fig.5-2(a). By applying KVL to C-E loop of this circuit we get,

$$V_{CC} - I_C R_L - V_{CE} = 0 \quad (5-1)$$

$$\text{or,} \quad \boxed{V_{CE} = V_{CC} - I_C R_L} \quad (5-2)$$

Equ.(5-2) is an equation of straight line or the load line. To draw this line on the output characteristics of the transistor we have to determine two points. Let's do that.

Point A:

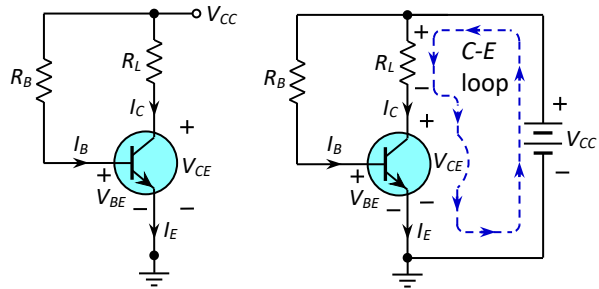


Fig.5-2: Transistor Circuit for determining load line.

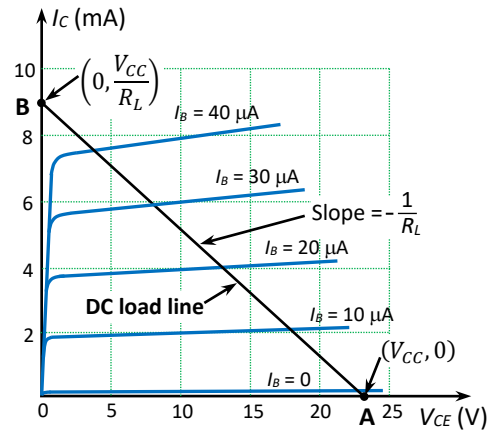


Fig.5-3: Load line drawn on the output characteristic curves.

When, $I_C = 0$ from Equ.(5-2) we get,

$$V_{CE} = V_{CC}$$

Point B:

When, $V_{CE} = 0$, again from the same equation we get,

$$I_C = \frac{V_{CC}}{R_L}$$

Now, if we put these points on the output characteristics of the transistor and connect them we get the load line as shown in Fig.5-3.

Equ.(5-2) can be written as,

$$I_C = -\frac{1}{R_L}V_{CE} + \frac{V_{CC}}{R_L} \quad (5-3)$$

This equation is comparable with a standard equation of straight line,

$$y = mx + c \quad (5-4)$$

Comparing Equ.(5-3) and (5-4) we find that, the slope of the load line is $-1/R_L$ and V_{CC}/R_L is the y-intercept. Thus, the shape of the load-line completely depends on the values of V_{CC} and R_L .

If the values of V_{CC} and R_L are fixed the operating point of the transistor can change only along this load line depending on the values of I_B . For example, if the value of I_B increases (with constant values of V_{CC} and R_L) the operating point will move toward the point B and if I_B decreases the operating point will move to point A. But, for a fixed value of I_B , the operating point also be fixed as shown in Fig.5-4.

Thus, the intersecting point of the DC load line and the output characteristic curve of a transistor is called the **operating point**. It is also called the **quiescent point** or simply **Q-point**. Therefore, a Q-point gives a fixed value of output voltage and output current which are represented as V_{CEQ} and I_{CQ} . If we draw a vertical line through the Q-point it will intersect the x-axis and this is the value of V_{CEQ} . Similarly, a horizontal line through the Q-point will intersect the y-axis and that will give the value of I_{CQ} . The value of the base current for which the operating point is set, is called quiescent base current and is denoted as I_{BQ} .

It is not always necessary to have the device characteristics in order to draw DC load line. A simple graph of I_C versus V_{CE} can be draw for any value of V_{CC} and R_L . But if the load line is drawn on the output characteristics, knowing any value of I_B , I_C or V_{CE} the other two can easily be determined.

5.3.1 Variation of Load Line and Operating Point

We know that the slope and position of the load line depends only on the values of V_{CC} and R_L . It will not depend on the properties of the transistor. The position of the Q-point depends on the load line and the value of I_{BQ} . Keeping V_{CC} constant, if we increase the value of R_L , the value of V_{CC}/R_L

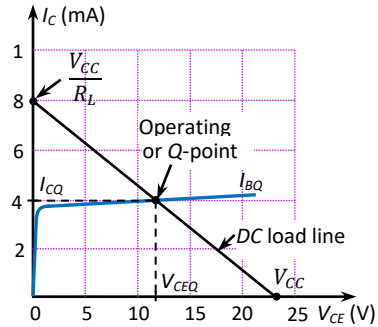


Fig.5-4: Operating point (intersection of load line and a fixed output characteristic curve)

decreases. Then, the load lines will start from the same point on the x -axis, but on the y -axis they will go down, and for a fixed value of base current I_{BQ} , the operating point will move from right to left as shown in Fig.5-5. On the other hand, if the value of V_{CC} increases with a constant value of R_L , the load lines will be parallel to each other as shown in Fig.5-6. Because, if V_{CC} is made double V_{CC}/R_L will also be double. In this case, for a fixed value of I_{BQ} , the operating point will also move from left to right.

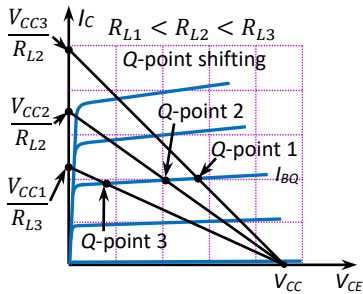


Fig.5-5: Variation of load line and Q-point for fixed V_{CC} and variable R_L .

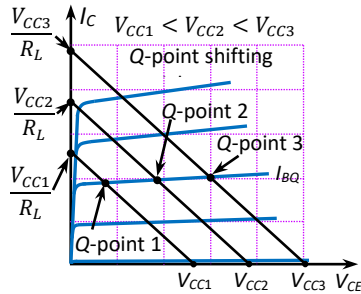


Fig.5-6: Variation of load line and Q-point for fixed R_L and variable V_{CC} .

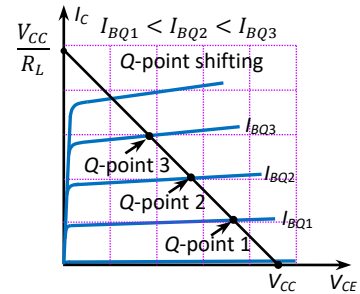


Fig.5-7: Variation of Q-point for fixed V_{CC} and R_L but variable β or I_{BQ} .

Another reason of variation of the operating point is the change of base current or the change of β of the transistor. In both the cases, the output collector current and C - E voltage change along the load line. This is illustrated in Fig.5-7.

Example 5-2

For the circuit of Fig.5-2 draw the load line and the operating point. Assume, $R_L = 2 \text{ k}\Omega$, $V_{CC} = 20 \text{ V}$ and $I_{BQ} = 30 \mu\text{A}$. Also determine the values of I_{CQ} and V_{CEQ} .

Solution:

To solve this problem, we will use the characteristic curves of Fig.5-3. We already know that the load line can be drawn connecting two points, $A(V_{CC}, 0)$ on the x -axis and $B(0, \frac{V_{CC}}{R_L})$ on the y -axis. For this example these points will be $(20 \text{ V}, 0)$ and $(0, \frac{20 \text{ V}}{2 \text{ k}\Omega}) = (0, 10 \text{ mA})$.

Now, these points are put on the output characteristic curve and connecting them the load line is drawn as shown in Fig.5-8. The intersecting point of this load line and the $I_{BQ} = 30 \mu\text{A}$ curve is the Q-point. From the Q-point the values of V_{CEQ} and I_{CQ} are found as 8.53 V and 5.7 mA , respectively.

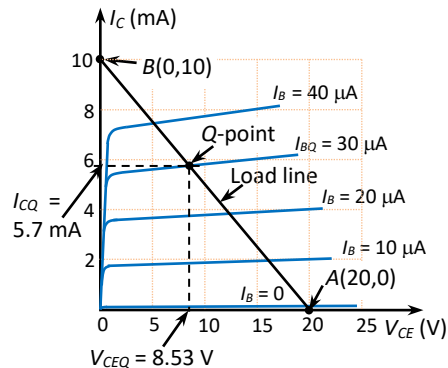


Fig.5-8: Load line and Q-point for Example 5-2

Comments: This method of determining the Q-point is called the graphical method, that can be used only when the output characteristics of the transistor are given.

5.3.2 Swing of Operating Point with AC signal

Already we have found that the DC operating point determines the transistor collector current and collector-emitter voltage. Thus, it defines the DC conditions of the circuit. But, when a transistor is used as an amplifier, an AC signal is applied to the input circuit as described in Chapter 4. For common emitter amplifier the AC signal is applied to the base through a capacitor as shown in Fig.5-9. Due to this AC signal, I_B changes according to the instantaneous value of this signal, which is shown in Fig.5-10. This variation in I_B causes I_C to vary, and consequently produces a variation in V_{CE} . During the positive half-cycle of the signal, I_B increases and for this reason I_C increases also. Due to this increase in I_C , the value of V_{CE} decreases. On the other hand, during the negative half-cycle of the input signal, I_B decreases and consequently I_C decreases. As I_C decreases, the value of V_{CE} increases. In this way, an AC swing is produced in the output current, I_C , and in the output voltage, V_{CE} . To make the process clearer, let us draw the DC load line and the operating point for $R_C = 2\text{ k}\Omega$, $V_{CC} = 20\text{ V}$, and $I_B = 30\text{ }\mu\text{A}$. The load line and the operating point (10 V, 5 mA) are shown in Fig.5-10. When, due to the positive swing of the input signal, I_B is increased from DC quiescent value $30\text{ }\mu\text{A}$ to $50\text{ }\mu\text{A}$, I_C increases to approximately 8.3 mA and V_{CE} decreases to 3.4 V as illustrated by point C on the load line. Thus, the swing (change) of the output voltage due to the positive cycle of the input is,

$$\Delta V_{CE} = 10\text{ V} - 3.4\text{ V} = 6.6\text{ V}$$

And the swing in the output current is,

$$\Delta I_C = 8.3\text{ mA} - 5\text{ mA} = 3.3\text{ mA}$$

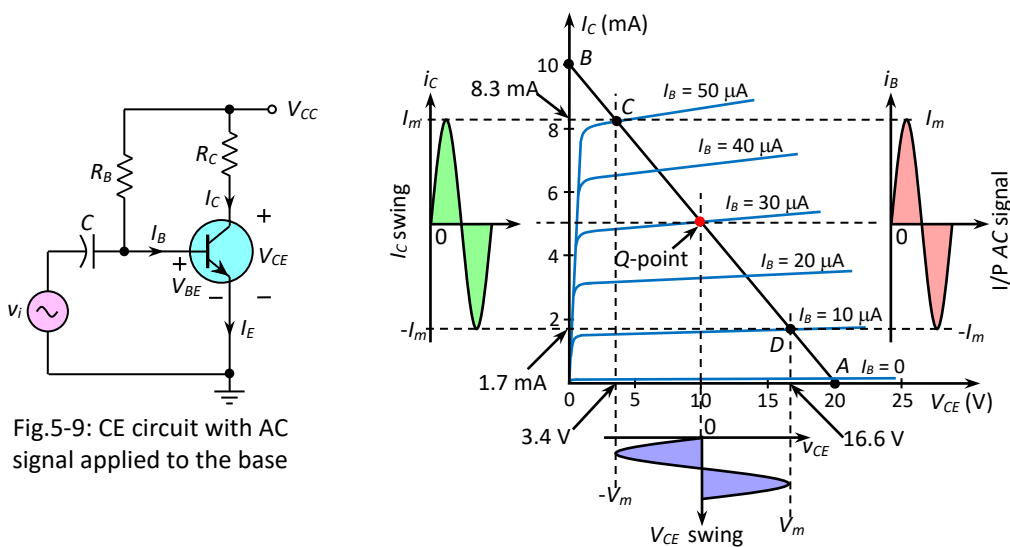


Fig.5-10: Swing of Q-point along the load line

Now, due to the negative swing of the input signal, I_B is decreased from $30\ \mu\text{A}$ to $10\ \mu\text{A}$, I_C decreases to approximately $1.7\ \text{mA}$ and V_{CE} increases to $16.6\ \text{V}$ as illustrated by point D on the load line. Thus, the swing of the output voltage (for $-ve$ half-cycle of input) is,

$$\Delta V_{CE} = 16.6\ \text{V} - 10\ \text{V} = 6.6\ \text{V}$$

And the swing in the output current is,

$$\Delta I_C = 5\ \text{mA} - 1.7\ \text{mA} = 3.3\ \text{mA}$$

Therefore, we find that, with the Q -point set at $I_{CQ} = 5\ \text{mA}$ and $V_{CEQ} = 10\ \text{V}$ a variation of $\pm 20\ \mu\text{A}$ in the base current (I_B) produces an output voltage (V_{CE}) swing of $\pm 6.6\ \text{V}$ and also a an output current (I_C) swing of $\pm 3.3\ \text{mA}$. The higher the swing in V_{CE} the greater the voltage gain can be achieved from the transistor circuit.

The maximum possible $C-E$ voltage swing (ΔV_{CE}) for a given circuit, can be determined without using the transistor characteristics. For convenience, if we neglect the effect of saturation and cutoff region, it may be assumed that I_C can be driven to zero at one extreme and to V_{CC}/R_L at the other extreme. For this changes, the $C-E$ voltage will also change from ($V_{CE} =$) V_{CC} to ($V_{CE} =$) 0 as illustrated in Fig.5-11. Thus, if we set the Q -point at the centre of the load line it is possible to achieve the maximum collector voltage swing and the maximum collector current swing. Then, the value of the maximum swing in V_{CE} will be $V_{CC}/2$ and the value of the maximum swing in I_C will be $V_{CC}/2R_L$.

5.3.3 Effect of Cutoff and Saturation Region on Q-Point Swing

The cutoff and saturation region of a transistor hamper to produce the maximum swing of V_{CE} and I_C . We have already learnt in Chapter 4 that in saturation region both the $B-E$ and $B-C$ junctions are forward biased and in cutoff region both of them are reverse biased. Hence, when a transistor is used as an amplifier, these regions are avoided. That is, V_{CE} is never allowed to fall below the saturation voltage of the transistor, $V_{CE(sat)}$. Similarly, I_C is never allowed to fall below, I_{CEO} . Thus, in practical transistor circuit the maximum swing of the operating point will not touch the x -axis and y -axis, rather its swing will be as shown in Fig.5-12. But, as the value of I_{CEO} is very small compared to the level I_C , in the first approximation, it can be neglected. For low power transistor the value

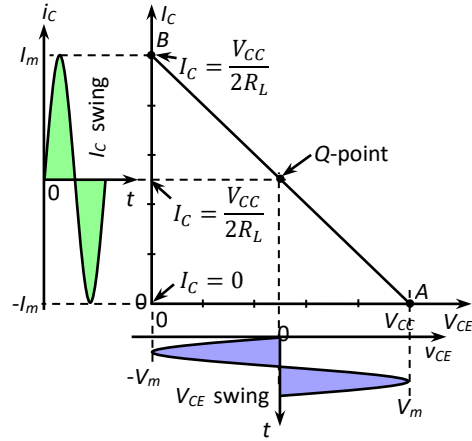


Fig.5-11: Swing of Q -point along the load line.

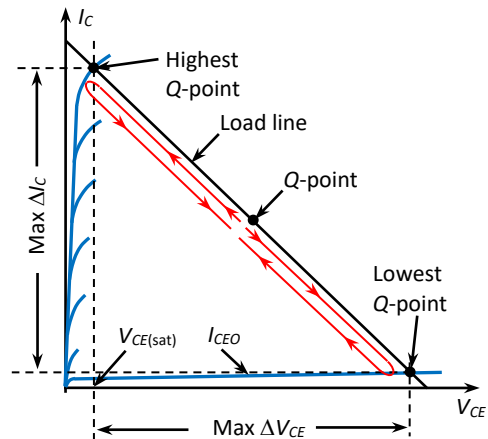


Fig.5-12: Possible max. swing of Q -point considering $V_{CE(sat)}$ and I_{CEO}

of $V_{CE(sat)} \approx 0.3 \text{ V}$. Hence, in the second approximation this value can also be neglected. Then, we will get the swing of the Q-point as shown in Fig.5-11

5.3.4 Selection of DC Operating Point

It is wise to set the DC operating point at the middle of the maximum swing span. Then, the circuit can amplify comparatively a larger input signal as shown in Fig.5-11. But, if we do not set the Q-point at the middle of the swing range, the peak swing will be limited by the lower portion of the load line as shown in Fig.5-13(a). Q-point can be set either by the value of I_C or by the value of V_{CE} . If one parameter is set at the middle of the load line the other parameter will automatically be set.

For ideal circuit (neglecting $V_{CE(sat)}$ and I_{CEO}) the Q-point will be set at the middle of the load line by any one of the following equations.

$$I_{CQ} = \frac{V_{CC}}{2R_L} = \frac{V_{CC}}{2R_C} \quad (5-5)$$

$$V_{CEQ} = \frac{V_{CC}}{2} \quad (5-6)$$

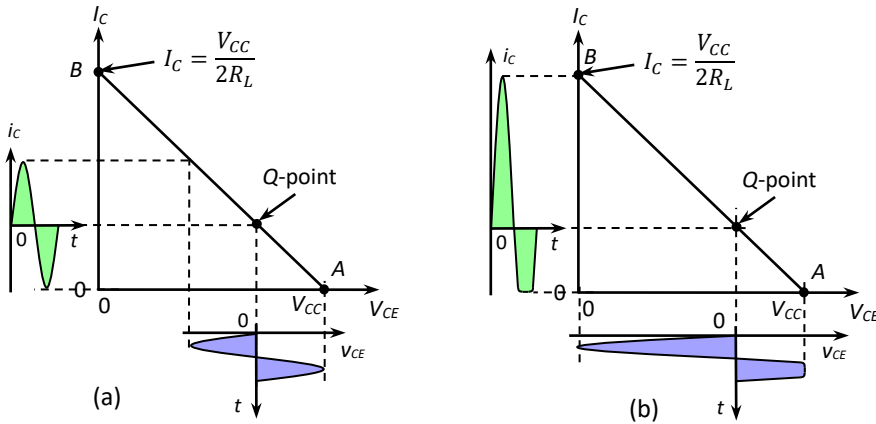


Fig.5-13: Result of setting Q-point not at the middle of the swing span, (a) Small swing of I_C and V_{CE} , and (b) One half-cycle is distorted for large swing of I_C and V_{CE}

On the other hand, for a practical circuit (considering $V_{CE(sat)}$ and I_{CEO}), we have to use either of the following equations to set the Q-point at the middle of swing span.

$$I_{CQ} = I_{CEO} + \frac{1}{2} \times \left(\frac{V_{CC} - V_{CE(sat)}}{R_C} - I_{CEO} \right) \quad (5-7)$$

$$V_{CEQ} = V_{CE(sat)} + \frac{(V_{CC} - I_{CEO}R_C) - V_{CE(sat)}}{2} \quad (5-8)$$

Example 5-3

TIP 2955 and TIP 3055 are two power transistors. The values of I_{CEO} and $V_{CE(sat)}$ are 0.7 mA and 3 V, respectively. Considering $V_{CC} = 30$ V and $R_L = 30 \Omega$, (i) Draw the load line, (ii) Maximum swing span and (iii) Position of the operating point. Show them in a sketch.

Solution:

(i) By plotting the points, $A(V_{CC}, 0) = (30 \text{ V}, 0)$ on the x -axis and $B(0, \frac{V_{CC}}{R_L}) = (0, \frac{30 \text{ V}}{30 \Omega}) = (0, 1 \text{ A})$ on the y -axis and joining them the load line is drawn on the Fig.5-14.

(ii) The swing span of V_{CE} :

Minimum value of V_{CE} , $V_{CE(min)} = V_{CE(sat)} = 3 \text{ V}$

Maximum value of V_{CE} , $V_{CE(max)} = V_{CC} - I_{CEO} R_L$

or, $V_{CE(max)} = 30 \text{ V} - (0.7 \text{ mA})(30 \Omega) = 29.98 \text{ V}$

Thus, the swing span of V_{CE} will be, $\Delta V_{CE} = V_{CE(max)} - V_{CE(min)}$

$\therefore \Delta V_{CE} = 29.98 \text{ V} - 3 \text{ V} = 26.98 \text{ V}$ [Ans.]

The swing span of I_C :

Minimum value of I_C ,

$$I_{C(min)} = I_{CEO} = 0.7 \text{ mA}$$

Maximum value of I_C

$$I_{C(max)} = \frac{V_{CC} - V_{CE(sat)}}{R_L} = \frac{30 \text{ V} - 3 \text{ V}}{30 \Omega} = 0.9 \text{ A}$$

Thus, the swing span of I_C will be,

$$\Delta I_C = I_{C(max)} - I_{C(min)} = 900 \text{ mA} - 0.7 \text{ mA} = 899.3 \text{ mA}$$
 [Ans.]

(iii) To get maximum swing, the Q-point must be set at the middle point. So the value of I_{CQ} and V_{CEQ} will be,

$$I_{CQ} = I_{CEO} + \frac{\Delta I_C}{2} = 0.7 \text{ mA} + \frac{899.3 \text{ mA}}{2} = (0.7 + 449.65) \text{ mA} = 450.35 \text{ mA}$$
 [Ans.]

$$V_{CEQ} = V_{CE(sat)} + \frac{\Delta V_{CE}}{2} = \frac{3 \text{ V} + 26.98 \text{ V}}{2} = (3 \text{ V} + 13.49 \text{ V}) = 16.49 \text{ V}$$
 [Ans.]

All of the results are shown in Fig.5-14.

Comments: The effect of I_{CEO} is insignificant and can be neglected.

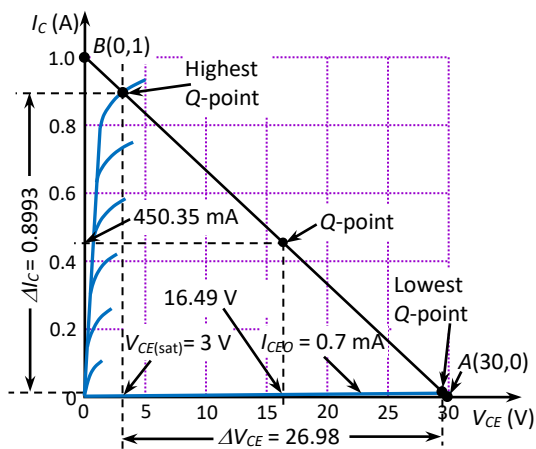


Fig.5-14: Graph for Example 4.3

5.4 Bias Stability (Stability of Operating Point)

The target of a biasing circuit is to provide proper bias of the transistor junctions and to keep the bias conditions fixed irrespective of the change of related parameters (temperature, β etc.). The bias point or the operating point is determined by the circuit components as well the parameters of the transistor. But the transistor parameters do not remain constant, rather they change due to the variation of temperature. Even, transistors with same number may have different parameters due to manufacturing tolerance. The parameter β is a prime example: it increases with temperature as well as with collector current. Again, every transistor may have any value of β within a range $\beta_{(min)}$ to $\beta_{(max)}$. All the variable parameters cause instability of the biasing condition and shift the operating point. To design a better biasing circuit we must have the knowledge of the parameters that are sensitive to temperature. These are listed below:

Beta (β): Beta increases with the increase in temperature.

Barrier voltage of B-E junction (V_{BE}): V_{BE} decreases with the increase in temperature. For Ge transistor its value decreases at the rate of about 1.6 mV/°C and for Si transistor the decrease rate is about 2.5 mV/°C.

Leakage current (I_{CBO}): For both Ge and Si transistor the value of I_{CBO} doubles for every 10°C rise in temperature.

The changes of these parameters, from the reference temperature (25°C) to (45°C), are illustrated in Fig.5-15.

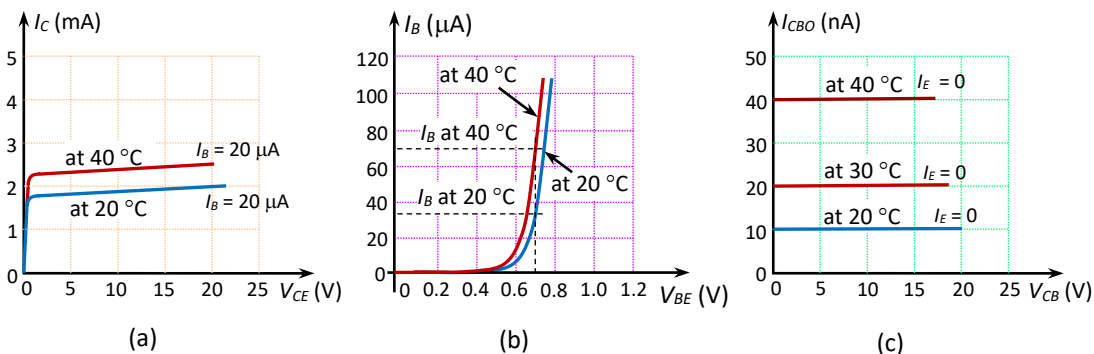


Fig.5-15: Effect of temperature (a) Effect of β increase, (b) Effect of V_{BE} decrease, and (c) Increase of I_{CBO} .

The value of I_C of a transistor depends on all of these parameters. To understand this, look at the following equation.

$$I_C = \beta I_B + (\beta + 1) I_{CBO} \quad (5-9)$$

I_C directly depends on β and I_{CBO} . Moreover, if V_{BE} decreases due to increases in temperature, I_B will increase which will further increase I_C .

5.4.1 Thermal Runaway:

The flow of collector current produces heat in the transistor. This rises the temperature of the transistor. With the rise in temperature, the values of β and I_{CBO} increase, and the value of V_{BE}

decreases. According to equation (5-9) the change in these parameters increase the value of I_C . The increase in I_C increases the temperature of the transistor further. If the biasing point of the transistor is not properly stabilized, this regenerative effect may increase I_C to an excessively high level. This excessive value of I_C produces a lot of heat, which is not possible for the transistor to withstand. Consequently, the transistor is destroyed. This type of self-destruction of an unstaibilzed transistor is called **thermal runaway**.

5.4.2 Stability Factor

Stability factor is a parameter that expresses how much the operating condition of a transistor circuit is stable. That is, a stability factor is a measure of how sensitive the collector bias current (I_{CQ}) is to the change in I_{CBO} , V_{BE} , and β due to temperature or any other reason. Thus, stability factor is defined for each of these parameters as:

$$S(I_{CBO}) = \left. \frac{\Delta I_C}{\Delta I_{CBO}} \right|_{V_{BE} \text{ and } \beta \text{ constant}} \quad (5-10)$$

$$S(V_{BE}) = \left. \frac{\Delta I_C}{\Delta V_{BE}} \right|_{I_{CBO} \text{ and } \beta \text{ constant}} \quad (5-11)$$

$$S(\beta) = \left. \frac{\Delta I_C}{\Delta \beta} \right|_{I_{CBO} \text{ and } V_{BE} \text{ constant}} \quad (5-12)$$

To illustrate, if the value of $S(I_{CBO})$ for a certain circuit is 20, then a change in I_{CBO} from 1 μA to 2 μA ($\Delta I_{CBO} = 1\mu\text{A}$) will cause a change in I_C equal to $\Delta I_C = S(I_{CBO})\Delta I_{CBO} = 20 \times (1\mu\text{A}) = 20\mu\text{A}$. An ideal circuit will have the value of stability factor of *zero*, implying no change in I_C for any change in the related parameters. The actual value of a practical bias circuit depends on the components used in the circuit and never be 0.

Each of the equations (5-10) to (5-12) gives the stability factor related to variation in one parameter only, so the total change in collector current over a certain temperature range can be calculated approximated by,

$$\Delta I_C \approx \Delta I_{CBO} \times S(I_{CBO}) + \Delta V_{BE} \times S(V_{BE}) + \Delta \beta \times S(\beta) \quad (5-13)$$

The expression is an approximation because all three parameters change simultaneously with temperature.

5.5 Biasing Circuits

In section 5.2 we have seen how a transistor can be biases properly using a single voltage source (battery or power supply). This is the simplest form of biasing circuit. To achieve the fixed operating point, there are many other biasing circuits. The names of different biasing circuits are:

- Fixed bias or base bias
- Base bias with emitter feedback or emitter bias
- Collector-to-base bias
- Voltage divider bias

5.6 Fixed Bias Circuit

Well, the simplest biasing circuit discussed in section 5.2 is called the *fixed bias*. For proper understanding of the calculation, the circuit is redrawn in Fig.5-16(a). The collector resistor used in transistor circuit is also called the load resistance. So, for biasing only a fixed resistance is used in the base circuit. That's why, this circuit is called *fixed bias* or *base bias*. The value of the supply voltage and R_B sets a base current, I_B . Depending on β of the transistor a collector current is developed. V_{CE} is determined by I_C , R_C and V_{CC} . As the target of the biasing circuit is to set a DC operating point or bias point, we will derive the equations that determine the position of the operating point.

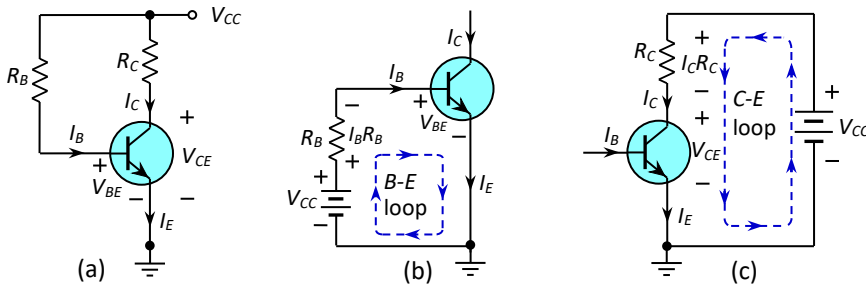


Fig.5-16: Fixed bias circuit, (a) Complete circuit, (b) B-E loop, and (c) C-E loop

Applying KVL in the base-emitter circuit [Fig.5-16(b)] we get,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

or,

$$I_B R_B = V_{CC} - V_{BE}$$

$\therefore$

$$I_B = I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} \quad (5-14)$$

Using this equation we can easily calculate the value of I_B . As, this value determines the position of Q-point, it is also represented by I_{BQ} .

If we know the value of β of the transistor, the collector current can be calculated as,

$$I_C = I_{CQ} = \beta I_B + I_{CEO} = \beta I_B + (\beta + 1) I_{CBO} \approx \beta I_B = \beta I_{BQ} \quad (5-15)$$

Now, applying KVL in the collector-emitter circuit [Fig.5-16(c)] we get,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$\therefore$

$$V_{CE} = V_{CEQ} = V_{CC} - I_C R_C \quad (5-16)$$

Using the equations (5-14) to (5-16), we can calculate the value of I_B , I_C , and V_{CE} corresponding the operating point of the base bias circuit. That's why, these collector current is represented by I_{CQ} and the collector-emitter voltage is represented by V_{CEQ} .

5.6.1 Determination of Load Line of Fixed Bias Circuit

Determination process of load line for fixed bias circuit has already been discussed in Section 5.2. Load line can be drawn using Equ.(5-16). When $I_C = 0$, $V_{CE} = V_{CC}$ and when $V_{CE} = 0$, $I_C = V_{CC}/R_C$. Thus, the load line starts from V_{CC} on the x -axis and ends to V_{CC}/R_C on the y -axis.

5.6.2 Saturation of Transistor in Fixed Bias Circuit

In fixed bias circuit, if a resistor of any value is connected to the base, the transistor will never go to cutoff. This is true for all basing circuits that will be discussed later. That is, by using any inaccurate biasing components, there is no chance of driving the transistor to cutoff. But if we use incorrect values of biasing resistors, transistor may become saturated. That's why in transistor biasing, we should be aware of saturation of the transistor, because saturation causes distortion. Transistors in base bias circuit may be saturated due to very large value of β and R_C , and very small value of R_B (considering V_{CC} constant). In servicing (repair) electronic circuit with base bias, we may replace the transistor. If the new transistor has a very large value of β the transistor may be saturated. Similarly, if we replace R_B by a very low value resistor, or R_C by a very high value resistor the transistor may also be saturated. To make the concept clear refer to Example 5-4. The saturation current of base bias will be approximately,

$$I_{C(sat)} = \frac{V_{CC}}{R_C} \quad (5-17)$$

Example 5-4

Assuming $V_{CC} = 12\text{ V}$, $R_B = 180\text{ k}\Omega$, $R_C = 1\text{ k}\Omega$ and a silicon transistor with $\beta = 100$ for the circuit of Fig.5-16(a), calculate the operating point and determine the load line.

Solution:

Using Equ. (5-14), the base current of the bias point is,

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{180\text{ k}\Omega} \approx 62.8\text{ }\mu\text{A}$$

Using Equ. (5-15), collector current at the bias point is,

$$I_{CQ} \approx \beta I_{BQ} = 100 \times (62.8\text{ }\mu\text{A}) \approx 6.28\text{ mA [Ans.]}$$

Using Equ.(5-16), collector-emitter voltage at the operating point is,

$$\therefore V_{CEQ} = 12\text{ V} - (6.28\text{ mA}) \times (1\text{ k}\Omega) = 5.72\text{ V [Ans.]}$$

The calculated operating point and the load line is drawn in Fig.5-17.

The saturation current can be calculated from Equ.(5-16) as,

$$V_{CEQ} = V_{CC} - I_C R_C \quad \text{or,} \quad 0.5\text{ V} = 12\text{ V} - I_C \times 1\text{ k}\Omega$$

$$\therefore I_C = \frac{12\text{ V} - 0.5\text{ V}}{1\text{ k}\Omega} = 11.5\text{ mA [Ans.]}$$

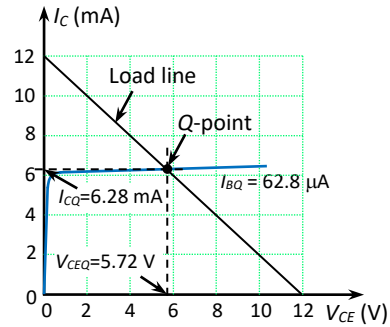


Fig.5-17: Load line and Q-point of example 5-4

Example 5-5

A fixed bias circuit is designed with $V_{CC} = 12\text{ V}$ and a transistor with $\beta = 100$ which is shown in Fig.5-18. In this design the operating point is set at the middle of the load line. Calculate the operating point. To check how higher values of β and R_C and lower value of R_B saturate the transistor, calculate the operating points considering $\beta = 200$, $R_B = 270\text{ k}\Omega$ and $R_C = 6.2\text{ k}\Omega$, one by one.

Solution:

Using Equ.(5-14),
$$I_{BQ} = \frac{12\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} \approx 20.1786\text{ }\mu\text{A}$$

Using Equ.(5-15),
$$I_{CQ} = 100 \times 20.1786\text{ }\mu\text{A} \approx 2.0179\text{ mA}$$

Using Equ.(5-16),
$$V_{CEQ} = 12\text{ V} - (2.0179\text{ mA}) \times (3\text{ k}\Omega) \approx 5.95\text{ V}$$

As, $V_{CEQ} \approx 1/2 (V_{CC})$, the operating is at the middle of the load line.

Operating point for $\beta = 200$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} \approx 20.1786\text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_B = 200 \times 20.1786\text{ }\mu\text{A} \approx 4.0357\text{ mA}$$

$$\therefore V_{CEQ} = 12\text{ V} - (4.0357\text{ mA}) \times (3\text{ k}\Omega) \approx -0.11\text{ V [less than } 0.3\text{ V means saturation]}$$

Operating point for $R_B = 270\text{ k}\Omega$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{270\text{ k}\Omega} \approx 41.8519\text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_B = 100 \times 41.8519\text{ }\mu\text{A} \approx 4.1852\text{ mA}$$

$$\therefore V_{CEQ} = 12\text{ V} - (4.1852\text{ mA}) \times (3\text{ k}\Omega) \approx -0.56\text{ V [saturation]}$$

Operating point for $R_C = 6.2\text{ k}\Omega$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} \approx 20.1786\text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_B = 100 \times 20.1786\text{ }\mu\text{A} \approx 2.0179\text{ mA}$$

$$\therefore V_{CEQ} = V_{CC} - I_C R_C = 12\text{ V} - (2.0179\text{ mA}) \times (6.2\text{ k}\Omega) \approx -0.51\text{ V [saturation]}$$

Comments: From this example, it is clear that higher value of β and R_C , and lower value of R_B can saturate a transistor in fixed bias circuit.

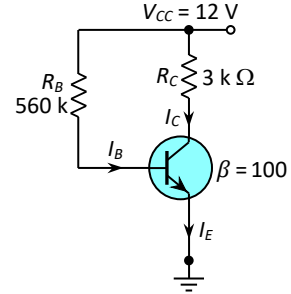


Fig.5-18: Fixed bias circuit for Example 5-5

5.6.3 Calculation of Stability Factors of Fixed Bias Circuit

To calculate the stability factor we need the exact expression of the collector current,

From Equ.(5-15),
$$I_C = \beta I_B + (\beta + 1)I_{CBO} \quad (5-18)$$

Putting the value of I_B from Equ.(5-14) into Equ.(5-18) we get,

$$I_C = \frac{\beta(V_{CC} - V_{BE})}{R_B} + (\beta + 1)I_{CBO} \quad (5-19)$$

$S(I_{CBO})$ can be calculated by differentiating Equ.(5-19) with respect to I_{CBO} considering V_{BE} and β constant,

$$S(I_{CBO}) = \left. \frac{\delta I_C}{\delta I_{CBO}} \right|_{V_{BE}, \beta} = \frac{\delta}{\delta I_{CBO}} \frac{\beta(V_{CC} - V_{BE})}{R_B} + \frac{\delta}{\delta I_{CBO}} (\beta + 1)I_{CBO}$$

$$\therefore \boxed{S(I_{CBO}) = 0 + (\beta + 1) = (\beta + 1)} \quad (5-20)$$

Now, differentiating Equ.(5-19) with respect to V_{BE} for constant values of I_{CBO} and β ,

$$S(V_{BE}) = \left. \frac{\delta I_C}{\delta V_{BE}} \right|_{I_{CBO}, \beta} = \frac{\delta}{\delta V_{BE}} \frac{\beta(V_{CC} - V_{BE})}{R_B} + \frac{\delta}{\delta V_{BE}} (\beta + 1)I_{CBO}$$

$$\text{or,} \quad S(V_{BE}) = \frac{\delta}{\delta V_{BE}} \frac{\beta V_{CC}}{R_B} - \frac{\delta}{\delta V_{BE}} \frac{\beta V_{BE}}{R_B} + \frac{\delta}{\delta V_{BE}} (\beta + 1)I_{CBO}$$

$$\text{or,} \quad S(V_{BE}) = 0 - \frac{\beta}{R_B} + 0$$

$$\therefore \boxed{S(V_{BE}) = -\frac{\beta}{R_B}} \quad (5-21)$$

Again, differentiating Equ.(5-19) with respect to β for constant I_{CBO} and V_{BE} ,

$$S(\beta) = \left. \frac{\delta I_C}{\delta \beta} \right|_{I_{CBO}, V_{BE}} = \frac{\delta}{\delta \beta} \frac{\beta(V_{CC} - V_{BE})}{R_B} + \frac{\delta}{\delta \beta} (\beta + 1)I_{CBO}$$

$$\text{or,} \quad S(\beta) = \frac{\delta}{\delta \beta} \frac{\beta V_{CC}}{R_B} - \frac{\delta}{\delta \beta} \frac{\beta V_{BE}}{R_B} + \frac{\delta}{\delta \beta} (\beta + 1)I_{CBO}$$

$$\text{or,} \quad S(\beta) = \frac{\delta}{\delta \beta} \frac{\beta V_{CC}}{R_B} - \frac{\delta}{\delta \beta} \frac{\beta V_{BE}}{R_B} + \frac{\delta}{\delta \beta} (\beta + 1)I_{CBO}$$

$$\text{or,} \quad S(\beta) = \frac{V_{CC}}{R_B} - \frac{V_{BE}}{R_B} + I_{CBO}$$

$$\text{or,} \quad S(\beta) = \frac{V_{CC}}{R_B} - 0 + I_{CBO} \quad [\because V_{BE} \text{ is very small}]$$

$$\therefore I_{CBO} \text{ is very small,} \quad \boxed{S(\beta) = \frac{V_{CC}}{R_B} + I_{CBO} \approx \frac{V_{CC}}{R_B} \approx I_B = \frac{I_C}{\beta}} \quad (5-22)$$

Example 5-6

The following table shows the parameter values for a typical transistor at temperature 10°C and 50°C. Find the total change in DC collector current over the temperature range for the base bias circuit of Fig.5-16(a). Assume the values of $V_{CC} = 16 \text{ V}$ and $R_B = 260 \text{ k}\Omega$.

	10°C	50°C	Change
I_{CBO}	0.01 μA	0.16 μA	$\Delta I_{CBO} = 0.15 \mu\text{A}$
V_{BE}	0.74 V	0.64 V	$\Delta V_{BE} = -0.1 \text{ V}$
β	60	90	$\Delta\beta = 30$

Solution:

Using Equ.(5-14), base current at 10°C, $I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 \text{ V} - 0.74 \text{ V}}{260 \text{ k}\Omega} \approx 58.7 \mu\text{A}$

Using Equ.(5-15), collector current at 10°C is,

$$I_C = \beta I_B + (\beta + 1)I_{CBO} = 60 \times 58.7 \mu\text{A} + (60 + 1) \times (0.01 \mu\text{A}) \approx 3.52 \text{ mA}$$

Using Equ.(5-13), $\Delta I_C \approx \Delta I_{CBO} \times S(I_{CBO}) + \Delta V_{BE} \times S(V_{BE}) + \Delta\beta \times S(\beta)$

Using equations (5-20) to (5-22), $\Delta I_C \approx \Delta I_{CBO} \times (\beta + 1) + \Delta V_{BE} \times \left(-\frac{\beta}{R_B}\right) + \Delta\beta \times \frac{I_C}{\beta}$

$$\text{or,} \quad \Delta I_C = (0.15 \mu\text{A})(60 + 1) + (-0.1 \text{ V}) \times \left(-\frac{60}{260 \text{ k}\Omega}\right) + 30 \times \frac{3.52 \text{ mA}}{60}$$

$$\text{or,} \quad \Delta I_C = 9.15 \mu\text{A} + 23.1 \mu\text{A} + 1760 \mu\text{A}$$

$$\therefore \Delta I_C = 1792.5 \mu\text{A} [\text{Ans.}]$$

Comments: As the change in β is large, the change in collector current mainly depends on it.

5.6.4 Effect of β_{min} and β_{max}

When the transistor's DC current gain (β) is known, it is easy to determine the circuit bias condition as in Example 5-4. Similarly, if the value of (β) is given, we can easily design a bias circuit. However, in practical situations the manufacturer's datasheet of transistor does not mention a single value of β . We know, transistors are usually identified by type numbers. For a specific number of a transistor, datasheet mentions the value of β in a range. That is, the minimum value of β and the maximum value of β (written as Min. h_{FE} and Max. h_{FE}) and sometimes a typical value of β . In circuit analysis, it is sometimes convenient to use the typical value of β of the transistor. But, we should be aware of the effect of β_{min} and β_{max} .

To illustrate the situation, let us consider the transistor BD145. The values of $\beta_{(min)}$ and $\beta_{(max)}$ are 40 and 250, respectively (shown in the Datasheet in Chapter 4). We will first design a fixed bias circuit considering the value of β in between the range, say $\beta = 145$ and then, will see the effect of β_{min} and β_{max} . To design the bias circuit, assume $I_{CQ} = 2 \text{ mA}$, $V_{CC} = 12 \text{ V}$.

To set the Q-point at the middle of the load line, the value of the collector-emitter voltage should be,

$$V_{CEQ} = \frac{12 \text{ V}}{2} = 6 \text{ V}$$

the voltage drop across the collector resistor will be,

$$V_{RC} = V_{CC} - V_{CEQ} = 12 \text{ V} - 6 \text{ V} = 6 \text{ V}$$

Therefore, the value of collector resistor will be,

$$R_C = \frac{V_{RC}}{I_{CQ}} = \frac{6 \text{ V}}{2 \text{ mA}} = 3 \text{ k}\Omega$$

The value of base current will be,

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2 \text{ mA}}{145} = 13.793 \text{ }\mu\text{A}$$

So, the value of base resistor should be (using Equ.5-14),

$$R_B = \frac{V_{CC} - V_{BE}}{I_{BQ}} = \frac{12 \text{ V} - 0.7 \text{ V}}{13.793 \text{ }\mu\text{A}} = 819 \text{ k}\Omega \text{ (use standard } 820 \text{ k}\Omega)$$

The designed biasing circuit is shown in Fig.5-19(a).

For the designed value of R_B , actual base current will be,

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{820 \text{ k}\Omega} = 13.781 \text{ }\mu\text{A}$$

Hence, the actual operating will be,

$$I_{CQ} = I_{BQ}\beta = (13.781 \text{ }\mu\text{A})145 = 1.998 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C = 12 \text{ V} - (1.998 \text{ mA})(3 \text{ k}\Omega) \approx 6 \text{ V}$$

The load line and the operating point are shown in Fig.5-19(b).

Now, to see the effect β_{min} and β_{max} , we will recalculate the operating point considering, $\beta_{min} = 40$ and $\beta_{max} = 250$. These can be done using the above two equations.

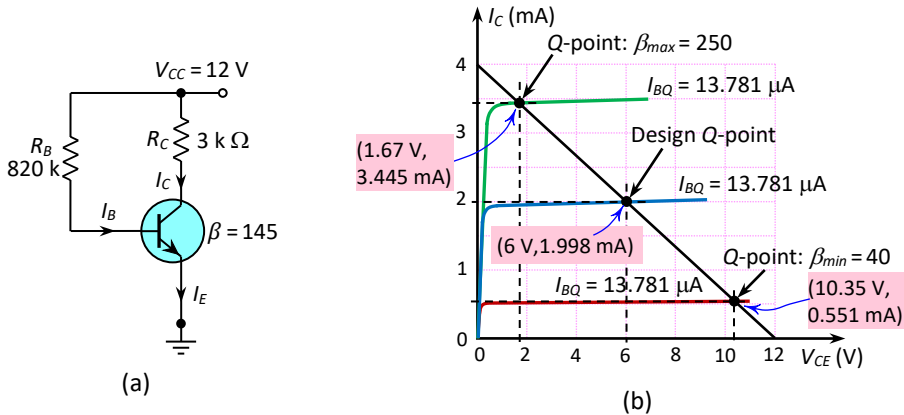


Fig.5-19: Effect of β_{min} and β_{max} operating point, (a) Designed circuit with $\beta = 145$, and (b) Load line and operating points

For $\beta_{min} = 40$:

$$I_{CQ} = (13.781 \text{ }\mu\text{A})40 = 0.551 \text{ mA}$$

$$V_{CEQ} = 12 \text{ V} - (0.551 \text{ mA})(3 \text{ k}\Omega) \approx 10.35 \text{ V}$$

For $\beta_{max} = 250$:

$$I_{CQ} = (13.781 \text{ }\mu\text{A}) 250 = 3.445 \text{ mA}$$

$$V_{CEQ} = 12 \text{ V} - (3.445 \text{ mA})(3 \text{ k}\Omega) \approx 1.67 \text{ V}$$

The variation of the Q -point for β_{min} and β_{max} is also shown in the same figure.

The variation is so large that the circuit will not work properly with β_{min} and β_{max} .

5.7 Fixed Bias with Emitter Feedback

This biasing circuit is exactly same as fixed bias, but an additional resistor (R_E) is used in the emitter terminal as shown in Fig.5-20(a). This biasing circuit is also called **emitter bias**. The function of R_E is to provide better stability of the DC operating point than the fixed bias network. The more stable a configuration, the less its response will change due to undesirable changes in temperature and device parameter variations. The emitter resistor applies a voltage to the input circuit which is proportional to the output current. This mechanism is called **feedback**, which will be studied later. For this reason R_E is called emitter feedback resistor.

Calculation of operating point:

From the forward biased B - E circuit, using KVL [Fig.5-20(b)] we get,

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\text{or,} \quad V_{CC} - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0 \quad [\because I_E = (\beta + 1)I_B]$$

$$\text{or,} \quad I_B [R_B + (\beta + 1)R_E] = V_{CC} - V_{BE}$$

$$\therefore \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \quad (5-23)$$

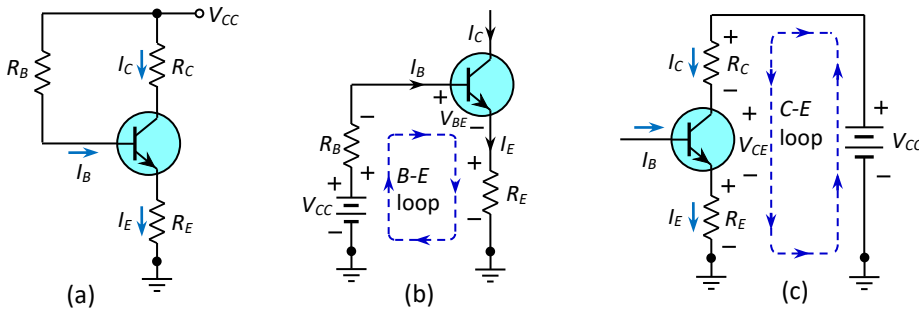


Fig.5-20: Fixed bias with emitter resistor, (a) Complete circuit, (b) B - E loop, and (c) C - E loop

Once, I_B is found, the collector current can be calculated as,

$$I_C = \beta I_B + I_{CEO} \approx \beta I_B \quad (5-24)$$

We can also calculate I_C , without calculating I_B as,

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\text{or,} \quad V_{CC} - \frac{I_C}{\beta} R_B - V_{BE} - I_C R_E = 0 \quad [\because I_B = \frac{I_C}{\beta} \text{ and } I_E \approx I_C]$$

$$\text{or,} \quad \left(\frac{R_B}{\beta} + R_E \right) I_C = V_{CC} - V_{BE}$$

$$\therefore \quad I_C = \frac{V_{CC} - V_{BE}}{R_B/\beta + R_E} \quad (5-25)$$

Now, applying KVL, in the C - E loop of Fig.5-20(c) we get,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\text{or,} \quad V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0 \quad [\because I_E \approx I_C]$$

$$\therefore \quad V_{CE} = V_{CC} - I_C (R_C + R_E) \quad (5-26)$$

Using equations (5-23), (5-24) and (5-26), the operating point of the circuit can easily be calculated.

5.7.1 Stabilization mechanism

From the forward biased B - E circuit we have found,

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0 \quad (5-27)$$

$$\text{or,} \quad I_B = \frac{V_{CC} - V_{BE} - I_E R_E}{R_B} \quad (5-28)$$

For fixed values of V_{CC} , V_{BE} , and R_E the value of I_B will be fixed. Now, if due to the increase in temperature collector current I_C increases, I_E will also increase. Hence, according to Equ.(5-27), I_B will also decrease. With this decrease in I_B collector current I_C will decrease. In this way, any attempt to increase I_C will automatically be adjusted. On the other hand, if I_C decrease from the DC bias point due to decrease in temperature, $I_E R_E$ will decrease, I_B will increase and I_C will also increase.

5.7.2 Emitter Resistance Feedback to Input

From Equ.(5-27), we see that the emitter resistor R_E also affects the input (base) current. Actually, R_E is fed back to the input circuit. This equation can be represented by an equivalent circuit as in Fig.5-21(a) and (b). Fig.5-21(b) will produce the same result as Equ.(5-27). From this figure it is clear that R_E is reflected to the base circuit being multiplied by a factor $(\beta + 1)$. In other words, the emitter resistor, which is part of the collector-emitter loop, appears as $(\beta + 1)R_E$ in the base-emitter loop. Because, β has a large value (typically 50 or more), the emitter resistor, multiplied by a large number, appears to the base circuit. Therefore, the resistance in the base terminal is increased. In general, for the configuration of Fig.5-21(b), the input resistance (R_i) seen from the base terminal will be,

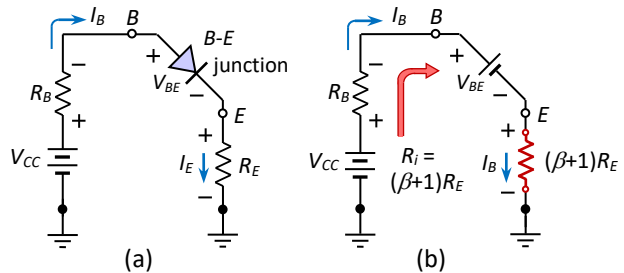


Fig.5-21: Equivalent circuit of B - E junction side,
(a) B - E junction is represented by a diode,
(b) B - E junction is represented by its barrier

$$R_i = (\beta + 1)R_E + r_D \approx (\beta + 1)R_E \quad (5-29)$$

Actually, the static resistance (r_D) of the forward biased B - E junction will be effective in series with $(\beta + 1)R_E$. But, as its value is very low, in the analysis of bias circuit it can be neglected.

5.7.3 Determination of Load Line of Emitter Bias Circuit

Determination process of load line for emitter bias circuit is a little different from the fixed bias circuit. Load line can be drawn using Equ.(5-26) as described in Section 5.3.

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (5-30)$$

Point A: When, $I_C = 0$, $V_{CE} = V_{CC}$ and

Point B: When $V_{CE} = 0$, $I_C = V_{CC}/(R_C + R_E)$. Point A($V_{CC}, 0$) lies on the x -axis and the point B($0, V_{CC}/(R_C + R_E)$) lies on the y -axis. Connecting these two points, we can draw the load line.

5.7.4 Saturation of Transistor in Emitter Bias Circuit

We have already learnt that there is very little chance to drive a transistor in cutoff, by using any inaccurate values of biasing resistors or by changing β unless the base terminal is open. If we need to change components of this circuit, we should be aware of saturation of the transistor. We will try to use components of designed values. But in any case, if we use inaccurate value of components, transistor in emitter bias circuit may also be saturated. Here, the transistors may be saturated for very large value of β and R_C , and very small value of R_B . Any change in R_E cannot saturate the transistor. These saturation effects are explained further in Example 5-7. The value of current that will saturate the transistor is,

$$I_{C(sat)} = \frac{V_{CC}}{R_C + R_E} \quad (5-31)$$

Example 5-7

An emitter bias circuit is designed with $V_{CC} = 12$ V and a transistor with $\beta = 100$ which is shown in Fig.5-22. In this design the operating point is set at the middle of the load line. Calculate the operating point. To check how the higher values of β and R_C and the lower value of R_B saturate the transistor, calculate the operating points considering $\beta = 260$, $R_B = 180$ k Ω and $R_C = 6.2$ k Ω , one by one. Also to prove that R_E cannot saturate the transistor repeat the problem for $R_E = 0$ and $R_E = 500$ k Ω .

Solution:

Operating point of the given circuit:

$$\text{Using Equ.(5-23), } I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (100 + 1) 1 \text{ k}\Omega} \approx 19.7898 \mu\text{A}$$

$$\text{Using Equ.(5-24), } I_{CQ} = \beta I_B = 100 \times 19.7898 \mu\text{A} \approx 1.979 \text{ mA}$$

$$\text{Using Equ.(5-26), } V_{CEQ} = 12 \text{ V} - (1.979 \text{ mA}) \times (2 \text{ k}\Omega + 1 \text{ k}\Omega) \approx 6.06 \text{ V}$$

As, $V_{CEQ} = 6.06 \text{ V} \approx V_{CC}/2$, the operating point of the given circuit must be at the middle of the load line.

Operating point for $\beta = 260$:

Using the same equations,

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (260 + 1) 1\text{k}\Omega} \approx 15.4583 \mu\text{A}$$

$$\therefore I_{CQ} = \beta I_B = 260 \times 15.4583 \mu\text{A} \approx 4.0192 \text{ mA}$$

$$\text{and, } V_{CE} = 12 \text{ V} - (4.0192 \text{ mA}) \times (2 \text{ k}\Omega + 1 \text{ k}\Omega) \approx -0.057 \text{ V [Saturated]}$$

Operating point for $R_B = 180 \text{ k}\Omega$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{180 \text{ k}\Omega + (100 + 1) 1\text{k}\Omega} \approx 40.2135 \mu\text{A}$$

$$\therefore I_{CQ} = \beta I_B = 100 \times 40.2135 \mu\text{A} \approx 4.0214 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 12 \text{ V} - (4.0214 \text{ mA}) \times (2 \text{ k}\Omega + 1 \text{ k}\Omega) \approx -0.064 \text{ V [Saturated]}$$

Operating point for $R_C = 6.2 \text{ k}\Omega$:

$$I_{BQ} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (100 + 1) 1\text{k}\Omega} \approx 19.7898 \mu\text{A}$$

$$\therefore I_{CQ} = \beta I_B = 100 \times 19.7898 \mu\text{A} \approx 1.979 \text{ mA}$$

$$V_{CE} = 12 \text{ V} - (1.979 \text{ mA}) \times (6.2 \text{ k}\Omega + 1 \text{ k}\Omega)$$

$$\text{or, } V_{CE} \approx -2.2488 \text{ V} - 0.27 \text{ V [Saturated]}$$

Operating point for $R_E = 0$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (100 + 1) \times 0} \approx 24.0426 \mu\text{A}$$

$$\therefore I_{CQ} = \beta I_B = 100 \times 24.0426 \mu\text{A} \approx 2.4043 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_C(R_C + R_E) = 12 \text{ V} - (2.4043 \text{ mA}) \times (2 \text{ k}\Omega + 0) \approx 7.19 \text{ V } 4.79 \text{ V}$$

Operating point for $R_E = 500 \text{ k}\Omega$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (100 + 1) 500 \text{ k}\Omega} \approx 0.2217 \mu\text{A}$$

$$\therefore I_{CQ} = \beta I_B = 100 \times 0.2217 \mu\text{A} \approx 0.0222 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_C(R_C + R_E) = 12 \text{ V} - (0.0222 \text{ mA}) \times (2 \text{ k}\Omega + 500 \text{ k}\Omega) \approx 11.93 \text{ V}$$

Comments: Although, from Equ.(5-26) it is apparent that the higher values of R_E will saturate the transistor. But from this example, we find that higher values of R_E tend to shift the transistor to cutoff, as the base current is reduced drastically!!

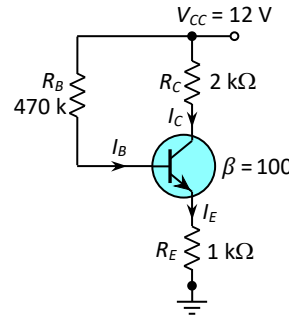


Fig.5-22: Emitter bias for Example 5-7

5.7.5 Calculation of Stability Factors of Emitter Bias Circuit

From the B - E circuit of Fig.5-20(b), we have found that,

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\text{or, } V_{CC} - I_B R_B - V_{BE} - I_B R_E - I_C R_E = 0 [\because I_E = I_C + I_B]$$

or,

$$I_B(R_B + R_E) = V_{CC} - V_{BE} - I_C R_E$$

∴

$$I_B = \frac{V_{CC} - V_{BE} - I_C R_E}{R_B + R_E} \quad (5-32)$$

From the *B-C* circuit we have also found,

$$I_C = \beta I_B + (\beta + 1)I_{CBO} \quad (5-33)$$

Putting the value of I_B from Equ.(5-32) into Equ.(5-33) we get,

$$I_C = \frac{\beta(V_{CC} - V_{BE} - I_C R_E)}{R_B + R_E} + (\beta + 1)I_{CBO}$$

or,

$$I_C(R_B + R_E) = \beta(V_{CC} - V_{BE} - I_C R_E) + (R_B + R_E)(\beta + 1)I_{CBO}$$

or,

$$I_C(R_B + R_E + \beta R_E) = \beta(V_{CC} - V_{BE}) + (R_B + R_E)(\beta + 1)I_{CBO}$$

∴

$$I_C = \frac{\beta(V_{CC} - V_{BE}) + (R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \quad (5-34)$$

Differentiating Equ.(5-34) with respect to I_{CBO} for constant V_{BE} and β , we can determine the stability factor $S(I_{CBO})$ as,

$$S(I_{CBO}) = \left. \frac{\delta I_C}{\delta I_{CBO}} \right|_{V_{BE}, \beta} = \frac{\delta}{\delta I_{CBO}} \left[\frac{\beta(V_{CC} - V_{BE}) + (R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

or,

$$S(I_{CBO}) = \frac{\delta}{\delta I_{CBO}} \left[\frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E} \right] + \frac{\delta}{\delta I_{CBO}} \left[\frac{(R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

or,

$$S(I_{CBO}) = 0 + \frac{(R_B + R_E)(\beta + 1)}{R_B + (\beta + 1)R_E} = \frac{(R_B + R_E)(\beta + 1)}{R_B + (\beta + 1)R_E}$$

∴

$$S(I_{CBO}) = \frac{(R_B + R_E)(\beta + 1)}{R_B + (\beta + 1)R_E} \quad (5-35)$$

Now, differentiating Equ.(5-34) with respect to V_{BE} for constant I_{CBO} and β we get,

$$S(V_{BE}) = \left. \frac{\delta I_C}{\delta V_{BE}} \right|_{I_{CBO}, \beta} = \frac{\delta}{\delta V_{BE}} \left[\frac{\beta(V_{CC} - V_{BE}) + (R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

or,

$$S(V_{BE}) = \frac{\delta}{\delta V_{BE}} \left[\frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E} \right] + \frac{\delta}{\delta V_{BE}} \left[\frac{(R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

or,

$$S(V_{BE}) = \frac{\delta}{\delta V_{BE}} \left[\frac{\beta V_{CC}}{R_B + (\beta + 1)R_E} \right] - \frac{\delta}{\delta V_{BE}} \left[\frac{\beta V_{BE}}{R_B + (\beta + 1)R_E} \right] + 0$$

or,

$$S(V_{BE}) = 0 - \frac{\delta}{\delta V_{BE}} \left[\frac{\beta V_{BE}}{R_B + (\beta + 1)R_E} \right]$$

or,

$$S(V_{BE}) = - \frac{\beta}{R_B + (\beta + 1)R_E}$$

∴

$$S(V_{BE}) = - \frac{\beta}{R_B + (\beta + 1)R_E} \quad (5-36)$$

Again, differentiating the same equation with respect to β for constant I_{CBO} and V_{BE} ,

$$S(\beta) = \left. \frac{\delta I_C}{\delta \beta} \right|_{I_{CBO}, V_{BE}} = \frac{\delta}{\delta \beta} \left[\frac{\beta(V_{CC} - V_{BE}) + (R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

$$\text{or,} \quad S(\beta) = \frac{\delta}{\delta \beta} \left[\frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E} \right] + \frac{\delta}{\delta \beta} \left[\frac{(R_B + R_E)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_E} \right]$$

Since, the differentiating terms have variables in numerator and denominator, we have to use u/v method. Let, $[\beta(V_{CC} - V_{BE})]/[R_B + (\beta + 1)R_E] = f_1$, and $[(R_B + R_E)(\beta + 1)I_{CBO}]/[R_B + (\beta + 1)R_E] = f_2$.

$$\therefore \quad S(\beta) = \frac{\delta}{\delta \beta} f_1 + \frac{\delta}{\delta \beta} f_2$$

$$\text{Now, let} \quad u = \beta(V_{CC} - V_{BE}) \quad \text{and} \quad v = R_B + R_E + \beta R_E$$

$$\therefore \quad \delta u / \delta \beta = (V_{CC} - V_{BE}) \quad \text{and} \quad \delta v / \delta \beta = R_E$$

$$\therefore \quad \frac{\delta}{\delta \beta} f_1 = \frac{v \left(\frac{\delta u}{\delta \beta} \right) - u \left(\frac{\delta v}{\delta \beta} \right)}{v^2} = \frac{(R_B + R_E + \beta R_E)(V_{CC} - V_{BE}) - \beta(V_{CC} - V_{BE})R_E}{[R_B + (\beta + 1)R_E]^2}$$

$$\therefore \quad \frac{\delta}{\delta \beta} f_1 = \frac{(R_B + R_E)(V_{CC} - V_{BE}) + \beta R_E(V_{CC} - V_{BE}) - \beta(V_{CC} - V_{BE})R_E}{[R_B + (\beta + 1)R_E]^2}$$

$$\therefore \quad \frac{\delta}{\delta \beta} f_1 = \frac{(R_B + R_E)(V_{CC} - V_{BE})}{[R_B + (\beta + 1)R_E]^2} \approx \frac{(R_B + R_E)V_{CC}}{[R_B + (\beta + 1)R_E]^2}$$

$$\text{Again, let} \quad u = (R_B + R_E)(\beta + 1)I_{CBO} \quad \text{and} \quad v = R_B + (\beta + 1)R_E$$

$$\therefore \quad \delta u / \delta \beta = (R_B + R_E)I_{CBO} \quad \text{and} \quad \delta v / \delta \beta = R_E$$

$$\therefore \quad \frac{\delta}{\delta \beta} f_2 = \frac{v \left(\frac{\delta u}{\delta \beta} \right) - u \left(\frac{\delta v}{\delta \beta} \right)}{v^2} = \frac{[R_B + (\beta + 1)R_E](R_B + R_E)I_{CBO} - (R_B + R_E)(\beta + 1)I_{CBO}R_E}{[R_B + (\beta + 1)R_E]^2}$$

$$\therefore, \quad \frac{\delta}{\delta \beta} f_2 = \frac{R_B(R_B + R_E)I_{CBO} + (\beta + 1)R_E(R_B + R_E)I_{CBO} - (R_B + R_E)(\beta + 1)I_{CBO}R_E}{[R_B + (\beta + 1)R_E]^2}$$

$$\text{or,} \quad \therefore \quad \frac{\delta}{\delta \beta} f_2 = \frac{R_B(R_B + R_E)I_{CBO}}{[R_B + (\beta + 1)R_E]^2}$$

$$\therefore \quad S(\beta) = \frac{\delta}{\delta \beta} f_1 + \frac{\delta}{\delta \beta} f_2 = \frac{(R_B + R_E)(V_{CC} - V_{BE})}{[R_B + (\beta + 1)R_E]^2} + \frac{R_B(R_B + R_E)I_{CBO}}{[R_B + (\beta + 1)R_E]^2}$$

$$\text{or,} \quad S(\beta) = \frac{(R_B + R_E)(V_{CC} - V_{BE}) + R_B(R_B + R_E)I_{CBO}}{[R_B + (\beta + 1)R_E]^2}$$

$$\text{or,} \quad S(\beta) = \frac{(R_B + R_E)(V_{CC} - V_{BE} + R_B I_{CBO})}{[R_B + (\beta + 1)R_E]^2} \approx \frac{(R_B + R_E)(V_{CC} - V_{BE})}{[R_B + (\beta + 1)R_E]^2}$$

$$\therefore \quad \boxed{S(\beta) = \frac{(R_B + R_E)(V_{CC} - V_{BE})}{[R_B + (\beta + 1)R_E]^2}} \quad (5-37)$$

If we compare the three stability factors of this circuit with those of base bias circuit, we find that the stability factors for this circuit have been reduced by a term $(\beta + 1)R_E$. This is due to the feedback of the emitter resistor to the input circuit.

Example 5-8

Fig.5-23(a) shows a fixed bias circuit with emitter resistor. The circuit has been designed to set the operating point at the middle of load line considering a typical β value of 145. Calculate the load line and operating point of this circuit. If the transistor has $\beta_{min} = 40$ and $\beta_{max} = 250$, calculate the lowest and the highest operating point. Compare the results with the results of Fig.5-19.

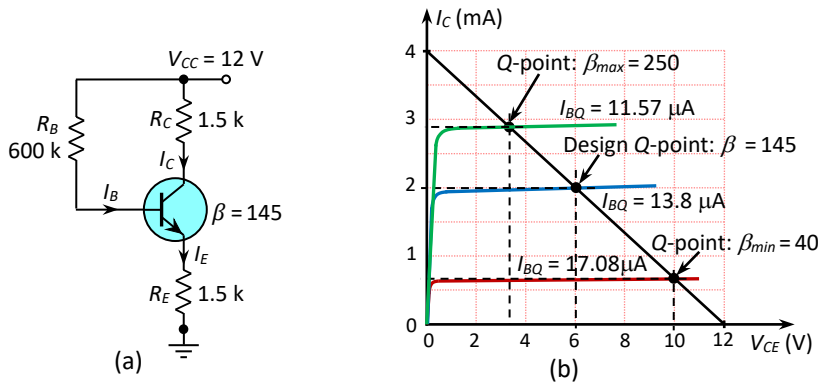


Fig.5-23: Circuit diagram and graph for Example 4.8

Solution:

$$\text{Using Equ.(5-23), } I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (145 + 1)1.5 \text{ k}\Omega} = 13.8 \mu\text{A}$$

$$\text{Using Equ.(5-24), } \therefore I_{CQ} = I_{BQ}\beta = (13.8 \mu\text{A}) \times 145 = 2 \text{ mA}$$

$$\text{Using Equ.(5-26), } V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = 12 \text{ V} - (2 \text{ mA})(1.5 \text{ k}\Omega + 1.5 \text{ k}\Omega) = 6 \text{ V}$$

For $\beta_{min} = 40$:

Using the same equations,

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (40 + 1)1.5 \text{ k}\Omega} = 17.082 \mu\text{A}$$

$$I_{CQ} = (17.082 \mu\text{A})40 = 0.683 \text{ mA}$$

$$V_{CEQ} = 12 \text{ V} - (0.683 \text{ mA})(1.5 \text{ k}\Omega + 1.5 \text{ k}\Omega) \approx 9.95 \text{ V}$$

For $\beta_{max} = 250$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (250 + 1)1.5 \text{ k}\Omega} = 11.572 \mu\text{A}$$

$$I_{CQ} = (11.572 \mu\text{A})250 = 2.89 \text{ mA}$$

$$V_{CEQ} = 12 \text{ V} - (2.893 \text{ mA})(1.5 \text{ k}\Omega + 1.5 \text{ k}\Omega) \approx 3.32 \text{ V}$$

The load line will start from 12 V on x -axis and ends to $12 \text{ V} / (1.5 \text{ k}\Omega + 1.5 \text{ k}\Omega) = 4 \text{ mA}$ on y -axis. The load line and the operating points are shown in Fig.5-23(b).

The values of operating points of the circuits of Fig.5-19(a) and 5.23(a) for different values of β are given in Table 5.3.

Table 5.3. Variation of operating points due to variation in β

β	Without R_E		With R_E	
	I_{CQ}	V_{CEQ}	I_{CQ}	V_{CEQ}
40	0.551	10.35	0.683	9.95
145	1.998	6.00	2.000	6.00
250	3.445	1.67	2.893	3.32

Comments: From the table 5.3 we see that, the change in operating point is less in the emitter bias circuit compared to that of fixed bias. In this circuit, base current decrease with the increase in β . Explanation of this decrease is given in section (**Stabilization mechanism**). Here, only a 1.5 k Ω resistor is used in the emitter. By using larger resistors, we could have achieved more stabilized operating point.

5.8 Base Bias for PNP Transistor

The base bias and the emitter bias have been discussed for NPN transistors. Here, we will study the

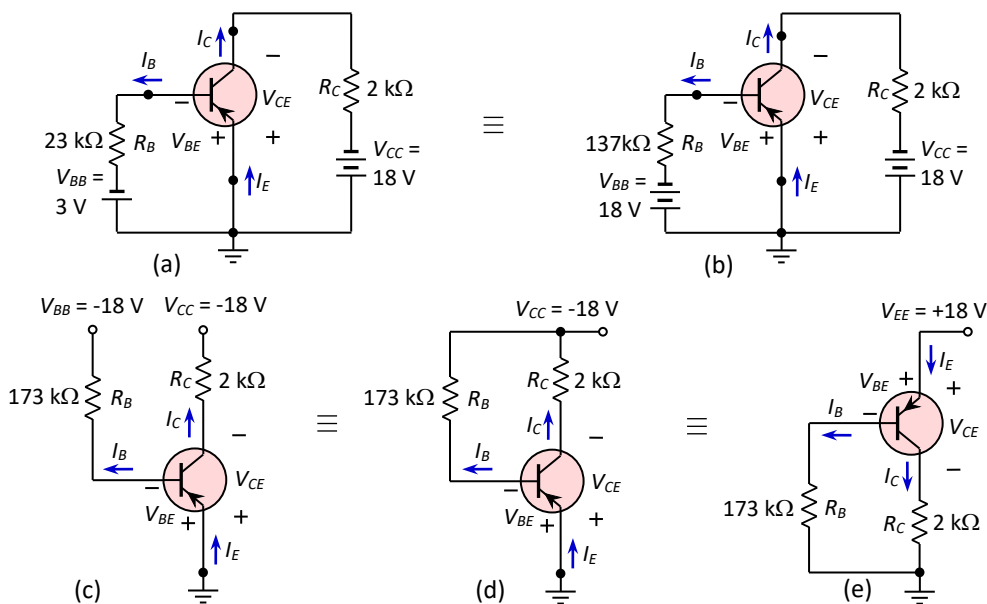


Fig.5-24: Base bias for PNP transistor, (a) Biasing using 3 V and 18 V sources, (b) Biasing using two 18 V sources, (c)-(e) Alternate representations of Fig.(b)

same bias circuits for PNP transistors. In Chapter 4, we have already known how to bias a PNP transistor. Let us start from that circuit. It is redrawn in Fig.5-24(a). Now applying the same concept as NPN transistor, we can use resistors to control the currents [Fig.5-24(b)]. Using suitable values of R_B exactly same amount of voltage can be applied to both in base and collector [Fig.5-24(c)]. Now connecting the same potential terminals we get the circuit of Fig.5-24(d). It can also be represented as in Fig.5-24(e). This is the base bias circuit for PNP transistor.

Here also we can use an emitter resistor to stabilize the Q-point as in Fig.5-25. This circuit is the base bias with emitter feedback resistor (or emitter bias) for PNP transistor.

All the equations, derived so far for base bias and emitter bias using the NPN transistors are also applicable for the biasing circuits of PNP transistor.

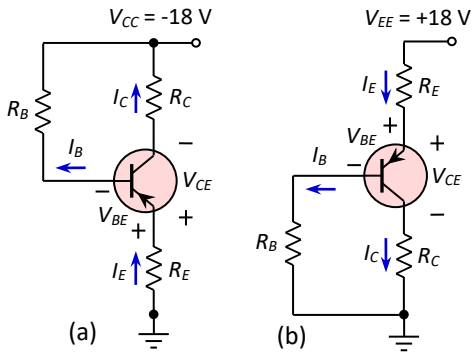


Fig.5-25: Emitter feedback bias for PNP transistor, (a) For negative voltage system, and (b) Positive voltage system

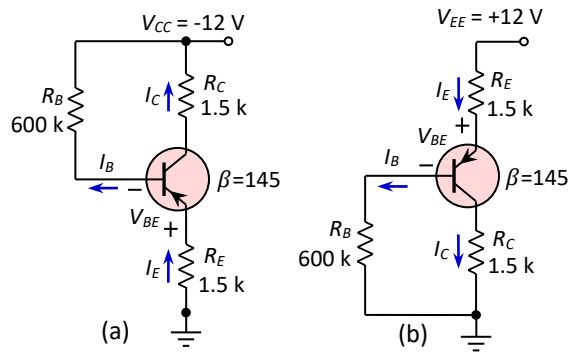


Fig.5-26: Emitter feedback bias for PNP transistor, (a) For negative voltage system, and (b) Positive voltage system

Example 5-9

Calculate the values of I_{BQ} , I_{CQ} , and V_{CEQ} for the emitter bias circuits of PNP transistor given in Fig.5-26.

Solution:

Negative voltage system [Fig.5-26(a)]:

$$\text{Using Equ.(5-23), } I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{-12 \text{ V} + 0.7 \text{ V}}{600 \text{ k}\Omega + (145 + 1)1.5 \text{ k}\Omega} = -13.8 \mu\text{A}$$

$$\text{Using Equ.(5-24), } I_{CQ} = I_{BQ}\beta = (-13.8 \mu\text{A}) \times 145 = -2 \text{ mA}$$

$$\text{Using Equ.(5-26), } V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = -12 \text{ V} - (-2 \text{ mA})(1.5 \text{ k}\Omega + 1.5 \text{ k}\Omega) = -6 \text{ V}$$

Positive voltage system [Fig.5-26(b)]: Using the same equations,

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (145 + 1)1.5 \text{ k}\Omega} = 13.8 \mu\text{A}$$

$$I_{CQ} = I_{BQ}\beta = (13.8 \mu\text{A}) \times 145 = 2 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = 12\text{ V} - (2\text{ mA})(1.5\text{ k}\Omega + 1.5\text{ k}\Omega) = 6\text{ V}$$

Comments: In both the circuits, the direction of currents and the polarities of voltages are similar. But due to the negative voltage of the first circuit, the currents have been negative.

5.9 Collector-to-Base Bias

The **collector-to-base bias** circuit is shown in Fig.5-27(a). In this circuit, collector voltage is used to bias the B - E junction, that's why it is called **collector-to-base bias**. Like the emitter resistor, here, the collector resistor provides feedback for bias stabilization. For this reason, this biasing circuit is also called **base bias with collector feedback** or simply **collector feedback bias**. The stabilization mechanism is same as described for emitter feedback bias circuit. If collector current (I_C) increases due to increase in temperature, collector voltage (V_C) decreases. This decrease in collector voltage reduces the level of base current. In the same process, if (I_C) decreases due to decrease in temperature, collector voltage increases which increases the base current. In this way, this biasing circuit provides better stabilized operating point.

In the emitter feedback bias, we can increase the stability by increasing the value of R_E . In that case, the voltage drop across R_E will also increase and hence, the voltage drop across R_C will decrease (for a fixed supply voltage). So for a fixed value of I_C , the value of R_C will be low. But the voltage gain of an amplifier depends on the value of R_C . Therefore, the increased stability reduces the value of R_C and hence, the value of voltage gain. On the other hand, in collector feedback bias circuit, to get better stabilization, we have to increase the value of R_C . This increased value of R_C will also increase the voltage gain of the amplifier circuit, which is the main advantage of this bias circuit.

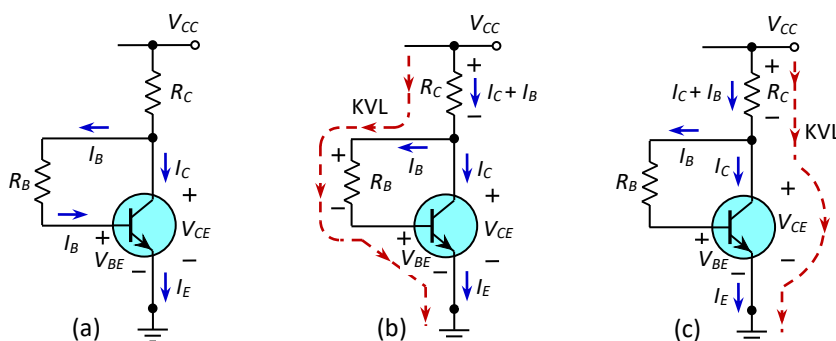


Fig.5-27: Circuit diagram of collector-to-base bias, (a) Biasing circuit, (b) KVL applied to input circuit, and (c) KVL applied to output circuit.

5.9.1 Calculation of Operating Point

Applying KVL in the forward biased B - E circuit as shown in Fig.5-27(b), we get,

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0 \quad (5-38)$$

or,
$$V_{CC} - (\beta I_B + I_B)R_C - I_B R_B - V_{BE} = 0 \quad [\because I_C = \beta I_B]$$

or,

$$I_B[R_B + (\beta + 1)R_C] = V_{CC} - V_{BE}$$

∴

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_C} \quad (5-39)$$

Another equation for the base current can be found by applying KVL in the loop- emitter-to-collector, collector-to base through R_B and then base-to-emitter as,

$$V_{CE} - I_B R_B - V_{BE} = 0$$

or,

$$I_B R_B = V_{CE} - V_{BE}$$

∴

$$I_B = \frac{V_{CE} - V_{BE}}{R_B} \quad (5-40)$$

Once, I_B is found, the collector current can be calculated as,

$$I_C = \beta I_B + I_{CEO} \approx \beta I_B \quad (5-41)$$

We can also calculate I_C , without calculating I_B from Equ.(5-38),

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0 \quad (5-42)$$

$$V_{CC} - I_C R_C - I_B R_B - V_{BE} = 0 \quad [\because I_C + I_B \approx I_C]$$

or,

$$V_{CC} - I_C R_C - \frac{I_C}{\beta} R_B - V_{BE} = 0 \quad [\because I_B = \frac{I_C}{\beta}]$$

or,

$$\left(\frac{R_B}{\beta} + R_C\right) I_C = V_{CC} - V_{BE}$$

∴

$$I_C = \frac{V_{CC} - V_{BE}}{R_B/\beta + R_C} \quad (5-43)$$

Now, applying KVL, in the reverse biased C-E junction we get [shown in Fig.5-27(c)],

$$V_{CC} - (I_C + I_B)R_C - V_{CE} = 0$$

∴

$$V_{CE} = V_{CC} - (I_C + I_B)R_C \quad (5-44)$$

Using equations (5-39), (5-41) and (5-44), the operating point of the circuit can easily be calculated.

Example 5-10

Calculate the operating point of the circuit of Fig.5-28. Assume $\beta = 100$. Verify the collector current using Equ.(5-37)

Solution:

Using Equ.(5-39), the value of base current is,

$$I_B = \frac{6 \text{ V} - 0.7 \text{ V}}{390 \text{ k}\Omega + (100 + 1)(5 \text{ k}\Omega)} = 5.92 \text{ }\mu\text{A}$$

Using Equ.(5-41), the value of collector current is,

$$I_C = \beta I_B = 100 \times 5.92 \text{ }\mu\text{A} = 0.592 \text{ mA [Ans.]}$$

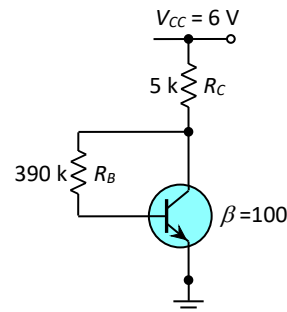


Fig.5-28: Circuit for Example 5-10

Using Equ.(5-44), the value of collector-emitter voltage is,

$$V_{CE} = 6 \text{ V} - (0.592 \text{ mA} + 0.00592)(5 \text{ k}\Omega) \approx 3 \text{ V [Ans.]}$$

Let us check I_C using Equ.(5-43),

$$I_C = \frac{V_{CC} - V_{BE}}{R_B/\beta + R_C} = \frac{6 \text{ V} - 0.7 \text{ V}}{(390 \text{ k}\Omega)/100 + 5 \text{ k}\Omega}$$

or, $I_C \approx 0.596 \text{ mA}$ [almost equal to actual value]

Comments: Operating point calculation method of this circuit is similar to that of emitter feedback biasing circuit. Here R_C is used instead of R_E .

5.9.2 Determination of Load Line of Collector-to-Base Bias Circuit

The load line for collector-to-base bias circuit can be determined using Equ.(5-38).

$$V_{CE} = V_{CC} - (I_C + I_B)R_C \approx V_{CC} - I_C R_C \quad (5-45)$$

When $I_C = 0$, $V_{CE} = V_{CC}$ and when $V_{CE} = 0$, $I_C = V_{CC}/R_C$. These points are exactly same as fixed bias circuit. Putting these points on an I_C vs. V_{CE} coordinate system, we can easily draw the load line.

5.9.3 Saturation of Transistor in Collector-to-Base Bias Circuit

As discussed for previous biasing circuits, use of inaccurate component values may also saturate transistors in collector-to-bias circuit. Here, the transistors can be saturated for very large value of β and R_C , and very small value of R_B . The effects are same as the emitter bias circuit. The proof of these saturations are left for the students in **Exercise no. 5..** The value of collector current that will saturate the transistor of collector-to-base bias will be approximately,

$$I_{C(sat)} \approx \frac{V_{CC}}{R_C} \quad (5-46)$$

5.9.4 Calculation of Stability Factor

From the forward biased $B-E$ circuit, we have already found Equ.(5-38) which is,

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0 \quad (5-47)$$

$$\text{or, } V_{CC} - I_C R_C - I_B R_C - I_B R_B - V_{BE} = 0$$

$$\text{or, } I_B (R_B + R_C) = V_{CC} - V_{BE} - I_C R_C = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE} - I_C R_C}{R_B + R_C} \quad (5-48)$$

$$\text{The collector current is, } I_C = \beta I_B + (\beta + 1)I_{CBO} \quad (5-49)$$

Putting the value of I_B from Equ. (5-48) into Equ. (5-49),

$$I_C = \frac{\beta(V_{CC} - V_{BE} - I_C R_C)}{R_B + R_C} + (\beta + 1)I_{CBO}$$

After solving the above equation for I_C , we find,

$$I_C = \frac{\beta(V_{CC} - V_{BE}) + (R_B + R_C)(\beta + 1)I_{CBO}}{R_B + (\beta + 1)R_C} \quad (5-50)$$

Comparing Equ.(5-50) with Equ.(5-34), we find that the only difference is R_C in place of R_E . So differentiating Equ.(5-50), we will get exactly the same results, but, R_C in place of R_E . Therefore, the stability factors for this biasing circuit will be,

$$S(I_{CBO}) = \frac{(R_B + R_C)(\beta + 1)}{R_B + (\beta + 1)R_C} \quad (5-51)$$

$$S(V_{BE}) = -\frac{\beta}{R_B + (\beta + 1)R_C} \quad (5-52)$$

and,

$$S(\beta) = \frac{(R_B + R_C)(V_{CC} - V_{BE})}{[R_B + (\beta + 1)R_C]^2} \quad (5-53)$$

Example 5-11

If the value of β of the transistor of Fig.5-28 increases from 100 to 150, calculate the change in collector current using normal procedure and using Equ.(5-53).

Solution:

Normal procedure:

Using Equ.(4.39), the value of base current is,

$$I_B = \frac{6\text{ V} - 0.7\text{ V}}{390\text{ k}\Omega + (150 + 1)(5\text{ k}\Omega)} = 4.629\text{ }\mu\text{A}$$

Using Equ.(5-41), the value of collector current for $\beta = 150$ will be,

$$I_{C(\beta=150)} = \beta I_B = 150 \times 4.629\text{ }\mu\text{A} = 0.694\text{ mA [Ans.]}$$

The value of collector current (I_C) for $\beta = 100$ is calculate in Example 5-10 and found $I_C = 0.592\text{ mA}$.

Using these values, the change in collector current is,

$$\Delta I_C = I_{C(\beta=150)} - I_{C(\beta=100)} = 0.694\text{ mA} - 0.592\text{ mA} = 0.102\text{ mA [Ans.]}$$

Using Equ.(5-53):

Here, change in current gain is, $\delta(\beta) = 150 - 100 = 50$

As Equ.(5-53) is applicable for a very small change in β , we have to use it in some small steps. Staring from $\beta = 100$, Equ.(5-53) is solved for every 5 increase in β . Their average value is determined. The data are shown in the following Table.

β	100	105	110	115	120	125	130	135	140	145	150	Average
$S(\beta)(\mu\text{A})$	2.61	2.47	2.34	2.22	2.11	2.01	1.92	1.83	1.75	1.67	1.6	2.048

Thus, 2.048 is the average value of the stability factor $S(\beta)$. According to the definition of $S(\beta)$,

$$S(\beta) = \frac{\delta I_C}{\delta(\beta)} = 2.048\text{ }\mu\text{A}$$

$$\therefore \delta I_C = S(\beta) \times \delta(\beta) = 50 \times (2.048 \mu\text{A}) = 0.102 \text{ mA [Ans.]}$$

Comment: Exactly the same results are found using normal procedure and the equation of the stability factor. As $S(\beta)$ has been calculated by differentiating Equ.(5-50), i.e., $S(\beta) = \delta I_C / \delta(\beta)$ it will work properly only for very small change in β . For this reason, here we have considered increase in β step by step.

Example 5-12

Fig.4.28(a) shows a collector-to-base bias circuit, designed to set the operating point at the middle of load line considering a typical β value of 145. Calculate the load line and operating point of this circuit. If the transistor has $\beta_{min} = 40$ and $\beta_{max} = 250$, calculate also the lowest and the highest operating point. Compare the results with the results of two other biasing circuits.

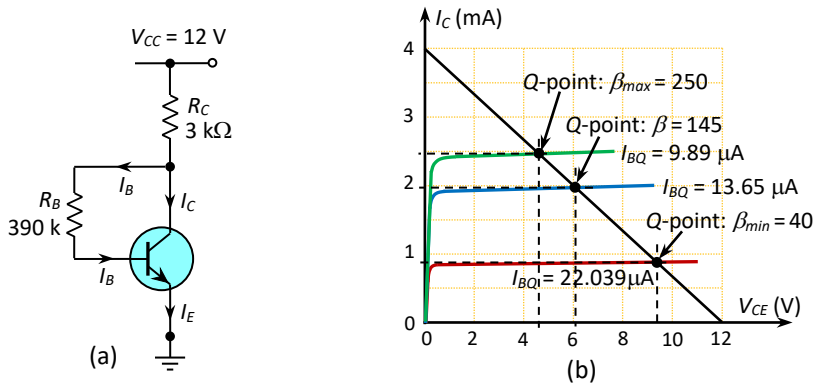


Fig.5-29: Circuit diagram and graph for Example 4.11

Solution:

Using Equ.(4.34),

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_C} = \frac{12 \text{ V} - 0.7 \text{ V}}{390 \text{ k}\Omega + (145 + 1)3 \text{ k}\Omega} = 13.65 \mu\text{A}$$

$$\therefore I_{CQ} = I_{BQ}\beta = (13.65 \mu\text{A}) \times 145 = 1.979 \text{ mA}$$

Using Equ.(4.38) [neglecting I_B],

$$V_{CEQ} \approx V_{CC} - I_{CQ}R_C = 12 \text{ V} - 1.98 \text{ mA} \times 3 \text{ k}\Omega = 6.06 \text{ V}$$

For $\beta_{min} = 40$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_C} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (40 + 1)3 \text{ k}\Omega} = 22.027 \mu\text{A}$$

$$\therefore I_{CQ} = I_{BQ}\beta = (22.027 \mu\text{A}) \times 40 = 0.881 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C = 12 \text{ V} - 0.881 \text{ mA} \times 3 \text{ k}\Omega = 9.36 \text{ V}$$

For $\beta_{max} = 250$:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_C} = \frac{12 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + (250 + 1)3 \text{ k}\Omega} = 9.886 \mu\text{A}$$

$$\therefore I_{CQ} = I_{BQ}\beta = (9.886 \mu\text{A}) \times 250 = 2.471 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ}R_C = 12 \text{ V} - 2.47 \text{ mA} \times 3 \text{ k}\Omega = 4.59 \text{ V}$$

The load line will start from 12 V on x-axis and ends to 12 V/3 k Ω = 4 mA on y-axis. The load line and the operating points are shown in Fig. 4.28(b).

The values of operating points of circuits this collector-to-base bias and other two biasing circuits for different value of β are given in Table 4.3.

Table 4.3. Variation of operating points due to variation in β .

β	Without R_E		With R_E		Collector-to-Base Bias	
	$I_{CQ}(\text{mA})$	$V_{CEQ}(\text{V})$	$I_{CQ}(\text{mA})$	$V_{CEQ}(\text{V})$	$I_{CQ}(\text{mA})$	$V_{CEQ}(\text{V})$
40	0.551	10.35	0.683	9.95	0.881	9.36
145	1.998	6.00	2.000	6.00	1.979	6.06
250	3.445	1.67	2.893	3.32	2.471	4.59

Comments: From the table we see that, the change in operating point is less in the collector-to-base bias circuit then other two biasing circuits. Here, only a 3 k Ω resistor is used in the collector. By using larger value of R_C , we can achieve more stabilized operating point.

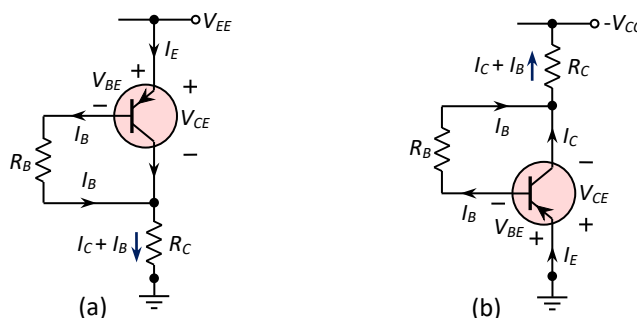


Fig.5-30: Collector feedback bias for PNP transistor, (a) For negative voltage system, and (b) Positive voltage system

5.10 Collector-to-Base Bias for PNP Transistor

The collector-to-base bias circuit for a PNP transistor is shown in Fig.5-29. The circuit is exactly same except the polarity of the biasing voltage and the directions of currents are reversed. All the equations derived for the NPN transistor can be used to analyze the circuits for PNP transistors.

5.10.1 Variation of Stability Factors

So far, we have studied three types of biasing circuits. Three stability factors have been calculated for all the biasing circuits. Before going to the next biasing circuit, we will study the variation of the

stability factors. Already we know that, to stabilize the operating point, we need to keep the stability factor as small as possible.

From the stability factors for fixed bias [Equ.(5-20), (5-21) and (5-22)], we find that they depend on the value of β , R_B and V_{CC} . β is the current gain of the transistor. We want a larger value of β of a transistor. On the other hand, V_{CC} is determined by the system voltage. Again, we cannot increase the value of R_B , because an exact value of R_B is necessary to set the value I_{BQ} . Hence, for the fixed bias circuit practically we have a little scope to improve the stability factors by changing the value of β , R_B and V_{CC} .

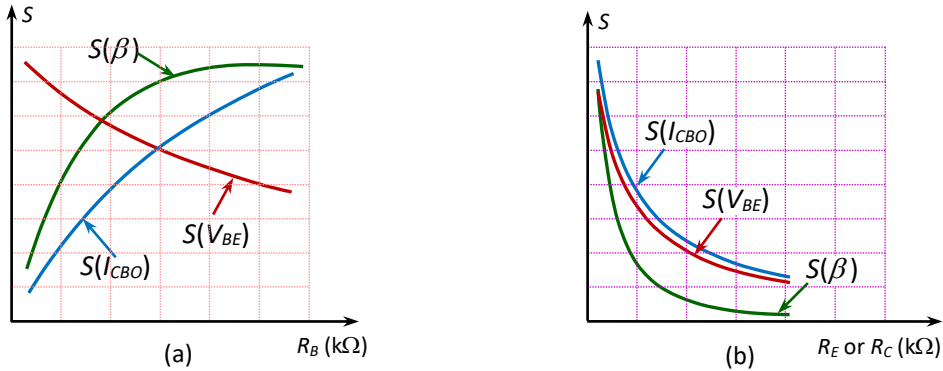


Fig.5-31: Variations of stability factors with $V_{CC} = 12\text{ V}$, $\beta = 100$,
(a) Variable R_B , (b) Variable R_E or R_C

The stability factors for the emitter feedback circuit are given by Equ.(5-35), (5-36) and (5-37) and the collector feedback circuit are given by Equ.(5-51), (5-52), and (5-53). The stability factors of emitter feedback circuit depend on β , R_B , R_E and V_{CC} . But the stability factors of collector feedback circuit depend on β , R_B , R_C and V_{CC} . While in the fixed bias circuit larger value of R_B gives lower stability factor, in the emitter bias lower value of R_B gives a better stability factor (lower values of stability factors) except $S(V_{BE})$. The variation of stability factors with the variations of R_B are shown in Fig.5-31(a). On the other hand, the higher values of R_C and R_E improve all the stability factors which are shown in Fig.5-31(b). Only $S(\beta)$ depends on the value of V_{CC} (biasing voltage of the base circuit) and lower value gives better stability.

In summary, we can say for better stability the values of R_B and biasing voltage of the base-emitter circuit V_{BB} should be smaller and the value of R_E and R_C should be larger.

5.11 Voltage Divider Bias

Voltage divider biasing circuit is the best among the four biasing circuits. It provides the scope to set any value of V_{BB} and R_B . The voltage divider biasing circuit is shown in Fig.5-32(a). Here, instead of supplying the full V_{CC} (as done in previous bias circuits), it is divided using the voltage divider network and then supplied to the base. For this reason, it is called **voltage divider bias**. It is also called **self bias**. By changing the ratio of R_1 and R_2 any value of voltage can be set for the base circuit. Similarly, by changing the values of R_1 and R_2 the value of R_B (here it will be R_{Th}) can be set to any desired value. Like the other two feedback biasing circuits, here also R_E produces

negative feedback and provide better stability of the circuit. For these reasons, we have a better control on the stability factor of the voltage divider bias circuit.

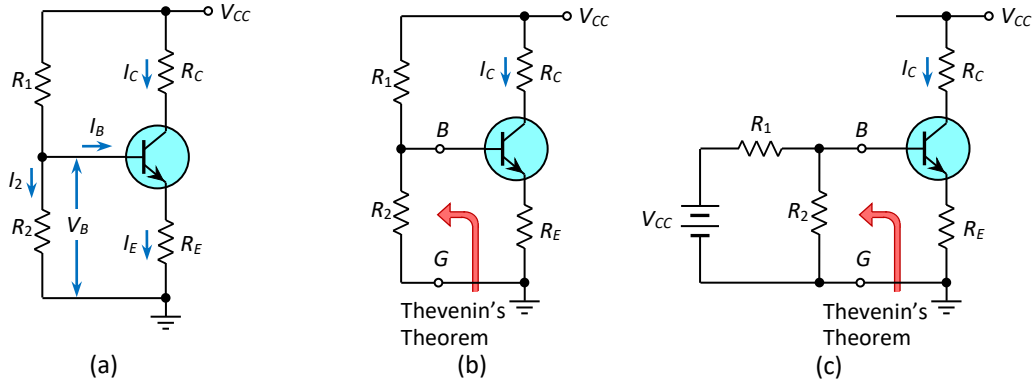


Fig.5-32: (a) Voltage divider bias circuit, (b) Location of Thevenin's theorem, (c) Fig. b is redrawn.

5.11.1 Calculation of the operating point

The circuit can be analyzed both in exact method and also in approximate method. We will see the exact analysis first.

Exact Analysis:

First, we have to determine the Thevenin's equivalent circuit of the voltage divider portion, i.e., at the base-emitter terminals as shown in Fig.5-32(b) and 4.32(c).

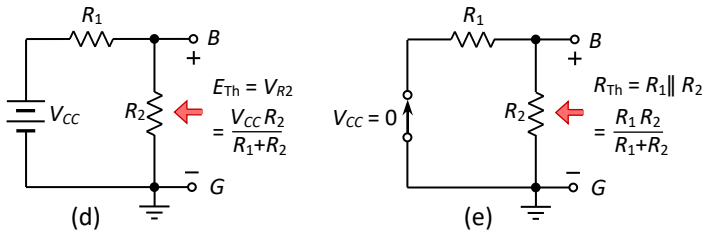


Fig.5-32: (d) Circuit for E_{Th} calculation,
 (e) Circuit for R_{Th} calculation.

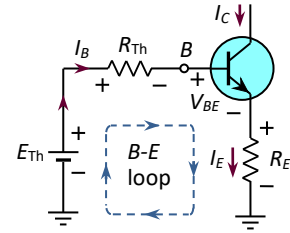


Fig.5-33: Base-emitter loop with Thevenin's equivalent circuit

E_{Th} calculation: Thevenin's voltage will be the open circuit voltage across R_2 [see Fig.5-32(d)]. Applying voltage divider rule we get,

$$E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} \quad (5-54)$$

R_{Th} calculation: To calculate R_{Th} , first we have to replace the voltage source by short circuit as shown in Fig.5-32(e). Looking at the B and G terminals we find that R_1 and R_2 are in parallel. Therefore the Thevenin's resistance will be,

$$R_{Th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \quad (5-55)$$

Now, if we redraw the biasing circuit using the Thevenin's equivalent circuit, we get the circuit of Fig.5-33 (only the input loop is shown).

To calculate the value of I_B (which is also I_{BQ}), let us apply KVL in the base-emitter circuit of Fig.5-33.

$$E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0 \quad (5-56)$$

$$\text{or,} \quad E_{Th} - I_B R_{Th} - V_{BE} - (\beta + 1)I_B R_E = 0 \quad [\because I_E = (\beta + 1)I_B]$$

$$\text{or,} \quad I_B [R_{Th} + (\beta + 1)R_E] = E_{Th} - V_{BE}$$

$$\therefore \quad I_B = I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \quad (5-57)$$

Therefore, the value of collector current will be,

$$I_C = I_{CQ} = \beta I_B = \frac{\beta(E_{Th} - V_{BE})}{R_{Th} + (\beta + 1)R_E} \quad (5-58)$$

The collector-emitter loop will be exactly same as in the Fig.5-20(c). Applying KVL to that loop,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\text{or,} \quad V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0 \quad [\because I_E \approx I_C]$$

$$\text{or,} \quad V_{CC} - I_C (R_C + R_E) - V_{CE} = 0$$

$$\therefore \quad V_{CE} = V_{CEQ} = V_{CC} - I_C (R_C + R_E) \quad (5-59)$$

Using Equ.(5-58) and (5-59) the exact position of the operating point of voltage divider bias circuit can be calculated.

5.11.2 Analysis Neglecting the Effect of R_{Th} (Approximate Analysis)

When we can neglect R_{Th} : The voltage divider network, and hence, the Thevenin's equivalent circuit works as the source of supply voltage to the base-emitter circuit. If very small amount of current is drawn by the base terminal of the transistor (i.e., I_{BQ} is very small), we can neglect the voltage drop across R_{Th} . Alternately, we can neglect the voltage drop of R_{Th} if the load connected to the Thevenin's circuit, is very large compared to R_{Th} . Let's set a limit. We can neglect the voltage drop across R_{Th} , if the load resistance is ten times of R_{Th} or even more. As shown in Fig.5-33, the emitter resistor, R_E , seen from the base circuit, i.e., $(\beta + 1)R_E$ will be the load resistance to the Thevenin's circuit. Thus, the condition to neglect the effect of R_{Th} will be:

$$(\beta + 1)R_E \geq 10 \times R_{Th} \geq 10 \times (R_1 \parallel R_2) \quad (5-60)$$

But, generally in a voltage divider bias circuit $R_1 \gg R_2$ [$\therefore R_{Th} = R_1 \parallel R_2 \approx R_2$] and for any transistor $\beta \gg 1$ [$\therefore \beta + 1 \approx \beta$]. Hence, we can write the Equ.(5-60) as,

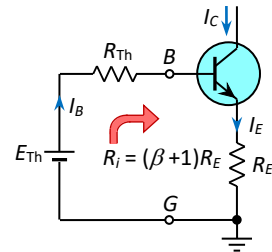


Fig.5-34: Load resistance of the Thevenin's circuit is $R_i = (\beta + 1)R_E$

$$\beta R_E \geq 10R_2 \quad (5-61)$$

If any voltage divider bias circuit fulfill the condition of Equ.(5-61), we can neglect the effect of R_{Th} and analyze the circuit very easily. This analysis is called **approximate analysis**.

Approximate Analysis:

If we neglect the voltage drop across R_{Th} , Equ.(5-66) can be written as,

$$E_{Th} - 0 - V_{BE} - I_E R_E = 0$$

$$\text{or,} \quad I_E R_E = E_{Th} - V_{BE}$$

$$\text{or,} \quad I_E = \frac{E_{Th} - V_{BE}}{R_E} \approx I_{CQ} \quad (5-62)$$

The value of E_{Th} will be same as before, i.e., $E_{Th} = V_{CC} \times (R_2)/(R_1 + R_2)$ and the value of V_{CEQ} can be calculated using Equ.(5-59).

Example 5-13

For the voltage divider biasing circuit of Fig.5-35, calculate (i) I_{BQ} , (ii) I_{CQ} , (iii) V_{CEQ} , (iv) V_E , and (v) V_{RC} using exact analysis and approximate analysis and compare the results.

Solution:

Exact Analysis:

Values of E_{Th} and R_{Th} can be calculated using Equ.(5-54) and Equ.(5-55) as,

$$E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{12 \text{ V} (4.7 \text{ k}\Omega)}{(22 + 4.7) \text{ k}\Omega} \approx 2.11 \text{ V}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{(4.7 \text{ k}\Omega) (22 \text{ k}\Omega)}{(22 + 4.7) \text{ k}\Omega} \approx 3.87 \text{ k}\Omega$$

The biasing base current can be calculated using Equ.(5-57),

$$I_{BQ} = \frac{2.11 \text{ V} - 0.7 \text{ V}}{3.87 \text{ k}\Omega + (100 + 1)1.5 \text{ k}\Omega}$$

$$\text{or,} \quad I_{BQ} \approx 9.09 \text{ }\mu\text{A [Ans.]}$$

Now the quiescent collector current will be,

$$I_{CQ} = \beta I_B = 100 \times (9.09 \text{ }\mu\text{A}) = 0.909 \text{ mA [Ans.]}$$

Using Equ.(5-59), we can calculate V_{CEQ} as,

$$V_{CEQ} = V_{CC} - I_C (R_C + R_E) = 12 \text{ V} - 0.909 \text{ mA} \times (5.6 \text{ k}\Omega + 1.5 \text{ k}\Omega) \approx 5.55 \text{ V [Ans.]}$$

V_E and V_{RC} can be calculated multiplying R_E and R_C by the current flowing through them.

$$V_E = V_{RE} = R_E \times I_E \approx R_E \times I_{CQ} = 1.5 \text{ k}\Omega \times 0.909 \text{ mA} \approx 1.36 \text{ V [Ans.]}$$

$$V_{RC} = R_C \times I_{CQ} = 5.6 \text{ k}\Omega \times 0.909 \text{ mA} \approx 5.1 \text{ V [Ans.]}$$

Approximate Analysis:

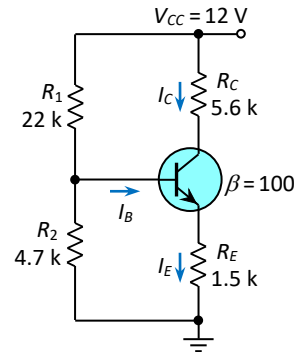


Fig.5-35: Voltage divider bias circuit for Example 5-13

Condition for approximate analysis: $\beta R_E \geq 10R_2$ or $100 \times 1.5 \text{ k}\Omega \geq 10 \times 4.7 \text{ k}\Omega$ or $150 \text{ k}\Omega \geq 47 \text{ k}\Omega \Rightarrow \text{True}$.

Values of I_{CQ} and V_{CEQ} can be calculated using Equ.(5-62) and Equ.(5-59),

$$I_{CQ} = \frac{E_{Th} - V_{BE}}{R_E} = \frac{2.11 \text{ V} - 0.7 \text{ V}}{1.5 \text{ k}\Omega} \approx 0.94 \text{ mA [Ans.]}$$

$$V_{CEQ} = V_{CC} - I_C(R_C + R_E) = 12 \text{ V} - 0.94 \text{ mA} \times (5.6 \text{ k}\Omega + 1.5 \text{ k}\Omega) \approx 5.32 \text{ V [Ans.]}$$

V_E and V_{RC} can be calculated multiplying R_E and R_C by the current flowing through them.

$$V_E = R_E \times I_E \approx R_E \times I_{CQ} = 1.5 \text{ k}\Omega \times 0.94 \text{ mA} \approx 1.41 \text{ V [Ans.]}$$

$$V_{RC} = R_C \times I_{CQ} = 5.6 \text{ k}\Omega \times 0.909 \text{ mA} \approx 5.27 \text{ V [Ans.]}$$

Now, dividing I_{CQ} by β we will get the value of I_{BQ} ,

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{0.94 \text{ mA}}{\beta} \approx 9.41 \mu\text{A [Ans.]}$$

Comparison: The results obtained using exact and approximate analysis are given in the following table with the % of variation.

	I_{BQ}	I_{CQ}	V_{CEQ}	V_E	V_{RC}
Exact	9.09 μA	0.909 mA	5.55 V	1.36 V	5.1 V
Approx.	9.41 μA	0.94 mA	5.32 V	1.41 V	5.27 V
Variation	3.52%	3.41%	-4.14%	3.68%	3.33%

Comment: As the condition for approximate analysis was true, the variation is less than 5%. But the variation will be different depending on the values of βR_E and R_2 , even the condition is true.

Example 5-14

Even the condition for approximate analysis is not true for the voltage divider bias circuit of Fig.5-36, repeat Example 5-13, to see the differences in the results.

Solution:

Here, $10R_2 = 10 \times 100 \text{ k}\Omega = 1000 \text{ k}\Omega$, and $\beta R_E = 100 \times 1.5 \text{ k}\Omega = 150 \text{ k}\Omega$. $\therefore \beta R_E \geq 10R_2 \Rightarrow \text{False}$.

Now applying the same formulas as used in Example 5-13 the values of I_{BQ} , I_{CQ} , V_{CEQ} , V_E , and V_{RC} have been calculated using exact method and approximate method. The results are shown in the following table:

	I_{BQ}	I_{CQ}	V_{CEQ}	V_E	V_{RC}
Exact	6.72 μA	0.67 mA	5.48 V	1.00 V	5.51 V
Approx.	10.43 μA	1.04 mA	1.89 V	1.56 V	8.55 V

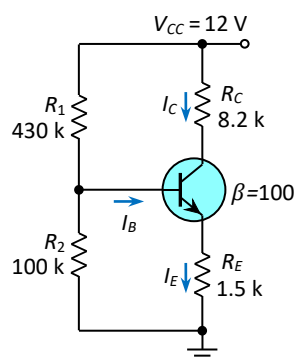


Fig.5-36: Voltage divider bias circuit for example 4.13

Variation	55.21%	55.22%	-65.51%	56.00%	55.17%
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Comment: As the condition is not true for this example, the differences between the exact and approximate method are very high. To get more detail idea how the values of R_2 and βR_E affect the results, refer to the graph of Fig.5-37. Here, the difference (%error) between I_{CQ} found in exact method and approximate method is shown for different values of the ratio of βR_E to $10R_2$.

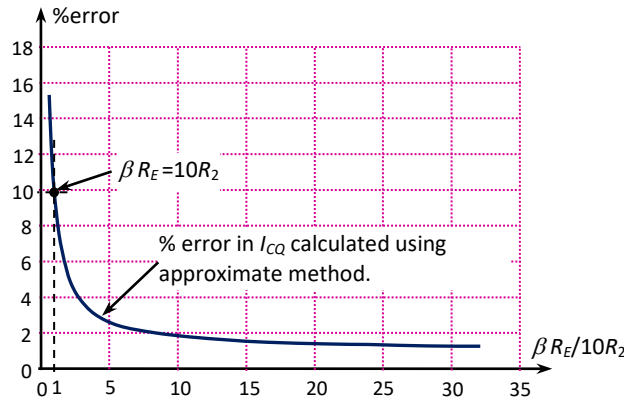


Fig.5-37: % error in I_{CQ} calculated using approximate method of voltage divider bias with $V_{CC} = 12\text{ V}$, $\beta = 100$, $R_C = 5.6\text{ k}$, $R_E = 1.5\text{ k}$ and variable R_1 and R_2

5.11.3 Determination of Load Line of Voltage Divider Bias Circuit:

Determination process of load line for voltage divider bias circuit is exactly the same as that of emitter bias circuit. Load line can be drawn using Equ.(5-59).

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (5-63)$$

When $I_C = 0$, $V_{CE} = V_{CC}$ and when $V_{CE} = 0$, $I_C = V_{CC}/(R_C + R_E)$. Now, we have to put these two points on a $I_C - V_{CE}$ coordinate system. Point $A(V_{CC}, 0)$ lies on x -axis and the point $B(0, V_{CC}/(R_C + R_E))$ lies on y -axis. Joining these two points by a straight line, we will get the load line.

Saturation of Transistor in Voltage Divider Bias Circuit:

We have already learnt the causes of saturation of transistors in different biasing circuits. Voltage divider biasing circuit may also be saturated by inaccurate component values. Here, the transistors can be saturated for very larger values of R_2 and R_C , and very small values of R_1 and R_E . **Any value of β will not saturate the transistor.** The proof of these saturations are left for the students in **Exercise no. 4..** The saturation collector current for this voltage divider circuit will be approximately,

$$I_{C(sat)} \approx \frac{V_{CC}}{R_C + R_E} \quad (5-64)$$

5.11.4 Calculation of Stability Factors of Voltage Divider Bias

From the forward biased input circuit ($B-E$ loop), we have already found Equ.(5-56), which is

$$E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

$$\text{or,} \quad E_{Th} - I_B R_{Th} - V_{BE} - I_C R_E - I_B R_E = 0 [\because I_E = I_C + I_B]$$

$$\text{or,} \quad I_B (R_{Th} + R_E) = E_{Th} - V_{BE} - I_C R_E = 0$$

$$\therefore \quad I_B = \frac{E_{Th} - V_{BE} - I_C R_E}{R_{Th} + R_E} \quad (5-65)$$

We know that the collector current of a transistor is,

$$I_C = \beta I_B + (\beta + 1) I_{CBO} \quad (5-66)$$

Putting the value of I_B from Equ.(5-65) into Equ.(5-66),

$$I_C = \frac{\beta(E_{Th} - V_{BE} - I_C R_E)}{R_{Th} + R_E} + (\beta + 1) I_{CBO}$$

$$\text{or,} \quad I_C = \frac{\beta(E_{Th} - V_{BE} - I_C R_E) + (R_{Th} + R_E)(\beta + 1) I_{CBO}}{R_{Th} + R_E}$$

$$\text{or,} \quad I_C (R_{Th} + R_E) = \beta(E_{Th} - V_{BE}) - \beta I_C R_E + (R_{Th} + R_E)(\beta + 1) I_{CBO}$$

$$\text{or,} \quad I_C [R_{Th} + (\beta + 1) R_E] = \beta(E_{Th} - V_{BE}) + (R_{Th} + R_E)(\beta + 1) I_{CBO}$$

$$\therefore \quad I_C = \frac{\beta(E_{Th} - V_{BE}) + (R_{Th} + R_E)(\beta + 1) I_{CBO}}{R_{Th} + (\beta + 1) R_E} \quad (5-67)$$

Now, if we compare Equ.(5-67) with Equ.(5-34), we find that the only difference is, E_{Th} in place of V_{CC} and R_{Th} in place of R_B . So differentiating Equ.(5-67), we will get exactly the same results, but, E_{Th} in place of V_{CC} , and R_{Th} in place of R_B . Therefore, the stability factors for the voltage divider bias circuit will be,

$$S(I_{CBO}) = \frac{(R_{Th} + R_E)(\beta + 1)}{R_{Th} + (\beta + 1) R_E} \quad (5-68)$$

$$S(V_{BE}) = -\frac{\beta}{R_{Th} + (\beta + 1) R_E} \quad (5-69)$$

$$\text{and,} \quad S(\beta) = \frac{(R_{Th} + R_E)(E_{Th} - V_{BE})}{[R_{Th} + (\beta + 1) R_E]^2} \quad (5-70)$$

The expressions for the stability factors for voltage divider bias are comparable with those of the emitter bias and collector feedback bias circuits. Already we have found that a lower value of R_B and V_{CC} give better stability. In voltage divider bias, the corresponding values are R_{Th} and E_{Th} . But the values of R_{Th} and E_{Th} are much lower than that of R_B and V_{CC} . So, the voltage divider bias provides the best stability of the operating point. The dependency of the stability factors on R_{Th} , R_E will be same as shown in Fig.5-31.

Lower values of R_{Th} and higher values of R_E give improved stability factor except $S(V_{BE})$ [higher values of R_{Th} and R_E give better $S(V_{BE})$]. However, lower value of R_{Th} causes power loss in voltage divider network, and higher value of R_E decrease the voltage gain of the transistor circuit.

For this reason, we cannot make R_{Th} very low, and R_E very high. We have to make a tradeoff between them. To make it clear, let us rewrite Equ.(5-68) as,

$$S(I_{CBO}) = \frac{(R_{Th} + R_E)(\beta + 1)}{R_{Th} + (\beta + 1)R_E} = (\beta + 1) \frac{(R_{Th} + R_E)}{R_{Th} + (\beta + 1)R_E}$$

$$\text{or,} \quad S(I_{CBO}) = (\beta + 1) \frac{1 + R_{Th}/R_E}{(\beta + 1) + R_{Th}/R_E} \quad (5-71)$$

From Equ.(5-71), if $R_{Th}/R_E < 1$ (i.e., $R_E > R_{Th}$), we can write, $S(I_{CBO}) \approx 1$. To get exactly $S(I_{CBO}) = 1$, R_{Th}/R_E must be zero, which is not possible. But, theoretically $S(I_{CBO})$ can be made equal to 2 by making $R_{Th} = R_E$. Again, practically it is not possible. Because, in practical circuit, R_{Th} must be greater than R_E .

If $1 < R_{Th}/R_E < (\beta + 1)$ (i.e., $(\beta + 1)R_E > R_{Th}$), then we get, $S(I_{CBO}) \approx R_{Th}/R_E$. That is, the value of $S(I_{CBO})$ will remain less than $(\beta + 1)$.

Finally, if $R_{Th}/R_E > (\beta + 1)$ (i.e., $(\beta + 1)R_E < R_{Th}$), then the value of the stability factor will be $S(I_{CBO}) \approx (\beta + 1)$. Although, here we have approximate three different values of $S(I_{CBO})$, actual values will change gradually with the increase in the ratio R_{Th}/R_E as shown in the Fig.5-38. This is a semi-log graph, that has been drawn with $V_{CC} = 12\text{ V}$, $\beta = 50$, $R_C = 5\text{ k}\Omega$, and different values of R_1 and R_2 . To keep the value of E_{Th} unchanged (and hence the operating point of the bias circuit) the ratio of R_1/R_2 was not changed. As practically $S(I_{CBO})$ cannot be 1, here the graphs has been started from 2 for $R_{Th} = R_E$. When $R_{Th}/R_E = (\beta + 1)$, the value of $S(I_{CBO})$ will be $(\beta/2 + 1)$.

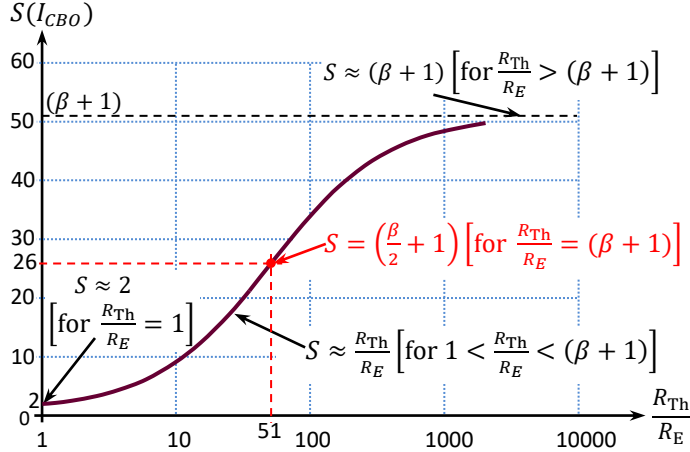


Fig.5-38: Variation of $S(I_{CBO})$ with the variation of R_{Th}/R_E

As the graph of Fig.5-38 has been drawn for $\beta = 50$, this point is shown by (51, 26) on the graph.

Although, the graph shows the variation of I_{CBO} for voltage divider bias, by using R_B in place of R_{Th} the result for emitter feedback bias and using R_C in place of R_E collector feedback bias will be found. The shape of the graphs for emitter feedback bias and collector feedback bias will be same and all the conditions and results discussed here, will be equally applicable for them.

Example 5-15

Fig.5.39(a) shows a voltage divider bias circuit considering a typical β value of 145. Calculate the load line and operating point of this circuit. If we consider BD135 transistor, that has $\beta_{min} = 40$ and $\beta_{max} = 250$, calculate the lowest and the highest operating point corresponding to these β values. Compare the results with the results of Fig.5-19, and Fig.5-23.

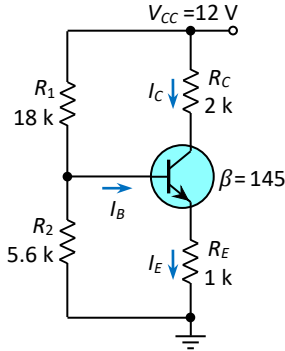


Fig.5-39(a): Voltage divider bias circuit for Example 4.14

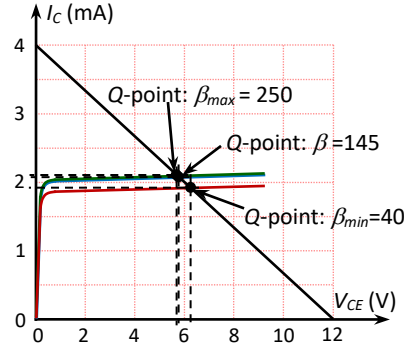


Fig.5-39(b): Operating points of example 4.14

Solution:**Exact Analysis:**

Using Equ.(5-54),
$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12\text{ V}(5.6\text{ k}\Omega)}{(18 + 5.6)\text{ k}\Omega} \approx 2.8475\text{ V}$$

Using Equ.(5-55),
$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{(18\text{ k}\Omega)(5.6\text{ k}\Omega)}{(18 + 5.6)\text{ k}\Omega} \approx 4.27\text{ k}\Omega$$

The base current can be calculated using Equ.(5-57),

$$\therefore I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.8475\text{ V} - 0.7\text{ V}}{4.27\text{ k}\Omega + (145 + 1)1\text{ k}\Omega} \approx 14.2906\text{ }\mu\text{A} \text{ [Ans.]}$$

Now the quiescent collector current will be [Equ.(5-58)],

$$I_{CQ} = \beta I_B = 145 \times (14.2906\text{ }\mu\text{A}) = 2.072\text{ mA} \text{ [Ans.]}$$

Using Equ.(5-59), we can calculate V_{CEQ} ,

$$V_{CEQ} = V_{CC} - I_C(R_C + R_E) = 12\text{ V} - 2.072\text{ mA} \times (2 + 1)\text{ k}\Omega \approx 5.78\text{ V} \text{ [Ans.]}$$

By repeating above process, the operating points for $\beta_{min} = 40$ and $\beta_{max} = 250$ have been calculated and the results are found as below:

$$\beta_{min} = 40: I_{CQ} \approx 1.897\text{ mA, and } V_{CEQ} \approx 6.31\text{ V}$$

$$\beta_{min} = 250: I_{CQ} \approx 2.103\text{ mA, and } V_{CEQ} \approx 5.69\text{ V}$$

The load line will start from 12 V on x-axis and ends to $12\text{ V} / (2\text{ k}\Omega + 1\text{ k}\Omega) = 4\text{ mA}$ on y-axis.

The load line and the operating points are shown in Fig.5-39(b).

To compare the performance of three types of biasing circuits, the variations of operating points for different β -values in Fig.5-19, 5-23 and 5-39 are shown in Table 4.5.

Table 4.5: Variation of operating points due to variation in β .

β	Fixed bias		Emitter feedback bias		Voltage divider bias	
	I_{CQ} (mA)	V_{CEQ} (V)	I_{CQ} (mA)	V_{CEQ} (V)	I_{CQ} (mA)	V_{CEQ} (V)
40	0.551	10.35	0.683	9.95	1.897	6.31
145	1.998	6.00	2.000	6.00	2.072	5.78
250	3.445	1.67	2.893	3.32	2.103	5.69

Comments: Look at the operating points of the voltage divider bias. In spite of a very large variation in β , there is a very small change in the operating points. Here the collector feedback bias is considered as it's seldom used.

5.12 Voltage Divider Bias for PNP Transistor

The voltage divider bias circuit for a PNP transistor is shown in Fig.5-40. Here, the collector is connected to lower potential and the emitter is connected to higher potential, which is just opposite to that of NPN transistor. Moreover, the polarity of the voltages across the terminals and the direction of currents are reversed. In case of negative voltage system, Thevenin's theorem has

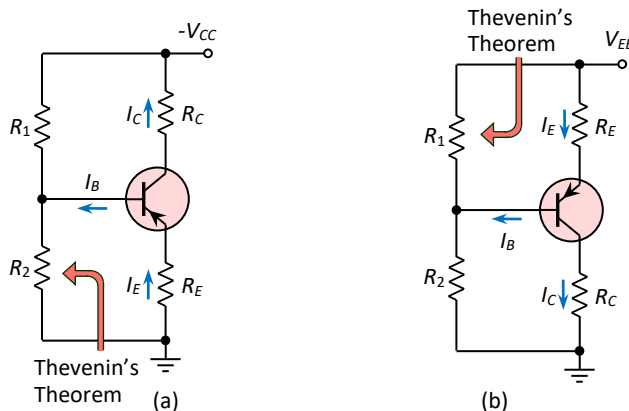


Fig.5-40: Voltage divider bias for PNP transistor, (a) Negative voltage system, and (b) Positive voltage system

to be applied in the same position as for NPN transistor and the operating point can be calculated by using the Equ.(5-54) to Equ.(5-59) with reverse polarities of voltages and reverse direction of currents. But for positive voltage system, Thevenin's voltage has to be calculated across R_1 instead of R_2 , as shown in Fig.5-40(b). That, the base voltage ($V_B = E_{Th}$) will be,

$$E_{Th} = \frac{V_{CC}R_1}{(R_1 + R_2)} \quad (5-72)$$

Apart from these difference, a PNP transistor voltage divider bias circuit with positive voltage can be analyzed in exactly using the same equations as derived for NPN transistor.

Example 5-16

For the voltage divider circuits of Fig.5-41, calculate the operating points.

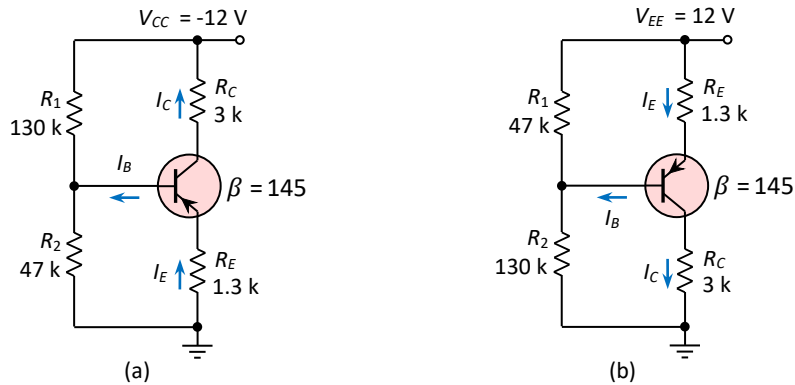


Fig.5-41: PNP transistor circuits for Example 5.15 (a) Negative voltage (b) Positive voltage

Solution:**Negative voltage system [Fig.5-41(a)]:**

Using Equ.(5-54),
$$E_{Th} = \frac{V_{CC}R_2}{(R_1 + R_2)} = \frac{(-12 \text{ V})(47 \text{ k}\Omega)}{(47 \text{ k}\Omega + 130 \text{ k}\Omega)} \approx -3.186 \text{ V}$$

Using Equ.(5-55),
$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{47 \text{ k}\Omega \times 130 \text{ k}\Omega}{(47 \text{ k}\Omega + 130 \text{ k}\Omega)} \approx 34.52 \text{ k}\Omega$$

Using Equ.(5-57),
$$I_B = I_{BQ} = \frac{-3.186 \text{ V} + 0.7 \text{ V}}{34.52 \text{ k}\Omega + (145 + 1)1.3 \text{ k}\Omega} \approx -11.084 \text{ }\mu\text{A}$$

Using Equ.(5-58),
$$I_{CQ} = 145 \times (-11.084 \text{ }\mu\text{A}) = -1.61 \text{ mA}$$

Using Equ.(5-59),
$$V_{CEQ} = -12 \text{ V} - (-1.61 \text{ mA}) \times (3 \text{ k}\Omega + 1.3 \text{ k}\Omega) \approx -5.1 \text{ V}$$

Positive voltage system [Fig. 4.40(b)]:

Using Equ.(5-5),
$$E_{Th} = \frac{V_{CC}R_1}{(R_1 + R_2)} = \frac{(12 \text{ V})(47 \text{ k}\Omega)}{(47 \text{ k}\Omega + 130 \text{ k}\Omega)} \approx 3.186 \text{ V}$$

Rest of the calculation is same and the same results will be found.

Comments: For negative voltage system negative currents and voltages are found. But for positive voltage the results will be same as for NPN transistors.

5.13 Biasing Circuit with Split Power Supply

We know a normal power supply or a voltage source has two terminals: a positive terminal and a negative terminal or ground same as any battery [Fig.-42(a)]. This type of power supply is called



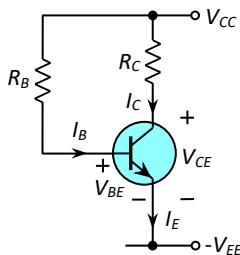
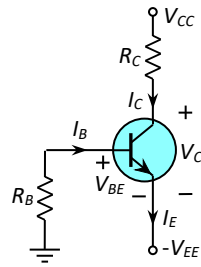
(a) Single-polarity variable DC power supply



(b) Dual polarity variable DC power supply

Fig.5-42: Single-polarity and dual polarity variable DC power supply

single-polarity power supply. But, there are some power sources that have three terminals: a positive terminal, a negative terminal and a ground [Fig.-42(a)]. This type of power source is called **split power supply** or **dual-polarity power supply**. Some electronic circuits need split power supplies, e.g., circuits consisting of operational amplifiers. In this section, we will see how to analyze basing circuits if a split voltage source (split power supply) is connected to the circuit.

(a) Base resistor is connected to V_{CC} 

(b) Base resistor is connected to ground

Fig.5-43: Base bias circuit with split power supply

5.14 Base Bias Circuit with Split Power Supply

If a base bias or fixed bias circuit uses a split supply as shown in Fig. 4.43(a), the circuit can be analyzed using the same formulae as single supply base bias circuit. Only the difference is, instead

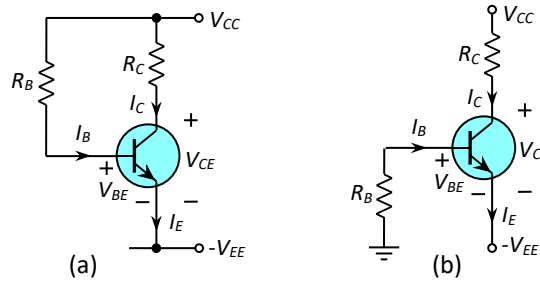


Fig.5-43: Base bias circuit with split power supply

of V_{CC} , we have to use $V_{CC} + V_{EE}$. The expressions for finding the operating point of this circuit will be:

$$I_{BQ} = \frac{V_{CC} + V_{EE} - V_{BE}}{R_B} \quad (5-73)$$

$$I_{CQ} = \beta I_B + (\beta + 1)I_{CBO} \approx \beta I_B \quad (5-74)$$

$$V_{CEQ} = (V_{CC} + V_{EE}) - I_{CQ}R_C \quad (5-75)$$

Base bias with split power supply is sometimes represented as in Fig.5-43(b). For this circuit, Equ. (5-73) will be different. Now the base-emitter circuit will be forward biased by V_{EE} . Therefore,

$$I_{BQ} = \frac{V_{EE} - V_{BE}}{R_B} \quad (5-76)$$

Equ.(5-74) and (5-75) will be applicable for this circuit, as well.

Example 5-17

Calculate the operating point of the base bias circuit of Fig.5-44. Also determine the operating point if a base resistor of 270 kΩ is connected from ground to the base.

Solution:

The biasing circuit of Fig.5-44(a) is similar to that of Fig.5-43(a). So we will use Equ.(5-73) to Equ.(5-75).

Using Equ.(5-73),
$$I_{BQ} = \frac{9\text{ V} + 9\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} \approx 30.8929\text{ }\mu\text{A} \text{ [Ans.]}$$

Using Equ.(5-74),
$$I_{CQ} \approx \beta I_B = 120 \times 30.8929\text{ }\mu\text{A} \approx 3.70714\text{ mA} \text{ [Ans.]}$$

Using Equ.(5-75),
$$V_{CEQ} = 9\text{ V} + 9\text{ V} - (3.70714\text{ mA}) \times 2.4\text{ k}\Omega \approx 9.1 \text{ [Ans.]}$$

In the circuit of Fig.5-44(b), 270 kΩ resistor is connected from ground to the base. So this circuit is same as that of Fig.5-43(b). So, we have to use Equ.(5-76) instead of Equ.(5-73).

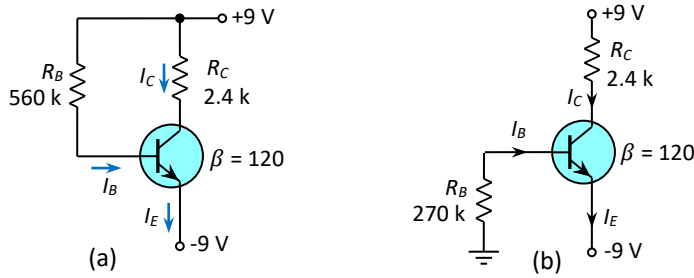


Fig.5-44: Circuit for Example 5-17

Using Equ.(5-73),
$$I_{BQ} = \frac{V_{EE} - V_{BE}}{R_B} = \frac{9\text{ V} - 0.7\text{ V}}{270\text{ k}\Omega} \approx 30.7407\text{ }\mu\text{A} \text{ [Ans.]}$$

Using Equ.(5-74),
$$I_{CQ} = \beta I_B = 120 \times 30.7407\text{ }\mu\text{A} \approx 3.69\text{ mA} \text{ [Ans.]}$$

Using Equ.(5-64),
$$V_{CEQ} = 9\text{ V} + 9\text{ V} - (3.69\text{ mA}) \times 2.4\text{ k}\Omega \approx 9.15 \text{ [Ans.]}$$

Comments: Using lower value resistor from ground to base same result can be found.

5.15 Emitter Bias with Split Voltage Supply

Emitter bias circuit with split voltage supply may also be constructed in two configurations. One configuration is same as previous emitter bias circuit [Fig.5-45(a)] and the other is with a base resistor connected to ground [Fig.5-45(b)]. For the circuit of Fig.5-45(a) the equations will be,

$$I_B = \frac{V_{EE} + V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \quad (5-77)$$

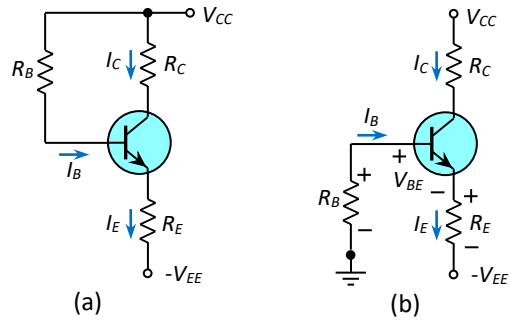


Fig.5-45: Emitter bias with split voltage supply

$$I_{CQ} = \beta I_B + (\beta + 1)I_{CBO} \approx \beta I_B$$

$$V_{CE} = (V_{CC} + V_{EE}) - I_C(R_C + R_E) \quad (5-78)$$

In circuit of Fig.5-45(b), the collector is connected to positive terminal and the emitter is connected to negative terminal of the split source. The base is connected to ground through a small resistor R_B . This circuit is also called **emitter current bias**. Now applying KVL in the base-emitter circuit, we get,

$$\begin{aligned} V_{EE} - I_B R_B - V_{BE} - I_E R_E &= 0 \\ V_{EE} - I_B R_B - V_{BE} - (\beta + 1)I_B R_E &= 0 \quad [\because I_E = (\beta + 1)I_B] \\ I_B [R_B + (\beta + 1)R_E] &= V_{EE} - V_{BE} \end{aligned}$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} \quad (5-79)$$

$$\therefore I_{CQ} = \beta I_B + (\beta + 1)I_{CBO} \approx \beta I_B$$

This circuit needs R_B with lower resistance compared to that of Fig.5-45(a). The lower value of R_B provides better stability. As R_B is very small, and the base current flowing through this resistance is small, the voltage drop across R_B will also be very small. If we neglect this voltage drop, we can assume the base is connected to the ground potential. That is, if R_B is selected so small that $I_{B(max)}R_B$ is very small compared to V_{BE} . A good approximation is $I_{B(max)}R_B = V_{BE}/10$. Then, the value of I_{CQ} can be calculated very simply as,

$$V_{EE} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\text{or,} \quad V_{EE} - V_{BE} - I_E R_E = 0 \text{ [neglecting } I_B R_B]$$

$$\therefore I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

If V_{EE} is very larger than V_{BE} , we can write,

$$I_E = \frac{V_{EE}}{R_E} \approx I_{CQ} \quad (5-80)$$

From Equ.(5-80), we find that I_{CQ} is independent of β and R_B . Thus, the circuit will give a very stable operating point.

For both the circuit the base-emitter voltage will be,

$$V_{CEQ} = (V_{CC} + V_{EE}) - I_{CQ}(R_C + R_E) \quad (5-81)$$

5.16 Collector-to-Base Bias with Split Voltage Supply

The collector-to-base bias with split power supply ($+V_{CC}$, $-V_{EE}$) is shown in Fig.5-46. All the equations derived for single polarity voltage source can be used to analyze this circuit by considering $(V_{CC} + V_{EE})$ instead of V_{CC} .

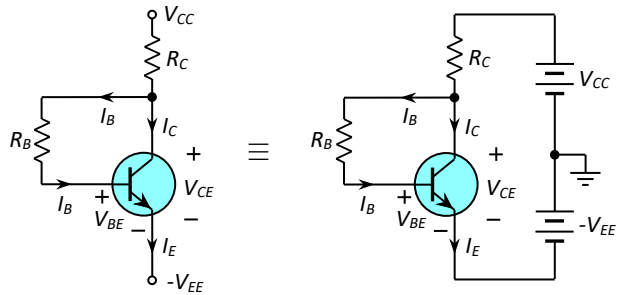


Fig.5-46: Collector-to-base bias with split power supply.

$$I_{BQ} = \frac{(V_{CC} + V_{EE}) - V_{BE}}{R_B + (\beta + 1)R_C} \quad (5-82)$$

$$I_{CQ} = \beta I_B + I_{CEO} \approx \beta I_B$$

$$V_{CEQ} = (V_{CC} + V_{EE}) - (I_{CQ} + I_{BQ})R_C \quad (5-83)$$

5.17 Voltage Divider Bias with Split Supply

The voltage divider bias with split power supply is shown in Fig.5-47(a). A positive voltage is supplied to the collector and a negative voltage is supplied to the emitter. The input and the output to this circuit will be with respect to ground as shown in the figure. All the equations derived for voltage divider bias with single voltage source can be used to analyze this circuit using $(V_{CC} + V_{EE})$ instead of V_{CC} .

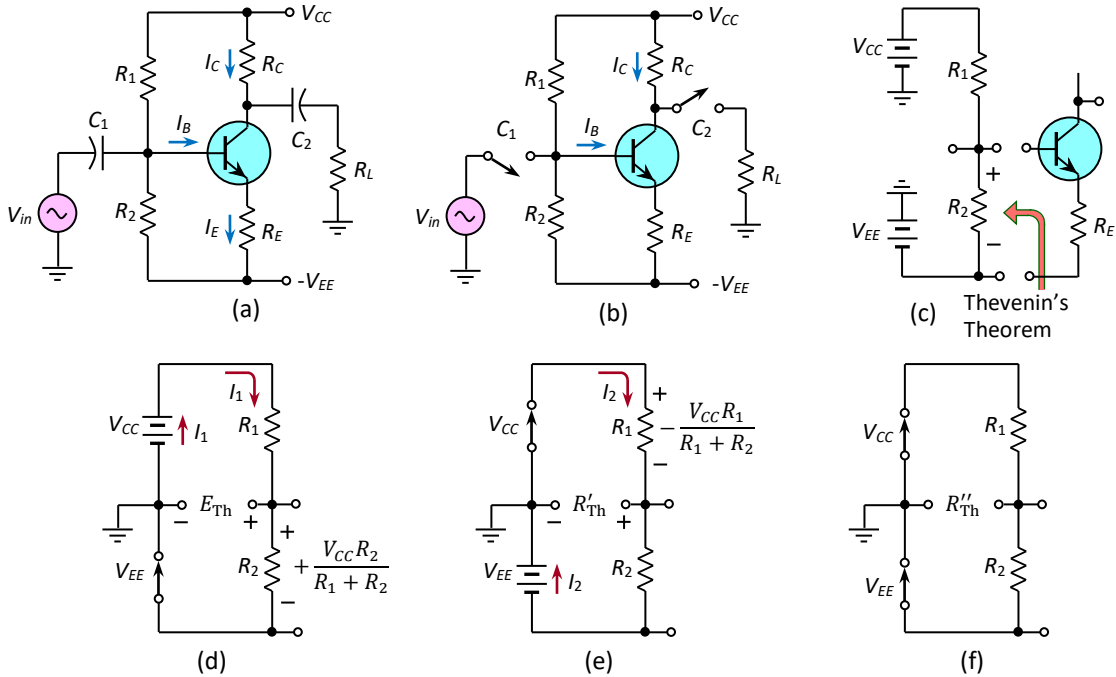


Fig.5-47: Voltage divider bias With split voltage source, (a) Biasing circuit with input and output (b) DC equivalent circuit, (c) Same circuit drawn with two batteries and separated the terminals, where Thevenin's Theorem is applied, (d,e) E_{Th} calculation, and (f) R_{Th}

However, in some literature, instead of using the same equations, E_{Th} is calculated in different way. To do that, look at the figures- Fig.5-47(b), which is the DC equivalent circuit. Fig.5-47(c) is the same figure with two voltage sources connected with respect to ground and the terminals are opened where Thevenin's theorem has to be applied. Since, there are two voltage sources, we have to use superposition theorem to calculate the value of E_{Th} . E_{Th} is the voltage to the base with respect to ground. If we replace V_{EE} by a short circuit and consider only V_{CC} , the value of E'_{Th} will be $V_{CC}R_2/(R_1 + R_2)$. As shown in Fig.5-45(d), this voltage is positive. If we replace V_{CC} by a short circuit and consider only V_{EE} , the value of E''_{Th} will be $-V_{CC}R_1/(R_1 + R_2)$ [as shown in Fig.5-55(e)]. According to the superposition theorem, the value of E_{Th} will be

$$E_{Th} = E'_{Th} + E''_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} - \frac{V_{EE}R_1}{R_1 + R_2} \quad (5-84)$$

If we short both the source of Fig.5-45(f), we can find the value of R_{Th} as,

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

Fig.5-48 shows the base-emitter loop of the biasing network with Thevenin's equivalent circuit. Now the value of I_B can be calculated by applying KVL in the base-emitter loop as,

$$\begin{aligned} E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E + |V_{EE}| &= 0 \\ E_{Th} - I_B R_{Th} - V_{BE} - (\beta + 1) I_B R_E + V_{EE} &= 0 \end{aligned}$$

$$\therefore \quad I_B = I_{BQ} = \frac{|V_{EE}| + E_{Th} - V_{BE}}{R_{Th} + (\beta + 1) R_E} \quad (5-85)$$

$$\therefore \quad I_{CQ} \approx \beta I_{BQ} \quad (5-86)$$

Value of V_{CE} can be calculated by apply KVL in the collector-emitter circuit. Its value will be,

$$\therefore \quad V_{CEQ} = (V_{CC} + |V_{EE}|) - (R_C + R_E) I_{CQ} \quad (5-87)$$

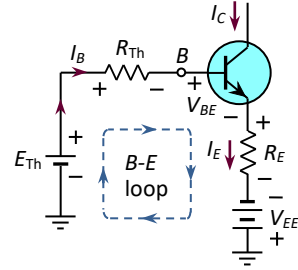


Fig.5-48: Base-emitter loop with Thevenin's equivalent circuit

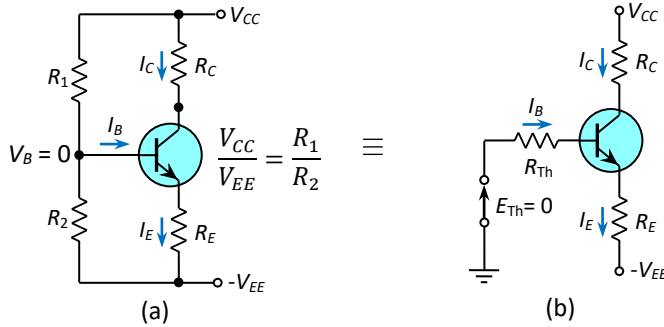


Fig.5-49: Voltage divider bias equivalent to emitter current bias when $V_B = 0$ V

5.18 Emitter Current Bias from Voltage Divider Bias:

The circuit of Fig.5-47(b) is redrawn in Fig.5-49(a) with the values of R_1 and R_2 selected in proportion to the negative and positive voltages, i.e., $V_{CC}/V_{EE} = R_1/R_2$. Now, from Equ.(5-84) the value of E_{Th} will be zero. However, the value of R_{Th} will remain same. So, E_{Th} of Fig.5-48 can be replaced by a short circuit and the circuit can be redrawn as in Fig.5-49(b), which is same as the emitter current bias circuit of Fig.5-45(b). Thus, we can say that the circuit of Fig.5-49(a) is the voltage divider equivalent of emitter current bias circuit with the condition $V_{CC}/V_{EE} = R_1/R_2$.

Example 5-18

For the voltage divider biasing circuit of Fig.5-50, determine the operating point, draw the load line and place the operating point on the load line.

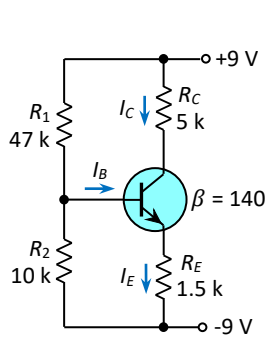


Fig.5-50: Voltage divider bias circuit for Example 5-18

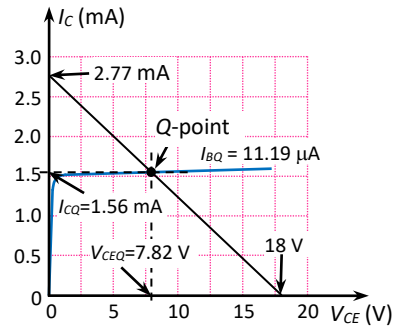


Fig.5-51: Load line and operating point of Example 5-18

Solution:

Using Equ.(5-84), $E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} - \frac{V_{EE}R_1}{R_1 + R_2} = \frac{9\text{ V} \times 10\text{ k}\Omega}{47\text{ k}\Omega + 10\text{ k}\Omega} - \frac{9\text{ V} \times 47\text{ k}\Omega}{47\text{ k}\Omega + 10\text{ k}\Omega} \approx -5.84\text{ V}$

and the value of R_{Th} , $R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{47\text{ k}\Omega \times 10\text{ k}\Omega}{47\text{ k}\Omega + 10\text{ k}\Omega} \approx 8.246\text{ k}\Omega$

Using Equ.(5-85), $I_{BQ} = \frac{9\text{ V} - 5.84\text{ V} - 0.7\text{ V}}{8.246\text{ k}\Omega + (140 + 1) \times 1.5\text{ k}\Omega} \approx 11.185\text{ }\mu\text{A}$ [Ans.]

$\therefore I_{CQ} \approx \beta I_{BQ} = 140 \times 11.185\text{ }\mu\text{A} \approx 1.566\text{ mA}$ [Ans.]

Using Equ.(5-83), $V_{CEQ} = (9\text{ V} + 9\text{ V}) - (47\text{ k}\Omega + 10\text{ k}\Omega)(1.566\text{ mA}) \approx 7.82\text{ V}$ [Ans.]

The load line:

Using Equ.(5-83), when $I_C = 0$, $V_{CE} = V_{CC} + V_{EE} = 9\text{ V} + 9\text{ V} = 18\text{ V}$

when $V_{CE} = 0$, $I_C = \frac{V_{CC} + V_{EE}}{R_C + R_E} = \frac{18\text{ V}}{5\text{ k}\Omega + 1.5\text{ k}\Omega} = 2.77\text{ mA}$

The load line is drawn by connecting two points: one is (18 V, 0) on the x -axis and the other is (0, 2.77) on the y -axis [Fig.5-51].

Comments: This problem can also be solved using the following equations:

$$E_{Th} = \frac{(V_{CC} + |V_{EE}|)R_2}{R_1 + R_2} \quad (5-88)$$

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} \quad (5-89)$$

$$I_{CQ} = \frac{(E_{Th} - V_{BE})\beta}{R_{Th} + (\beta + 1)R_E} \quad (5-90)$$

and,

$$V_{CEQ} = (V_{CC} + V_{EE}) - (R_C + R_E)I_{CQ} \quad (5-91)$$

5.19 Biasing Circuit for PNP Transistor with Split Power Supply

Like the NPN transistors, PNP transistors can be biased using split power supplies. All of the biasing circuits of NPN transistors using split supplies can be used for PNP transistors too, only changing the polarities of the supply voltages and the terminal voltages (like V_{BE} , V_{CE} etc.). So the directions of currents will also be opposite to those of NPN transistors. Here, we will discuss only the voltage divider biasing circuit as shown in Fig.5-52(a). The Thevenin's equivalent circuit can be determined as described for Fig.5-47. Now, changing the polarities of the supply voltages, Equ.(5-84) can be used as,

$$E_{Th} = -\frac{V_{CC}R_2}{R_1 + R_2} + \frac{V_{EE}R_1}{R_1 + R_2} \quad (5-92)$$

The value of E_{Th} should be any value less than $+V_{EE}$. The Thevenin's resistance will be same as before,

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

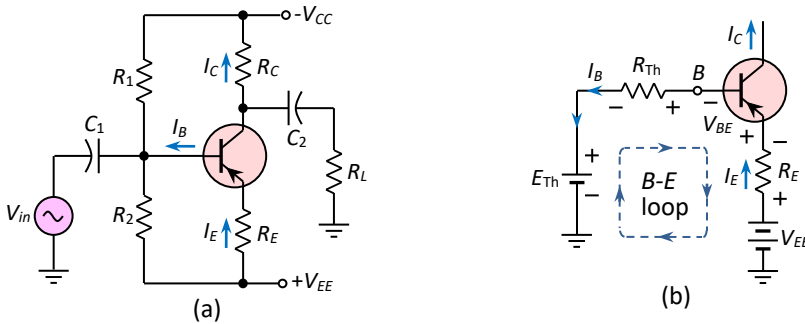


Fig.5-52: PNP transistor circuits for Example 5.15 (a) Negative voltage (b) Positive voltage

The circuit of Fig.5-52(a) can be redrawn as in Fig.5-52(b), using the Thevenin's equivalent circuit. Now applying KVL in this circuit, we can find the value of I_B as,

$$I_B = I_{BQ} = \frac{|V_{EE}| - E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \quad (5-93)$$

In deriving Equ.(5-93), E_{Th} has been considered positive as shown in Fig.5-52(b). But Equ.(5-91) may result negative value for E_{Th} . In that case, E_{Th} will positive in Equ.(5-93). Once the value of I_B is found other values of the operating point can be calculated as described for NPN transistor [Equ.(5-86) and Equ.(5-87)].

For proper biasing of the PNP transistors, the value of E_{Th} should be less than $+V_{EE}$. Hence, we may consider $E_{Th} = 0$. From this consideration, the voltage divider bias circuit of Fig.5-52(a) can also be designed just using only a single base resistor, as shown in Fig.5-53.

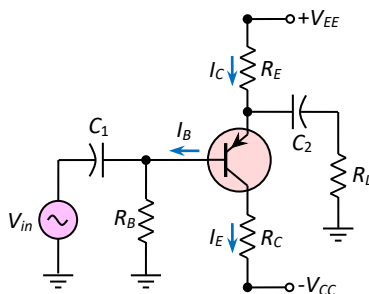


Fig.5-53: Biasing circuit for PNP transistor using a base resistor

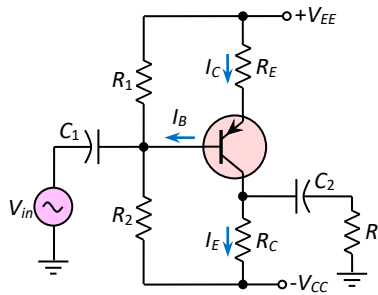


Fig.5-54: Alternate representation of Fig.5-52(a)

Circuit of Fig.5-52(a) can also be redrawn as in Fig.5-54.

Example 5-19

For the voltage divider bias circuit of Fig.5-55, draw the load-line and calculate the operating point.

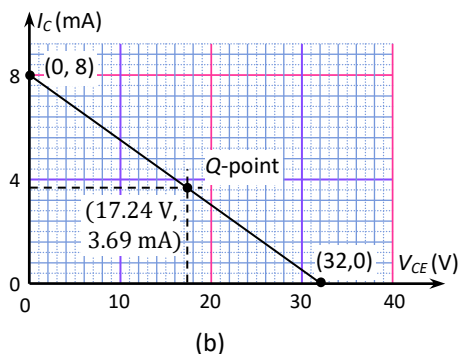
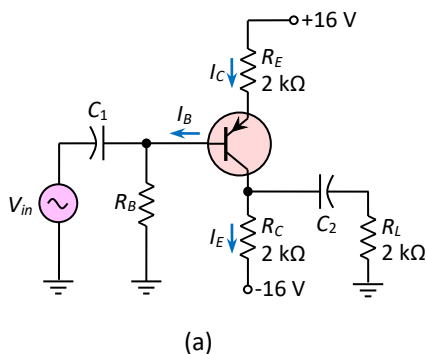


Fig.5-55: PNP transistor circuits for Example 5.15 (a) Negative voltage (b) Positive voltage

Solution:

Load line:

The load-line of this circuit can be calculated using the following equation:

$$V_{CE} = (V_{EE} - V_{CC}) - I_C(R_C + R_E)$$

$$\text{or,} \quad V_{CE} = (16 \text{ V} + 16 \text{ V}) - I_C(2 \text{ k}\Omega + 2 \text{ k}\Omega)$$

$$\text{or,} \quad V_{CE} = 32 \text{ V} - I_C(4 \text{ k}\Omega)$$

From this equation we find that when, $I_C = 0$, $V_{CE} = 32 \text{ V}$ and when $V_{CE} = 0$, $I_C = 32 \text{ V} / 4 \text{ k}\Omega = 8 \text{ mA}$. Putting these points on the I_C vs. V_{CE} coordinate system and connecting them we will find the load-line [shown in Fig.5-(b)]

Operating point:

The base current of this circuit can be calculated as,

Using Equ.(5-),
$$I_B = I_{BQ} = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{320 \text{ k}\Omega + (150 + 1)2\text{k}\Omega} \approx 23.6 \mu\text{A}$$

Using Equ.(5-57),
$$I_C = I_{CQ} = I_{BQ}\beta = 23.6 \mu\text{A} \times 150 \approx 3.69 \text{ mA}$$

Using Equ.(5-59),
$$V_{CE} = V_{CEQ} = (V_{EE} - V_{CC}) - I_C(R_C + R_E)$$

or,
$$V_{CEQ} = 32 \text{ V} - 3.69 \text{ mA}(4 \text{ k}\Omega) \approx 17.24 \text{ V}$$

The position of the operating point $(V_{CEQ}, I_{CQ}) = (17.24 \text{ V}, 3.69 \text{ mA})$ is shown on the load-line of Fig.5-

Comments: Analysis of this biasing circuit of PNP transistor is similar to that of NPN transistors.

5.20 Comparison of Biasing Circuits

It is already discussed that the main target of a biasing circuit is to provide a stable operating point irrespective of the change in temperature or the transistor current gain β . As the values of β for a particular transistor changes in a wide range (from $\beta_{\min}$ to $\beta_{\max}$), the biasing circuit must have the ability to compensate this wide change of β . Suppose an electronics company has designed a biasing circuit using the typical value of β for transistor. But when the circuit is manufactured, the transistor may have maximum value of β . If the biasing circuit doesn't have the ability of

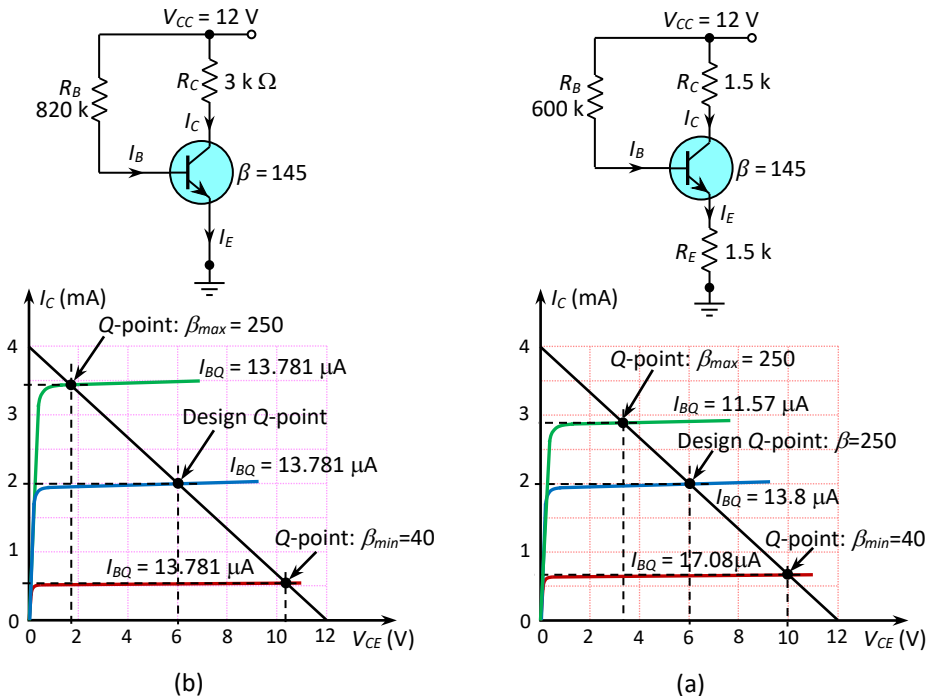


Fig.5-56: Comparison of biasing circuits, (a) Base bias, and (b) Emitter bias

compensate this change, the transistor may be saturated and the circuit will not work. In actual design, a circuit is tested whether it works for the minimum and maximum value of β . Thus, the performance of a biasing circuit is evaluated by the change in operating point due to the change in β . This is the main parameter of evaluation. Moreover, we can consider the number of components required, thermal stability, power loss etc. to compare the quality of biasing circuits.

So far we have studied four types of biasing circuits. We have already designed four biasing circuits considering $\beta = 145$. For every circuit the quiescent collector currents have been set approximately 2 mA and V_{CEQ} at the middle of the load line. The total resistance in the collector and emitter is kept 3 k Ω for every circuit. Then, the change in operating point for $\beta_{min} = 40$ and $\beta_{max} = 250$ has also been calculated. For better comparison the circuits and the graphs are reproduced here in Fig.5-56 and 5-57. The positions of the operating points are also plotted in a single load line in Fig.5-58.

From these graphs we can say that the voltage divider biasing circuit gives the highest stability and the base bias gives the lowest stability against the change of β . The emitter bias has stability greater than the base bias but lower than, collector-to-base bias and voltage divider bias. Although, here, the effect of change of β has been shown, the voltage divider bias gives the best thermal stability as well. That's why voltage divider bias is mostly used. The main disadvantages of voltage divider bias are, it needs more components, and the voltage divider network produces some power loss. On the other hand, the base bias needs the fewest components and produces minimum losses in biasing component. For this reason, the base bias is used for switching purpose.

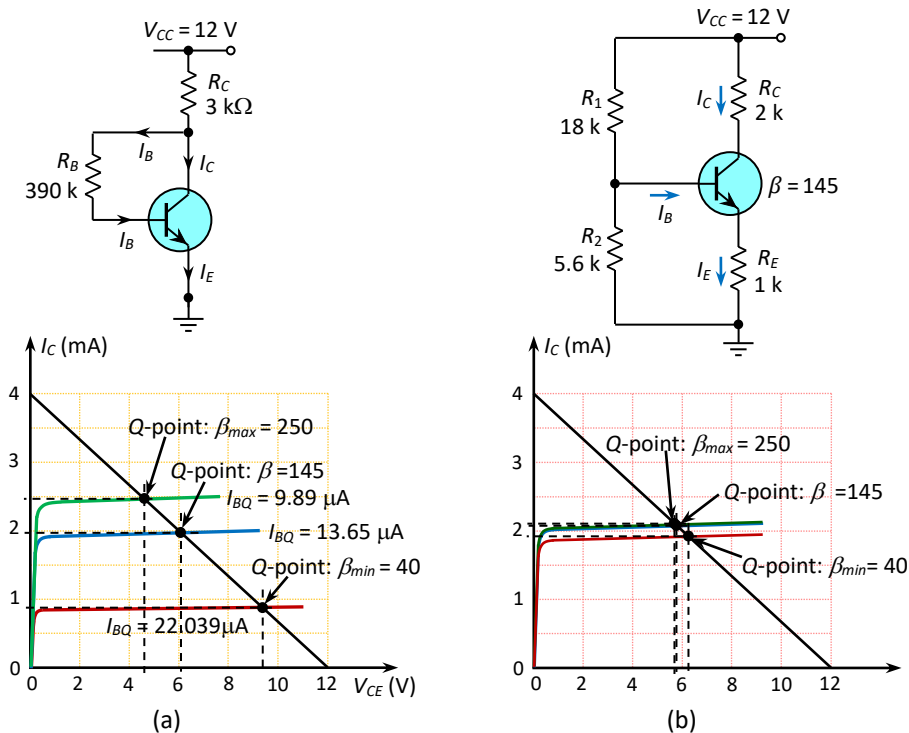


Fig.5-57: Comparison of biasing circuits, (a) Collector-to-base bias, and (b) Voltage divider bias

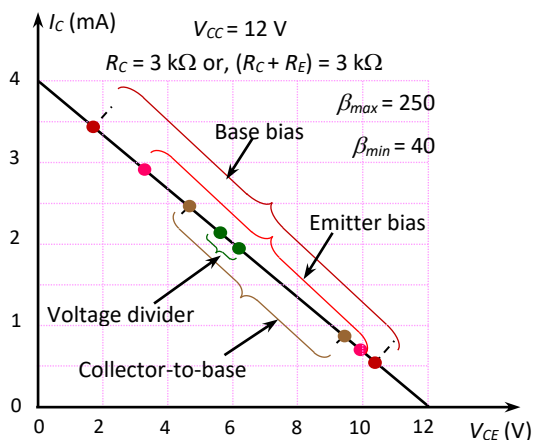


Fig.5-58: Change of operating points due to change in β in four biasing circuits

5.21 Biasing Circuit Design

Designing any circuit means to determine the appropriate values of the components and also selection of the devices. Designing a bias circuit depends on a lot of decisions, for example, supply voltage, the load to be driven, level of stability, position of operating point, level of collector current etc. However, for a given supply voltage, collector current and the transistor current gain β , the design process is quite simple. Unless otherwise specified normally the biasing circuit is designed to keep the operating point more or less middle of the load line. If the level of collector current is not given and the load is also not known, then the collector current level can be considered in the range of 1 mA to 5 mA. After selecting the value of I_C , the voltage levels of V_{RC} , V_{CE} and V_E are determined. Dividing the collector current by β , base current is calculated and solving standard equations of the bias circuit, values of different resistors are calculated. Once, we get the resistance values, we have to find the nearest standard values. Then we may have to recalculate the current or voltage using the standard values of the resistance. The list of the standard resistors is given in [Appendix](#). We have to calculate the currents and voltage levels using β_{min} and β_{max} as well.

5.22 Base Bias Circuit Design

If V_{CC} is known, to set the operating point at middle of the load line, we have to set $V_{CE} = V_{RC} = V_{CC}/2$, or if the value of V_{CE} is given, voltage drop across the collector resistor will be $V_{RC} = V_{CC} - V_{CE}$. Now, if the value of I_C is given or selected, dividing V_{RC} by I_C will give the value of R_C . Value of I_B can be calculated simply dividing I_C by β . Finally, the value of R_B can be calculated as $R_B = (V_{CC} - V_{BE})/I_B$.

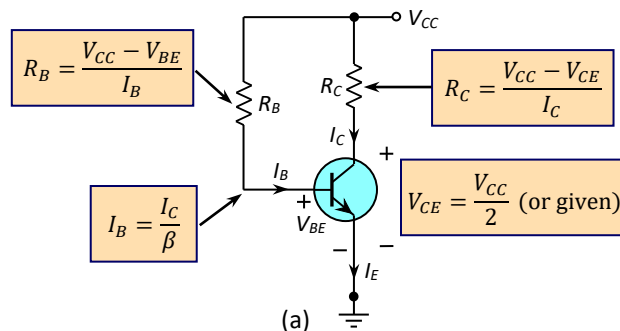


Fig.5-59: Base bias circuit design

Example 5-20

Design a base bias circuit that will be operated by a supply voltage of $V_{CC} = 12\text{ V}$. Assume the current gain of the transistor is $h_{FE} = 120$.

Solution:

As the collector current level is not given, let it be 2 mA.

Value of V_{CE} is not given as well. So,

$$V_{CE} = \frac{V_{CC}}{2} = \frac{12\text{ V}}{2} = 6\text{ V}$$

The value of base current will be,

$$I_B = \frac{I_C}{h_{FE}} = \frac{2\text{ mA}}{120} \approx 16.67\text{ }\mu\text{A}$$

Therefore,

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12\text{ V} - 0.7\text{ V}}{16.67\text{ }\mu\text{A}} \approx 678\text{ k}\Omega = 680\text{ k}\Omega$$

Value of R_C can be calculated as,

$$R_C = \frac{V_{CC} - V_{CE}}{I_C} = \frac{12\text{ V} - 6\text{ V}}{2\text{ mA}} \approx 3\text{ k}\Omega$$

The designed base bias circuit is shown in Fig.5-54.

Comments: Here, the operating point will be at the middle of the load-line. The gain of an amplifier depends on R_C . To increase the value of R_C , we can decrease the value of I_C to 1 mA, and $V_{CE} = 2\text{ V}$, as this type of biasing circuit is used as small signal amplifier.

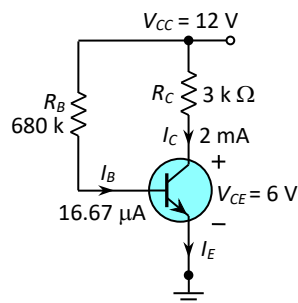


Fig.5-60: Designed base bias circuit of Example 5-19

5.23 Emitter Bias Circuit Design

The design procedure of emitter bias circuits is almost similar to that of base bias circuits except to determine the level of emitter voltage and hence the value of R_E . The values of collector current

should be decided by the current drawn by the load. To keep the voltage gain unchanged, I_C should be very large compared to load current (2 to 10 times). But, if the values of I_C and I_L are unknown, I_C can be set in the range of 1 mA to 5 mA. Once I_C is decided, next we have to determine the values of V_E and V_{CE} . Well, a larger value of V_E will result in a larger value of R_E and a larger R_E provides better stability. Question is how large should we set the value of V_E . A larger value of V_E or R_E produces smaller AC voltage swing across the load resistor R_C . Therefore, we can achieve stability at the cost of voltage gain. As the target of the

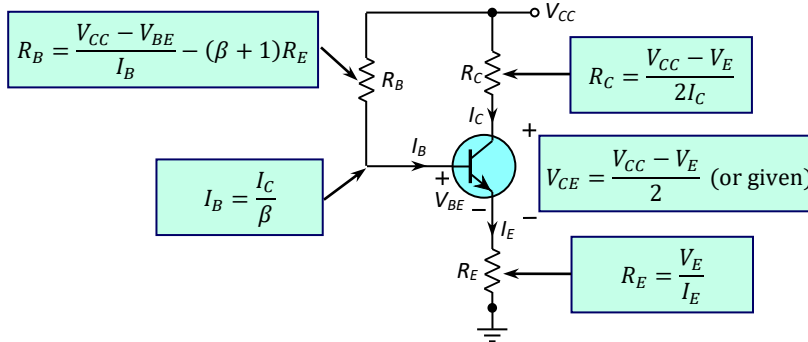


Fig.5-61: Emitter bias circuit design

biasing circuit is to provide voltage amplification, we must trade-off between the stability and the voltage gain. A thumb rule is to allocate more or less one-fourth [intuitively said by experience] of the supply voltage to the emitter, that is $V_E \approx V_{CC}/4$. The rest of the voltage should be equally divided between V_{CE} and V_{RC} . Now the values of R_C and R_E can easily be calculated by dividing their voltages by their current. Next by dividing I_C by β , the value of I_B can be calculated. Using this I_B , R_B can be found solving the equation of base current. The equations required to design an emitter bias circuit are shown in Fig.5-61. To make the concept clear, refer to the Example 5-20.

Example 5-21

Design an emitter bias circuit that will be operated by a supply voltage of $V_{CC} = 20$ V. Assume that the transistor has the current gain of 150.

Solution:

Let, $I_{CQ} = 2$ mA

and, $V_E = \frac{V_{CC}}{8} = \frac{20}{8} = 2.5$ V

$\therefore R_E = \frac{V_E}{I_E} \approx \frac{2.5}{I_{CQ}} = \frac{2.5}{2} = 1.25$ k Ω

Use $R_E = 1.2$ k Ω (standard).

$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2}{150} \approx 13.3333$ μ A

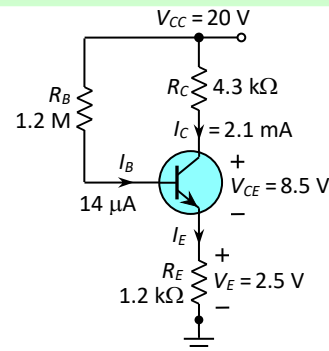


Fig.5-62: Emitter bias circuit designed in Example 5-20

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} - (\beta + 1)R_E = \frac{20 \text{ V} - 0.7 \text{ V}}{13.3333 \text{ } \mu\text{A}} - (150 + 1) \times 1.2 \text{ k}\Omega \approx 1.27 \text{ M}\Omega$$

Use $R_B = 1.2 \text{ M}\Omega$ standard. Actual value of I_B will be

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{1.2 \text{ M}\Omega + (150 + 1)1.2 \text{ k}\Omega} \approx 14 \text{ } \mu\text{A}$$

The value of actual $I_{CQ} = 14 \text{ } \mu\text{A} \times 150 \approx 2.1 \text{ mA}$.

$$V_E = 1.2 \text{ k}\Omega \times 2.1 \text{ mA} \approx 2.5 \text{ V}$$

$$\therefore V_{CE} = V_{RC} = \frac{1}{2}(V_{CC} - V_E) = \frac{1}{2}(20 \text{ V} - 2.5 \text{ V}) = 8.75 \text{ V}$$

$$\text{Let, } V_{CE} = 8.5 \text{ V} \quad \text{and} \quad V_{RC} = 9 \text{ V}$$

$$\therefore R_C = \frac{V_{RC}}{I_{CQ}} = \frac{9 \text{ V}}{2.1 \text{ mA}} \approx 4.3 \text{ k}\Omega$$

The designed circuit is shown in Fig.5-56.

Comments: Here, we have considered $V_E = 2.5 \text{ V}$, but if the swing of the amplified signal is small we can assign more voltage at emitter terminal for better stability.

5.24 Collector-to-Base Bias Circuit Design

The design procedure of collector-to-base bias circuits is almost similar to that of base bias circuits. But the voltage and current levels are different for calculating R_B and R_C . So, different equations have to be used to calculate their values. As found earlier, for this circuit the voltage across R_B is $V_{CE} - V_{BE}$ and the current through R_C is $I_B + I_C$. The design equations are shown in Fig.5-63. To make the concept clear, refer to the Example 5-21.

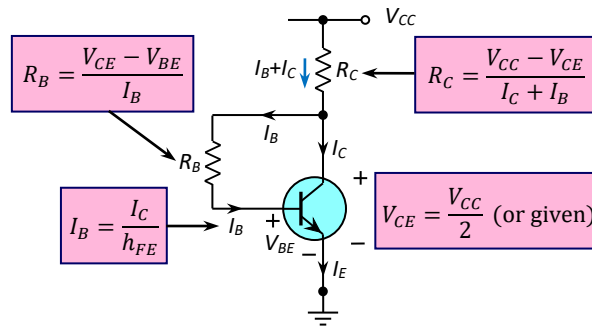


Fig.5-63: Collector-to-base bias circuit design

Example 5-22

Design a collector-to-base bias circuit to have $V_{CE} = 12 \text{ V}$ and $I_C = 5 \text{ mA}$. Assume that the supply voltage is 20 V and the transistor has $\beta = 120$.

Solution:

$$I_B = \frac{I_C}{\beta} = \frac{5 \text{ mA}}{120} \approx 41.67 \mu\text{A}$$

The value of collector resistor can be found as,

$$R_C = \frac{V_{CC} - V_{CE}}{I_C + I_B} = \frac{20 \text{ V} - 12 \text{ V}}{5 \text{ mA} + 41.67 \mu\text{A}} \approx 1.6 \text{ k}\Omega$$

Now, the value R_B can be calculated as,

$$\text{or, } R_B = \frac{V_{CE} - V_{BE}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{41.67 \mu\text{A}}$$

$$\text{or, } R_B \approx 271.18 \text{ k}\Omega \text{ (use standard } 270 \text{ k}\Omega)$$

The designed circuit is shown in Fig.5-58.

Comments: Here larger voltage has been assigned to V_{CE} , so the value of R_C is found lower.

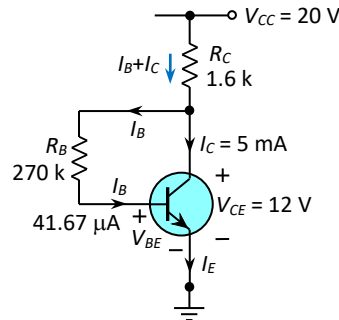


Fig.5-64: Collector-to-base bias circuit for Example 5-21

5.25 Voltage Divider Bias Circuit Design

The design procedure of voltage divider bias circuit is almost similar to that of emitter bias circuits except the calculation of R_1 and R_2 . Here also, the value of collector current should be decided depending on the current drawn by the load. To keep the voltage gain unchanged, I_C should be very large compared to load current (**2 to 10 times**). If the values of I_C and I_L are unknown, I_C can be set as before, i.e., in the range of 1 mA to 5 mA. Determination process of V_E is discussed in emitter bias circuit design. The same concept can be applied here. That is, $V_E \approx V_{CC}/8$. The rest of the voltage should be equally divided between V_{CE} and V_{RC} . Values of R_C and R_E can be calculated simply from Ohm's law. Next by dividing I_C by β , the value of I_B can be calculated. The base voltage will be $V_B = V_E + V_{BE}$.

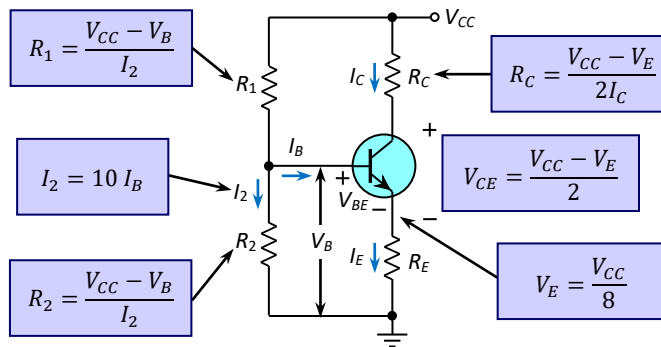


Fig.5-65: Voltage divider bias circuit design

The values of R_1 and R_2 should be as small as possible to keep V_B constant. Moreover, smaller values of R_1 and R_2 ensure better stability. But, too small R_1 and R_2 cause two problems. One is

loss of power, and the other is the lower input resistance. Hence, here we have to make another trade-off. There are a lot of processes to calculate R_1 and R_2 , that produce satisfactory results. We can assume,

$$I_2 \geq 10I_{BQ} \quad \text{or,} \quad I_2 = \frac{I_C}{10}$$

$$\text{or,} \quad R_2 \leq \frac{\beta R_E}{10}$$

Using the values of V_B and I_2 , R_2 can be calculated. Voltage across R_1 will be $V_{CC} - V_B$ and its current can also be assumed approximately I_2 . Using these value R_1 can be calculated. The equations required to design a voltage divider bias circuit are shown in Fig.5-65. To make the concept clear, refer to the Example 5-22.

Example 5-23

Design a voltage divider bias circuit considering the supply voltage 16 V and the transistor has $\beta = 100$.

Solution:

Let, $I_{CQ} = 2 \text{ mA}$ and

$$V_E = \frac{V_{CC}}{8} = \frac{16 \text{ V}}{8} = 2 \text{ V}$$

$$V_{CE} = V_{RC} = \frac{V_{CC} - V_E}{2} = \frac{16 \text{ V} - 2 \text{ V}}{2} = 7 \text{ V}$$

$$\therefore R_E = \frac{V_E}{I_E} = \frac{V_E}{I_{CQ}} = \frac{2 \text{ V}}{2} = 1 \text{ k}\Omega$$

$$R_C = \frac{V_{RC}}{I_{CQ}} = \frac{7 \text{ V}}{2} = 3.5 \text{ k}\Omega \text{ use standard } 3.6 \text{ k}\Omega$$

$$V_B = V_E + V_{BE} = 2 \text{ V} + 0.7 \text{ V} = 2.7 \text{ V}$$

$$\text{Let,} \quad I_2 = \frac{I_{CQ}}{10} = \frac{2 \text{ mA}}{10} = 0.2 \text{ mA}$$

$$\therefore R_2 = \frac{V_B}{I_2} = \frac{2.7 \text{ V}}{0.2 \text{ mA}} = 13.5 \text{ k}\Omega \text{ use standard } 13 \text{ k}\Omega$$

$$\text{Actual } I_2 \text{ will be} \quad I_2 = \frac{2.7 \text{ V}}{13 \text{ k}\Omega} \approx 0.208 \text{ mA}$$

$$R_1 = \frac{V_{CC} - V_B}{I_2} = \frac{16 \text{ V} - 2.7 \text{ V}}{0.208 \text{ mA}} \approx 64 \text{ k}\Omega \text{ use standard } 62 \text{ k}\Omega$$

The designed circuit is shown in Fig.5-66.

Comments: The actual value of I_1 can be calculated as $I_2 + I_{CQ}/\beta$.

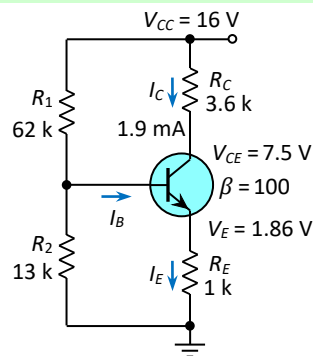


Fig.5-66: Voltage divider bias circuit designed in Example 5-22

5.26 Biasing Circuit for Common Base Amplifier

We have already learnt that in common-base amplifier, the input signal is applied to the emitter and base terminals and the output is taken from collector and base as shown in Fig.5-67(a). That is, the base is common to both the input and output circuit. The base-emitter junction is forward biased by V_{EE} and R_E and the collector-base junction is reverse biased by V_{CC} and R_C . To prevent the loading effect on the DC biasing condition, the input source and the output load are connected through two coupling capacitors. In electronics, these amplifiers have some special applications, because they have very small input impedance and very large output impedance and provide maximum voltage gain among all three combinations.

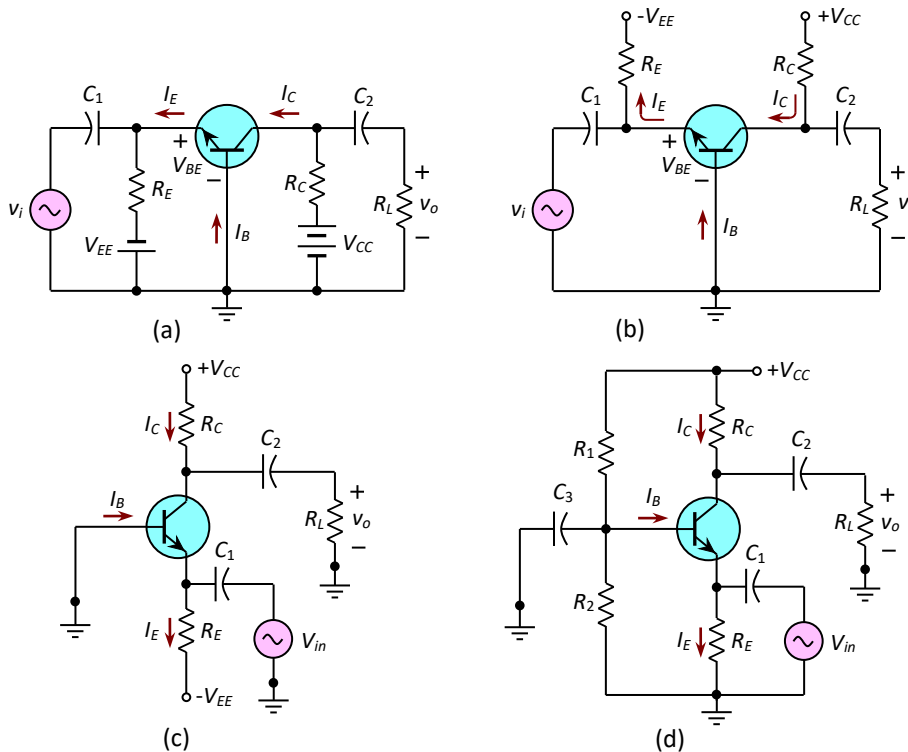


Fig.5-67: Biasing circuit for common–base amplifier, (a) Using two separate voltage source, (b) and (c) Same circuit represented alternately, (d) Voltage divider bias for CB amplifier.

The circuit of Fig.5-67(a) can easily be redrawn as in the Fig.5-67(b) and (c). Thus, using a split voltage source a CB amplifier can be biased as shown in Fig.5-67(c). This circuit is exactly same as emitter bias or emitter current bias with $R_B = 0$. However, we can also bias this circuit using a single voltage source as in Fig.5-67(d). In this circuit, the base is at a higher potential than the emitter which will be determined by R_1 and R_2 as CE voltage divider bias circuit. But, for the AC signal the base will be at the ground potential. In common base amplifier, the output current (I_C) is related to the input current (I_E) by α . The value of α is almost constant (≈ 1). Hence, the biasing circuit of CB amplifier is completely β independent.

Now, applying KVL in the input circuit (B-E loop in Fig.5-68(a), we get,

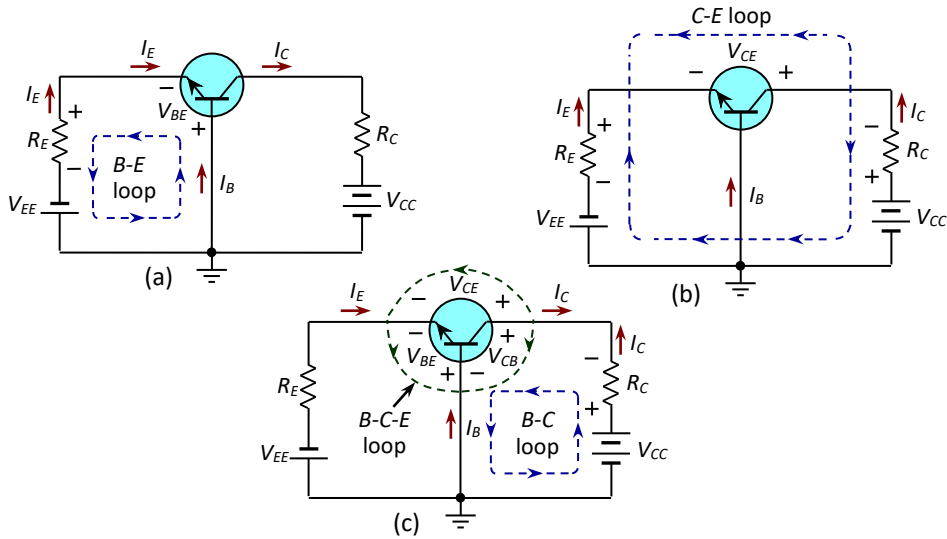


Fig.5-68: KVL applied to different loops of CB circuit (a) B-E loop, (b) C-E loop and, (c) B-C and B-C-E loops

$$-V_{BE} - I_E R_E + V_{EE} = 0$$

$$\therefore \boxed{I_E = \frac{V_{EE} - V_{BE}}{R_E}} \quad (5-94)$$

Again, applying KVL in the C-E loop of Fig.5-68(b), we get,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E + V_{EE} = 0$$

or,

$$V_{EE} + V_{CC} - I_C (R_C + R_E) - V_{CE} = 0 [\because I_E \approx I_C]$$

$$\therefore \boxed{V_{CE} = (V_{EE} + V_{CC}) - I_C (R_C + R_E)} \quad (5-95)$$

Applying KVL around the transistor, i.e., C-E-B loop in Fig.5-68(c), we get

$$V_{CB} - V_{CE} + V_{BE} = 0$$

$$\therefore \boxed{V_{CB} = V_{CE} - V_{BE}} \quad (5-96)$$

Again, from the output circuit [B-C loop of Fig.5-68.(c)] we get,

$$V_{CC} - I_C R_C - V_{CB} = 0$$

$$\therefore \boxed{V_{CB} = V_{CC} - I_C R_C} \quad (5-97)$$

In case of single power supply [Fig.5-67(d)], if we draw the DC equivalent circuit (all the capacitors open), then we will get the voltage divider biasing circuit as before. So, by using the equations of previous analysis, we can solve this circuit of CB configuration.

To see the variation of output voltage and current, we draw the load line on the output characteristic curve. For common-base configuration, we have seen in Chapter 4, the output characteristics is a I_C versus V_{CB} graph. Hence, the load line of CB amplifier has to be drawn using Equ.(5-95). From Equ.(5-95), we get,

$$V_{CB} = V_{CC} - I_C R_C \quad (5-98)$$

When, $I_C = 0, V_{CB} = V_{CC}$: this is represented as point A and lies on x -axis.

When, $V_{CB} = 0, I_C = V_{CC}/R_C$: this point is represented by B and lies on y -axis.

By joining these two points, the load line can be drawn.

Example 5-24

For the biasing circuit of common-base configuration in Fig.5-69, determine the load line and the operating point.

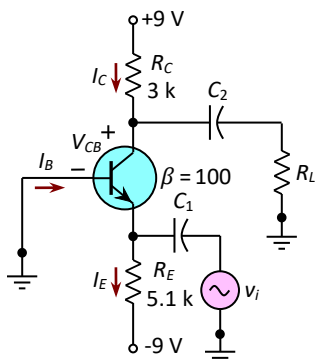


Fig.5-69: CB circuit for Example 5-23

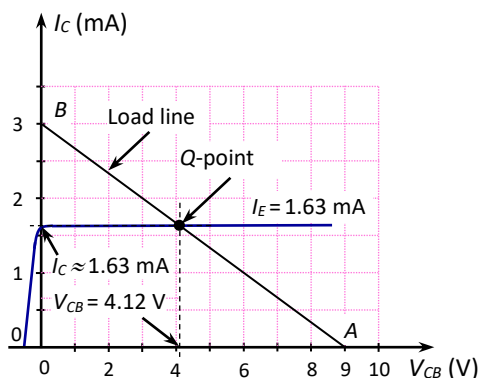


Fig.5-70: Load line and Q-point for Example 5-23

Solution:

Though the circuit has both positive and negative voltage, load line will use the positive one. So it will start from point $A(9\text{ V}, 0)$ on x -axis, and will end at point $B(0, 9\text{ V}/3\text{ k}\Omega) = (0, 3\text{ mA})$ on y -axis. It is shown in Fig.5-70.

Using Equ.(5-94) the emitter current is,

$$I_E \approx I_C = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6\text{ V} - 0.7\text{ V}}{5.1\text{ k}\Omega} \approx 1.63\text{ mA}$$

Using Equ.(5-95), the collector-base voltage will be,

$$V_{CB} = V_{CC} - I_C R_C = 9\text{ V} - 1.63\text{ mA} \times 3\text{ k}\Omega = 4.12\text{ V}$$

Comments: Swing of the Q -point will be limited by the value of V_{CC} . V_{EE} will only forward bias the input circuit.

5.27 Biasing Common Base Circuits Using PNP Transistor

The same biasing circuits can be used for PNP transistors. Only the supply voltage has to be reversed. For PNP transistors, the emitter has to be connected to higher potential and the collector to the lower potential. The base should be at higher potential than the collector and lower potential than the emitter. This can be easily done by either a split voltage source as in Fig.5-65 (a) or by using a **single-ended** voltage source as in Fig.5-65(b). Fig.5-65(a) is emitter bias and the Fig.5-

65(b) is the voltage divider bias. All the equations derived for the NPN transistor can also be used to analyze these circuits.

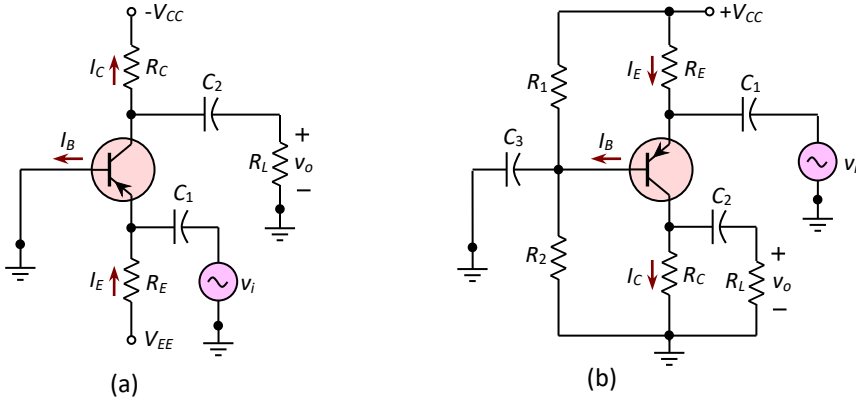


Fig.5-71: Biasing of PNP transistors in CB configuration, (a) Emitter bias with a split voltage supply, (b) Voltage divider bias with a single source

Example 5-25

Design a common-base amplifier with emitter bias using a PNP transistor of $h_{FE} = 150$. Assume the supply voltage is ± 12 V.

Solution:

Let, $I_{CQ} = 2 \text{ mA} \approx I_E$

$$\therefore I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2 \text{ mA}}{150} \approx 13.3333 \mu\text{A}$$

Using Equ.(5-94),

$$R_E = \frac{V_{EE} - V_{BE}}{I_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{2 \text{ mA}} \approx 5.6 \text{ k}\Omega$$

From Fig.5-72, for the base-emitter circuit we can write,

$$I_{BQ} = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

or, $R_B = \frac{V_{EE} - V_{BE}}{I_{BQ}} - (\beta + 1)R_E$

$$\therefore R_B = \frac{12 \text{ V} - 0.7 \text{ V}}{20.0 \mu\text{A}} - (100 + 1) 3 \text{ k}\Omega \approx 394 \text{ k}\Omega \text{ use standard } 390 \text{ k}\Omega$$

Let, $V_{CB} = V_{RC} = \frac{1}{2}(|V_{CC}|) = \frac{1}{2} 12 \text{ V} = 6 \text{ V}$

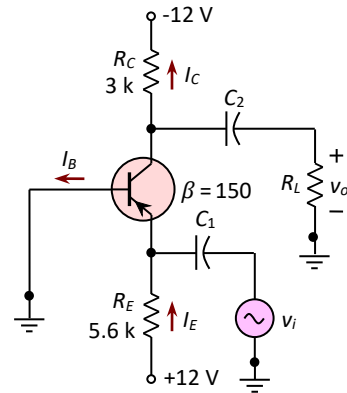


Fig.5-72: Designed circuit of Example 5-24

$$\therefore R_C = \frac{V_{RC}}{I_{CQ}} = \frac{6 \text{ V}}{2 \text{ mA}} \approx 3 \text{ k}\Omega$$

The designed circuit is shown in Fig.5-72.

Comments: The output voltage of this circuit is V_{CB} . Here, V_{CC} is divided between V_{CB} and R_C .

5.28 Biasing Circuits for Common Collector Amplifier

In Chapter 3, we have seen how to bias a transistor in common-collector configuration using two voltage sources. That biasing circuit is redrawn in Fig.5-73(a). Here, the voltage sources V_{BB} and V_{EE} with two series resistors (R_B and R_E) are used to bias the transistor. The input signal (v_i) is applied to the base through a coupling capacitor C_1 and the output is taken from the emitter through another coupling capacitor C_2 . The collector is common to both the input and output circuits. Fig.5-73(b) is just the alternate representation of the first figure.

If we carefully look at the Fig.5-73(a) and (b), we find that the base current is flowing from the negative terminal to the base of the transistor. We know that conventional current cannot flow from the negative terminal of a source. But, here it is possible because $-V_{BB}$ will be comparatively at higher potential than $-V_{EE}$. We know that the base of this circuit should be at higher potential than the emitter. Instead of using $-V_{EE}$, if we connect the base terminal to ground through a resistor, then we will find the same result, as ground is in higher potential than $-V_{EE}$. Hence, we can represent the circuit of Fig.5-73(b) as in Fig.5-73(c).

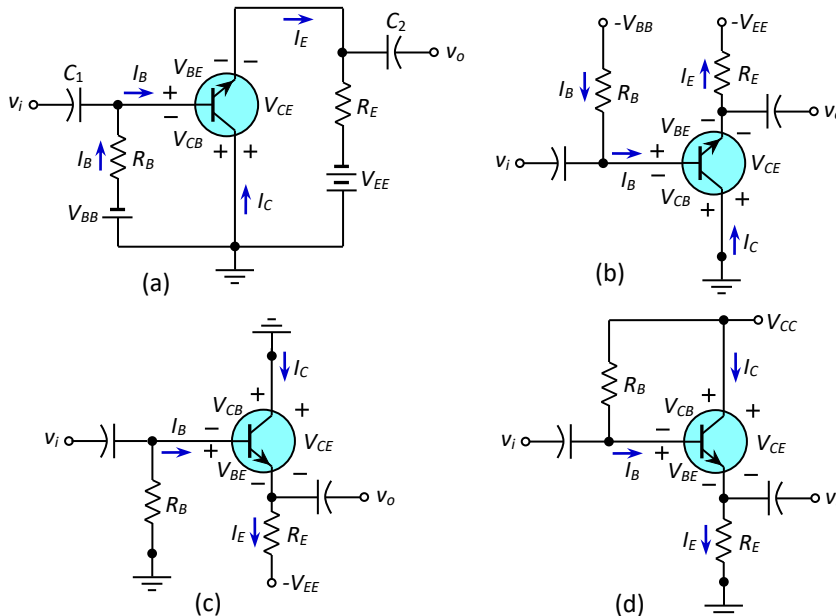


Fig.5-73: Biasing circuit for common-collector amplifier, (a) Using two separate voltage sources, (b) Same circuit represented alternately, (c) CC biasing circuit using negative voltage and (d) CC biasing circuit using positive voltage

Alternately, we can keep the base and collector at higher potential than the emitter, by connecting the positive terminal of a voltage source to the base and collector and the ground of the source to the emitter as shown in Fig.5-73(d). Finally, we can say that the circuit of Fig.5-73(c) is the common-collector amplifier with negative biasing system and the circuit of Fig.5-73(d) is that with positive voltage system. Here, R_E provides negative feedback and better stability as discussed for emitter bias circuit. These two circuits are called *emitter bias* for CC configuration. As shown in Fig.5-74, *voltage divider* bias can also be used for common-collector amplifier. For all of these biasing circuits, a relation between the input and the output voltage can be found, which is,

$$v_o = v_i - V_{BE} = v_i - 0.7 \text{ V} \quad (5-99)$$

From this equation, we see that the output voltage is slightly smaller than the input voltage. If we neglect the voltage drop of the base-emitter junction (V_{BE}), we can say the output is equal to the input. That is the voltage gain of this amplifier is (approximately) unity. For this reason, this circuit is called the **voltage follower** or **emitter follower**. In spite of its unity voltage gain, it provides a very large current gain. The input impedance of this amplifier is very high and the output impedance is low. Therefore, this amplifier is used at the final stage of a system for impedance matching or current amplification.

Analysis of Fig.5-73(a) to (c), will result in the same equations. Applying KVL in the input circuit of Fig.5-73(c) [redrawn in Fig.5-75(a)], we get

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

$$\text{or,} \quad -I_B R_B - V_{BE} - (\beta + 1)I_B R_E + V_{EE} = 0$$

$$\text{or,} \quad I_B [R_B + (\beta + 1)R_E] = V_{EE} - V_{BE}$$

$$\therefore I_B = \frac{|V_{EE}| - V_{BE}}{R_B + (\beta + 1)R_E} \quad (5-100)$$

Value of output current will be,

$$I_E = (\beta + 1)I_B \quad (5-101)$$

Applying Kirchhoff's voltage law at the output circuit, we get [Fig.5-75(b)],

$$V_{EE} - V_{CE} - I_E R_E = 0$$

$$\text{or,} \quad V_{CE} = V_{EE} - I_E R_E \quad (5-102)$$

For the positive voltage system, Fig.5-73(d) the base current will be,

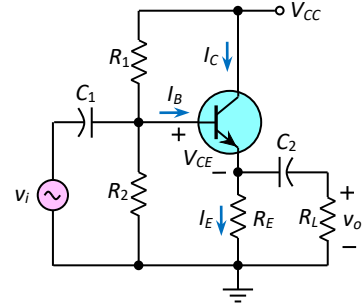


Fig.5-74: Voltage divider bias for CC configuration.

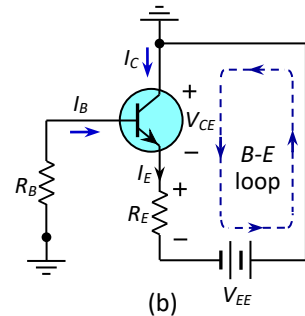
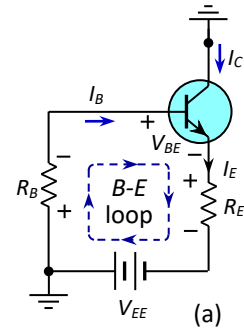


Fig.5-75: Input and output loops of common collector circuit.

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \quad (5-103)$$

$$V_{CE} = V_{CC} - I_E R_E \quad (5-104)$$

For the voltage divider biasing circuit, we have to find out the Thevenin's equivalent circuit as before and the equations will be,

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \quad (5-105)$$

$$V_{CE} = V_{CC} - I_E R_E \quad (5-106)$$

For common-collector configuration, we have seen in Chapter 3, the output characteristic is I_E versus V_{CE} graph. Hence, the load line of CC amplifier can be drawn using Equ.(5-106)

$$V_{CE} = V_{CC} - I_E R_E \quad (5-107)$$

When, $I_E = 0, V_{CE} = V_{CC}$: this is represented by point A and lies on x-axis.

When, $V_{CE} = 0, I_E = V_{CC}/R_E$: this point is represented by B and lies on y-axis [See Fig.5-76(b)].

By joining these two points, we will find the load line.

Example 5-26

For the common-collector biasing circuit in Fig.5-76(a) determine the load line and operating point.

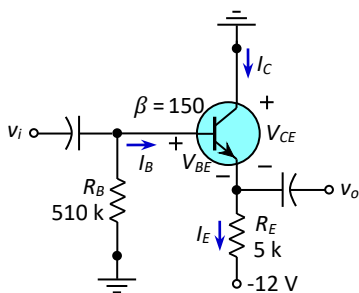


Fig.5-76(a): CC circuit for Example 5-25

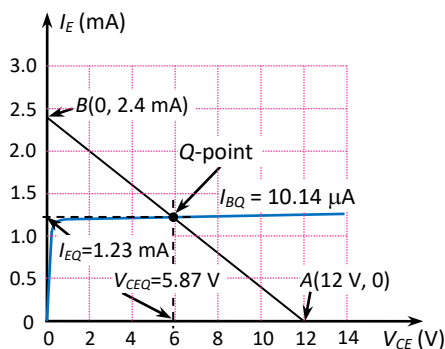


Fig.5-76 (b): Load line and Q-point for Example 5-25

Solution:

Load line can be drawn by joining point A(12 V, 0) on x-axis, and point B(0, 12 V/5 k) = (0, 2.4 mA) on y-axis. It is shown in Fig.5-76(b).

Using Equ.(5-100) the base current is,

$$I_{BQ} = \frac{|V_{EE}| - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega + (120 + 1)5 \text{ k}\Omega} \approx 10.1345 \text{ }\mu\text{A} \text{ [Ans.]}$$

Using Equ.(5-101), the output current is,

$$I_{EQ} = (\beta + 1)I_B = (120 + 1)10.1345 \text{ }\mu\text{A} \approx 1.2262 \text{ mA [Ans.]}$$

Using Equ.(5-102), the collector-emitter voltage will be,

$$V_{CE} = V_{EE} - I_E R_E = 12 \text{ V} - (1.2262 \text{ mA})5 \text{ k}\Omega \approx 5.87 \text{ V [Ans.]}$$

Comments: The process of determining load line and operating point of this circuit is same as those for emitter bias circuit.

5.29 Common Collector Biasing Circuits for PNP Transistor

The emitter bias and the voltage divider bias discussed for CC NPN transistors can also be used for CC PNP transistors. For PNP transistors, the emitter has to be placed at higher potential than the other two terminals. This can easily be done by either a positive voltage system as in Fig.5-77(a) or by using a negative voltage system as in Fig.5-77(b) and (c). Fig.5-77(a) and (b) are emitter bias and Fig.5-77(c) is voltage divider bias. We can also use positive voltage system for the voltage divider bias, connecting the ground of the source to upper terminal and the positive terminal to the lower terminal as shown in Fig.5-78 of Example 5-26. All the equations derived for the NPN transistor can also be used for the circuits with PNP transistors.

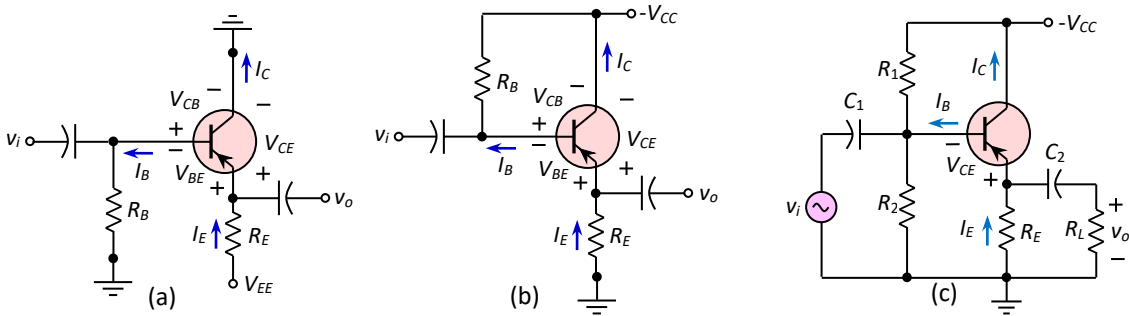


Fig.5-77: Biasing of PNP transistors in CC configuration, (a) Emitter bias with positive voltage supply, (b) Emitter bias with negative voltage supply, and (c) Voltage divider bias with a negative voltage source

Example 5-27

Design a common-collector amplifier with voltage divider bias using a PNP transistor of $h_{FE} = 120$. Assume the supply voltage is 12 V.

Solution:

Let,

$$I_{EQ} \approx I_{CQ} = 2 \text{ mA}$$

and,

$$V_E = \frac{1}{2} V_{CC} = 6 \text{ V}$$

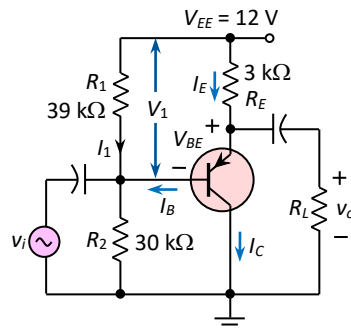


Fig.5-78: Common collector circuit designed in Example 5-26

$$\therefore R_E = \frac{V_E}{I_{EQ}} = \frac{6 \text{ mA}}{2 \text{ V}} = 3 \text{ k}\Omega$$

$$\text{and } V_1 = V_{RE} + V_{BE} = (V_{CC} - V_E) + V_{BE} = (12 \text{ V} - 6 \text{ V}) + 0.7 \text{ V} = 6 \text{ V} + 0.7 \text{ V} = 6.7 \text{ V}$$

$$\text{The value of base current will be } I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{2 \text{ mA}}{120} \approx 16.67 \text{ }\mu\text{A}$$

$$\text{Let, } I_1 = 10I_{BQ} = 166.7 \text{ }\mu\text{A}$$

$$\therefore R_1 = \frac{V_B}{I_2} = \frac{6.7 \text{ V}}{166.7 \text{ }\mu\text{A}} \approx 40.2 \text{ k}\Omega \text{ use } 39 \text{ k}\Omega \text{ standard}$$

$$\therefore R_2 = \frac{V_{CC} - V_B}{I_2 + I_{BQ}} = \frac{12 \text{ V} - 6.7 \text{ V}}{166.7 \text{ }\mu\text{A} + 16.67 \text{ }\mu\text{A}} \approx 29 \text{ k}\Omega \text{ use standard } 30 \text{ k}\Omega$$

The designed circuit is shown in Fig.5-72.

Comments: In common collector biasing circuit, the emitter voltage should be half of the supply voltage instead of **one eighth** as used for CE amplifier. Moreover, B-E junction will be forward biased by the voltage across R_1 not R_2 .

5.30 Transistor as a Switch

So far varieties of biasing circuits for different configurations of transistors have been discussed. The targets of these biasing circuits were to set the operating point of the transistor in the active region, so that the circuit can be used as an amplifier. However, in some applications a BJT may be used as a switch. Though for high power switching circuits, MOSFET (Metal-Oxide-Semiconductor. Field Effect Transistor) is most suitable, for low power switching BJT is also used. When used as a switch, a transistor is biased in such way that it can operate either in the cutoff region (OFF switch) or in the saturation region (ON switch). For a switch, we want that the load will consume power but the switch will consume zero power. How the transistor-switches consume ideally zero power (practically very small power) will be described next with a circuit. As the common emitter configuration gives both current gain and voltage gain, this configuration is mainly used as a switch. The schematic diagram of a switching circuit is given in Fig.5-79(a). This circuit is like a fixed bias configuration, but here the base resistor R_B is not directly connected to the supply voltage V_{CC} , rather it is connected to the system that will drive the switch ON/OFF. Here, R_C is the load for the switch. The driving system may be any kind of controlling circuit like a microcontroller based system, an Arduino system or a digital circuit or even an analog circuit. In our home, we use manual switch, but the transistor is a voltage controlled switch. When the driving system, gives high voltage (greater than 1 V), the B-E junction will be forward biased and current will flow through the load. The circuit will be designed such a way that the transistor will be saturated, i.e., the value of V_{CE} will be minimum ($V_{CE(sat)} \approx 0.2 \text{ V}$) and the transistor will work as an ON switch. Almost the full supply voltage will be dropped across the load, and it will be powered up. But if the driving system provides low voltage (less than 0.5 V), the B-E junction will not be forward biased, and hence, the transistor will work in the cutoff region and only the leakage current I_{CEO} will flow through the load. Almost the full supply voltage will be dropped across the transistor, i.e., $V_{CE} =$

V_{CC} . As the current is almost zero, the load will not get power. In this way, the transistor will operate either in saturation or cutoff region as shown in Fig.5-79.

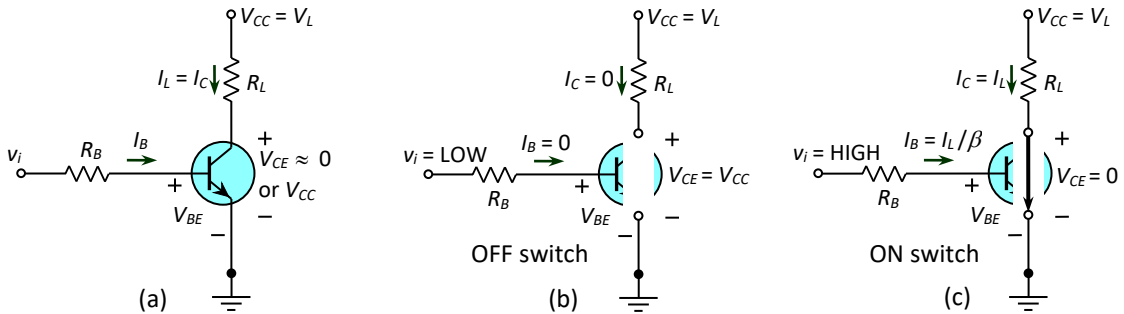


Fig.5-79: Switching action of transistor (a) Switching circuit, (b) Transistor is open (OFF), and (c) Transistor of closed (ON).

If we neglect $V_{CE(sat)}$ and I_{CEO} , when the transistor is ON, it carries maximum current but voltage drop across it (V_{CE}) is zero, so the power loss by the transistor will be zero. On the other hand, when the transistor is OFF, it drops the maximum voltage ($V_{CE} = V_{CC}$) but current through it is zero. So the power loss in the transistor will be zero. However, note that for very high power switching circuit, we have to use power transistor that has a metal body or plastic body with

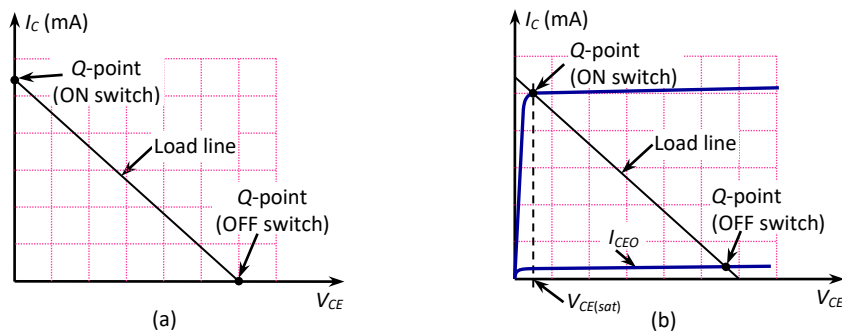


Fig.5-80: Load line of transistor switch (a) Ideal transistor, and (b) Practical switch

metal collector (TO-218, TO-220, TO-264 as shown in Chapter 3). In this case, the leakage current and the saturation voltage produce a substantial amount of power loss. So we have to use heat sink with the transistor. The heat sink is connected to the metal collector of the transistor. It absorbs heat and keeps the transistor-temperature in tolerable range. Here, it is described how a transistor can control power of a load (load ON/OFF). There is another application of a transistor switching circuit which is called the **inverter**.

The inverter circuit is shown in Fig.5-81. Here, R_C is not the load, rather the load is connected to the collector as shown in the figure. Well, we will see its operation with a square-wave input at the input terminal. When the input signal is high, the transistor will be ON and the output voltage will

be almost zero ($v_o = V_{CE(sat)}$). Again, when the input voltage is low, the transistor will be OFF and the output voltage will high ($v_o = V_{CC}$). In this way, when the input is HIGH (logic 1) the output is LOW (logic 0), and when the input is LOW (logic 0) the output is HIGH (logic 1) as shown in Fig.5-81(b). Therefore, we find that the circuit just inverts the input signal- hence it is called **inverter**.

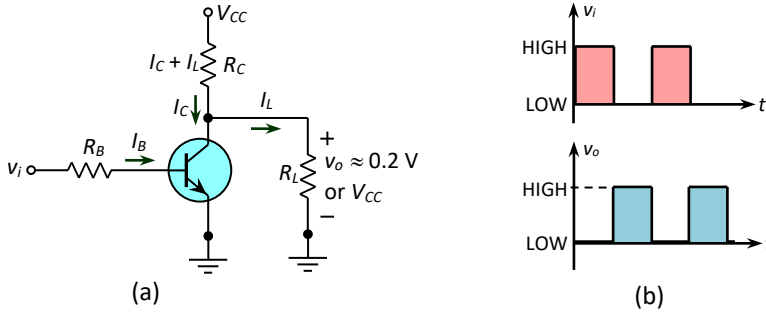


Fig.5-81 (a): Inverter circuit, (b) Input and output waveform.

5.31 Design of the Switch Circuit:

To design a power switch, we must know the specifications of the load, that is, its power and voltage ratings (P_L and V_L). Dividing the rated power (P_L) by the rated voltage (V_L), we will get the current consumed by the load (I_L). The value of V_{CC} of the switching circuit will be equal to the rated voltage (V_L) of the load. Now we have to select a BJT, that has $V_{CE(max)} \geq V_{CC}$ and $I_{C(max)} \geq I_L$. To calculate the value of I_B , $h_{FE(min)}$ has to be used. If we use the typical value of h_{FE} the transistor will not be saturated, if it is replaced by a transistor with lower h_{FE} .

From the load current I_L , base current has to be calculated as,

$$I_B = \frac{I_L}{h_{FE(min)}} = \frac{I_C}{h_{FE(min)}} \quad (5-108)$$

To calculate the base resistor, the HIGH voltage of the input signal has to be used,

$$\therefore R_B = \frac{V_{in(HIGH)} - V_{BE}}{I_B} \quad (5-109)$$

For R_B use the lower standard values, not the higher values, no matter which one are closest. If we use the higher standard values the transistors will not be saturated and will cause unnecessary power loss.

If the V_{CC} of a system is fixed, but the load needs lower voltage, then we have to use a resistor in series with the load. Its value can be calculated as,

$$R_C = \frac{V_{CC} - V_L}{I_C} \quad (5-110)$$

But, remember this collector resistor will produce power loss. So this type of arrangement can be used only in low power circuits.

If any inductive load is used in the collector, a PN junction diode has to be used across the load as shown in Fig.5-82. Otherwise, the induced voltage of the inductive load may destroy the transistor.

Example 5-28

A controlling circuit requires to ON/OFF a 220 V CFL light by a 12 V relay. If the output of the controlling circuit is TTL standard, design a transistor switching circuit to drive the relay and draw the schematic diagram of the full system.

Solution:

The controlling system will drive the transistor switch by TTL logic voltage, i.e., 2.5 V to 5.5 V in HIGH level and 0 to 0.8 V in LOW level. The transistor will drive the relay and the relay will drive the CFL lamp.

Let, $V_{CC} = V_L = 12\text{ V}$ and $I_L = 80\text{ mA}$ [typically a 12 V really needs 30 to 80 mA]

We need a transistor that has $V_{CE(max)} \geq 12\text{ V}$ and $I_{C(max)} > 80\text{ mA}$.

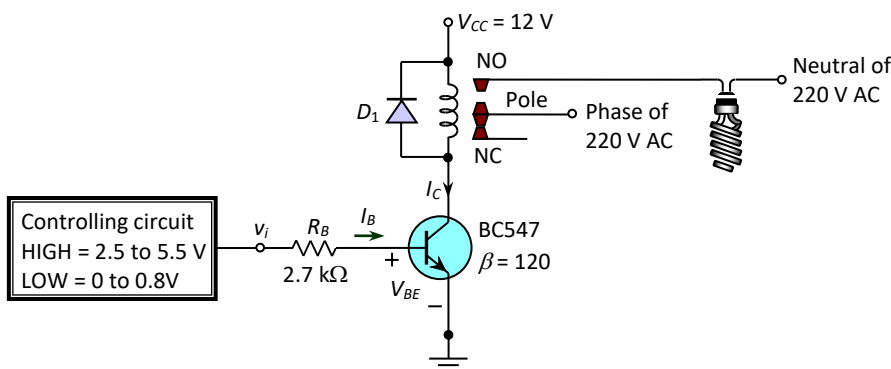


Fig.5-82: Light controlling circuit of Example 5-27

Small signal (low power) transistor like, BC547 has $V_{CE(max)} = 45\text{ V}$, $I_{C(max)} = 100\text{ mA}$ and $h_{FE(min)} = 120$. We can also use 2N2222 in place of BC547.

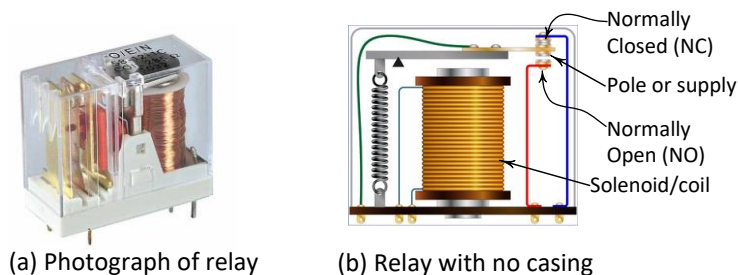


Fig.5-83: Photographs of relay

$$I_B = \frac{I_C}{h_{FE(min)}} = \frac{80\text{ mA}}{120} = 0.6667\text{ mA}$$

$$R_B = \frac{V_{in(HIGH)} - V_{BE}}{I_B} = \frac{2.5 \text{ V} - 0.7 \text{ V}}{0.6667 \text{ mA}} = 2.7 \text{ k}\Omega \text{ (is a standard value)}$$

The relay coil has to be connected to the collector and V_{CC} . One terminal of the CFL light has to be connected to the normally open (NO) terminal of the relay and the other terminal to the neutral line of the supply. The phase of the power line has to be connected to the pole of the relay. To protect the transistor a diode has to be connected across the relay coil as shown in Fig.5-76.

Comments: Relay is an electromechanical switch that has been discussed in Chapter 1. In place of the CFL any kind of load can be controlled using this circuit. Note that in calculating the base resistance the minimum value of HIGH level has to be considered to ensure the saturation of the transistor in ON state.

The inverter is used mainly for signal conditioning, so the required output current is comparatively low. If the load current is known, use $I_C = 5$ to 10 times of I_L . If the load current is not known, chose a collector current of 5 to 10 mA. The value of V_{CC} should be equal to HIGH logic value (e.g., for TTL it should be 5 V). As the values of I_C and V_{CC} are low, any transistor can tolerate these values. But here, the frequency of the transistor has to be large enough to follow the input signal. The value of I_B , and R_B can be calculated using Equ.(5-108) and (5-109). The value of R_C has to be calculated as,

$$\therefore R_C = \frac{V_{CC}}{I_C} \quad (5-111)$$

In selecting the value of R_C , we have to use the higher standard values, not the lower values, no matter which one is closest. If we use the lower standard values the transistors will not be saturated as before.

Example 5-29

Design an inverter circuit to drive a 5 k Ω load that requires TTL logic levels. Assume $h_{FE(min)} = 100$.

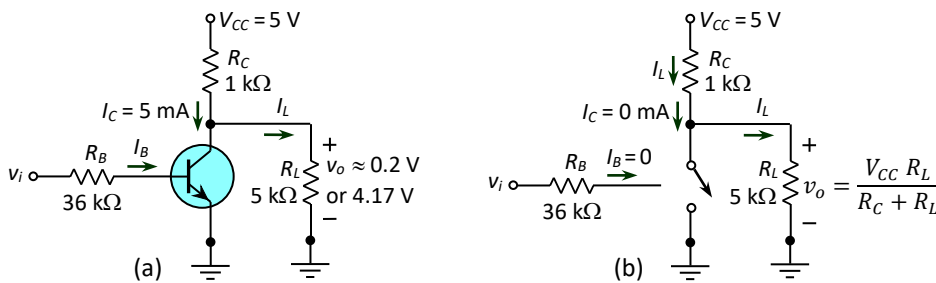


Fig.5-84: (a) Designed inverter of Example 5-28, (b) Output at HIGH state

Solution:

In TTL logic, the value of HIGH logic is 5.5 V to 2.5 V. So let, $V_{CC} = 5 \text{ V}$.

Load current,

$$I_L = \frac{5 \text{ V}}{R_L} = \frac{5 \text{ V}}{5 \text{ k}\Omega} = 1 \text{ mA}$$

Let,

$$I_C = 5 \times I_L = 5 \text{ mA}$$

$\therefore$

$$R_C = \frac{V_{CC}}{I_C} = \frac{5 \text{ V}}{5 \text{ mA}} = 1 \text{ k}\Omega$$

$$I_B = \frac{I_C}{h_{FE(\min)}} = \frac{(5 \text{ mA})}{100} = 50 \mu\text{A}$$

$\therefore$

$$R_B = \frac{V_{in(\text{HIGH})}}{I_B} = \frac{2.5 \text{ V} - 0.7 \text{ V}}{50 \mu\text{A}} = 36 \text{ k}\Omega$$

The designed circuit is shown in Fig.5-84.

Check for the HIGH logic level: When the transistor is OFF, according to the voltage divider rule the load voltage will be

$$V_L = \frac{V_{CC} \times R_L}{R_C + R_L} = \frac{5 \text{ V} \times 5 \text{ k}\Omega}{1 \text{ k}\Omega + 5 \text{ k}\Omega} \approx 4.17 \text{ V} > 3.5 \text{ V}(\text{checked})$$

Comments: Smaller value of I_C results in larger value of R_C and reduces the HIGH state output voltage. Whereas, very larger value of I_C produces larger power loss and reduces the switching speed. On the other hand, larger value of R_B will results in unsaturation of the transistor and produces power loss, while smaller value of R_B will produce hard saturation and lower switching speed.

5.32 Switching Circuit using PNP Transistor

Already we have studied that it is possible to ON/OFF the power of a load using an NPN transistor [Fig.5-79(a)]. Here the load is connected to V_{CC} and collector of the transistor. But, sometimes it is required that one terminal of the load must be connected to ground (grounded load). In that case we have to design the switch using a PNP transistor instead of an NPN transistor. Such a switching

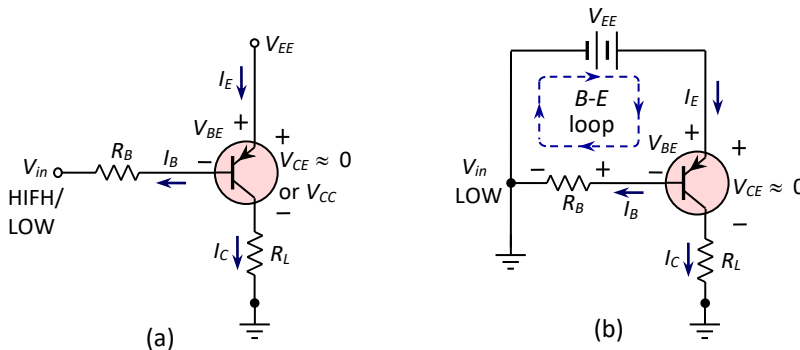


Fig.5-85: PNP transistor as a switch, (a) Switching circuit, and (b) B-E loop of the circuit.

circuit is shown in Fig.5-85. Here, the load is connected to the collector and ground terminals and the emitter is connected to supply voltage. The main difference of this switching circuit is, for the LOW input voltage the load will be ON and for the HIGH input voltage the load will be OFF. Any input voltage less than V_{EE} can turn the switch ON, but to turn the switch OFF the level of HIGH input should be equal to V_{EE} . Here, also $V_{EE} = V_L$ and $I_C = I_L$. The value of I_B will be $I_C/\beta(\min)$. From B-E loop of Fig.5-85(b), the value of the base resistor can easily be calculated as,

$$R_B = \frac{V_{EE} - V_{BE}}{I_B} \quad (5-112)$$

5.33 Limitation of Transistor as High Power Switch

If the current requirement of the load is very high, the $I_{C(max)}$ of the transistor should also be very high as the load current flows through the collector of the transistor. But a transistor with higher (collector) current capacity has lower value of β . For example TIP3055 transistor has $I_{C(max)} = 15$ A, but its current gain (β) is as low as 10. So if we design a 15 A switch using this transistor the value of base current will be $I_B = 1.5$ A. This base current will produce inefficiency of the switching circuit. If the value of supply voltage (V_{CC}) is 12 V, the load will consume $12 \text{ V} \times 15 \text{ A} = 180 \text{ W}$, while the power drawn from the source will be $12 \text{ V} \times (15 + 1.5) \text{ A} = 198 \text{ W}$. That is, 18 W is lost in the switching circuit as shown in Fig.5-86. This loss is due to the larger value of base current. For this reason, for high power switching instead of BJT, power MOSFET is used (will be described later).

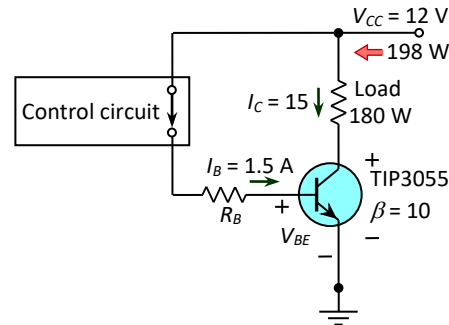


Fig.5-86: High power switch: 198 W is drawn from source, 180 W is used by load, 18 W loss in the circuit

5.34 Transistor as a Current Source

We all use voltage source either constant or variable. A constant voltage source provides a fixed level of voltage irrespective of the output current. That is, if we change the load resistance the value of load current will change but the voltage will remain fixed. A battery, a power supply are the examples of the constant voltage source. On the other hand, in solving problems of electrical currents, we use current source or constant current source. But do we know any source

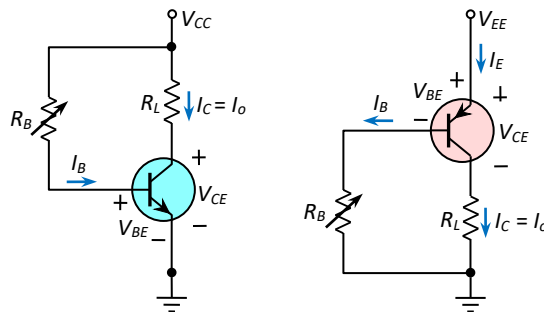


Fig.5-87: Transistor used as a current source

that can be used as a current source? Although, there are some current sources, like solar cell, they are not common. However, a transistor can be used as a constant current source if properly biased. For the common emitter configuration I_C is the output current that flows through the load R_C . The variation of R_C will not change the value of I_C as it depends only on I_B . Thus, we can say that a transistor circuit can behave as a constant current source. Generally, it is necessary to connect one terminal of the load to the current source and the other terminal to the ground. In that case, we have to design a current source using a PNP transistor. The schematic diagram of a current source is shown in Fig.5-87.

Example 5-30

For the constant current source of Fig.5-88, calculate the output current and output voltage. Determine the maximum value of R_L that can be used with this current source.

Solution:

The value of base current will be,

$$I_B = \frac{V_{EE} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{260 \text{ k}\Omega} \approx 43.46154 \text{ }\mu\text{A}$$

Here, the output current is the collector current and its value will be,

$$I_o = I_C = \beta I_B = 120 \times 43.46154 \text{ }\mu\text{A} \approx 15.2154 \text{ mA [Ans.]}$$

The output voltage will be,

$$V_o = I_o R_L = (15.2154 \text{ mA}) \times 100 \text{ }\Omega = 1.52 \text{ V [Ans.]}$$

The circuit will work as constant current source, before being saturated. Considering $V_{CE(sat)} = 0.5 \text{ V}$,

$$R_L = \frac{V_{EE} - V_{CE(sat)}}{I_o} = \frac{12 \text{ V} - 0.5 \text{ V}}{15.2154 \text{ mA}} \approx 756 \text{ }\Omega$$

Comments: For very low value of R_L , the value of V_{CE} will be high. In that case, the transistor will dissipate a very high power. Hence, in designing this circuit the power rating of the transistor has to be considered.

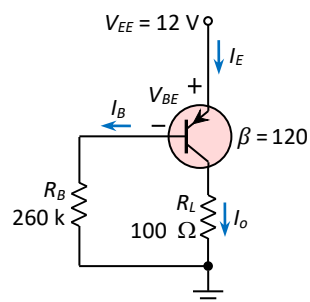


Fig.5-88: Constant current source for Example 5-30

5.35 Trouble-Shooting of Transistor Bias Circuits

It is a very common fact in laboratory that a biasing circuit is constructed using a transistor, but the circuit does not work. Then we have to find out the fault and make the circuit functioning. In servicing an electronic circuit, we may also need to check whether a transistor is ok or not. In a transistor circuit the supply voltage (V_{CC}) and the voltage levels at the transistor terminals (V_C , V_B , and V_E) are measured with respect to the ground terminal or negative terminal of the supply source as shown in Fig.5-89. Generally, the circuits are constructed in bread board using connecting wires (in laboratory). The connecting points of the bread board, the connecting wires, resistors, supply source or even the transistor itself may cause the fault. For a faulty circuit

different situations may be found. We may not find any voltage at any terminal of the transistor. In that case there must have an open-circuit. To find out the open-circuit, first connect a voltage meter to the positive and negative terminals of the supply source. The voltmeter should read the required voltage level for the transistor circuit. If not, fault is in the power supply. Then measure the voltage at the positive point of the bread board, then to resistor-terminals which are connected to the positive point of the bread board. Then check whether the resistors are properly connected to the transistor terminals. In this way, measure voltage and proceed to the terminals of the transistor. If any point does not give voltage reading, that point is not connected to proper position. In this way, we can find out the location of the open circuit. If the open circuit fault is cleared, we may get incorrect voltage levels at transistor terminals. The reasons of these inaccurate voltage levels depend on the types of biasing circuits as described below.

5.35.1 Base Bias

Fig.5-90 Illustrates typical error sources in a base biased circuit. Normal voltage levels in a base bias circuit are $V_{BE} \approx 0.7\text{ V}$, and V_C approximately half of V_{CC} or anywhere in the range of 2 V to $(V_{CC} - 2\text{ V})$. If $V_C = V_{CC}$ as shown in Fig.5-90(a), it means that the transistor is open. Either R_B is not supplying base current or emitter terminal is not connected to the ground, or the transistor itself is faulty (collector and emitter are open). Alternately, R_C may be short. When $V_{CE} \approx 0$, R_C could be of higher value than R_B . Other possibilities are either the collector and emitter terminals are short or R_C is short.

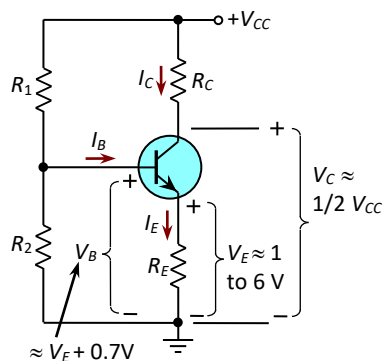


Fig.5-89: For testing a bias circuit, terminal voltages should be measured with respect to ground or negative supply

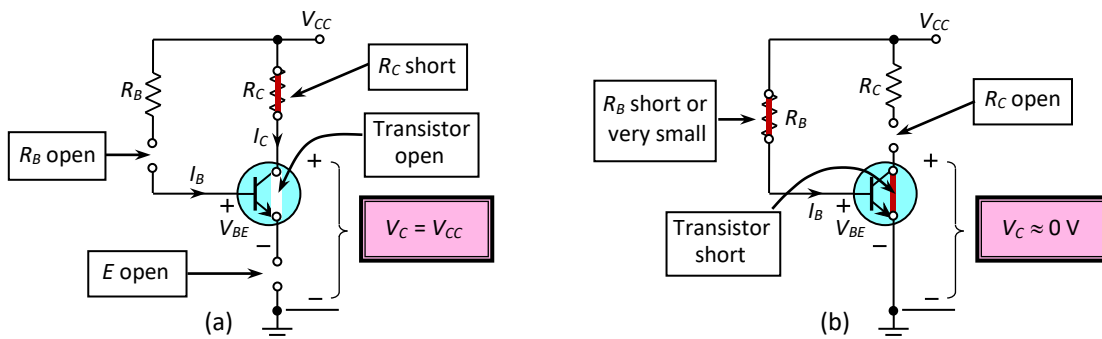


Fig.5-90: Incorrect voltage levels of base bias circuit and their probable reasons, (a) Probable reasons for $V_C = V_{CC}$, and (b) Probable reasons for $V_C = 0$

5.35.2 Emitter Bias

Some reasons for incorrect voltage levels across different terminals of a transistor in emitter bias are shown in Fig.5-91. If $V_C = V_{CC}$, the base resistor R_B and emitter resistor R_E may not be properly connected (open). The device may be open or the collector resistor may be short

circuited. The situation is same as depicted in Fig.5-91(a). If $V_C \approx 0$ or $V_{CE} \approx 0$, R_B may be short or very small. Alternately, R_C may be open or very large or the transistor may be internally short as shown in Fig.5-91(b).

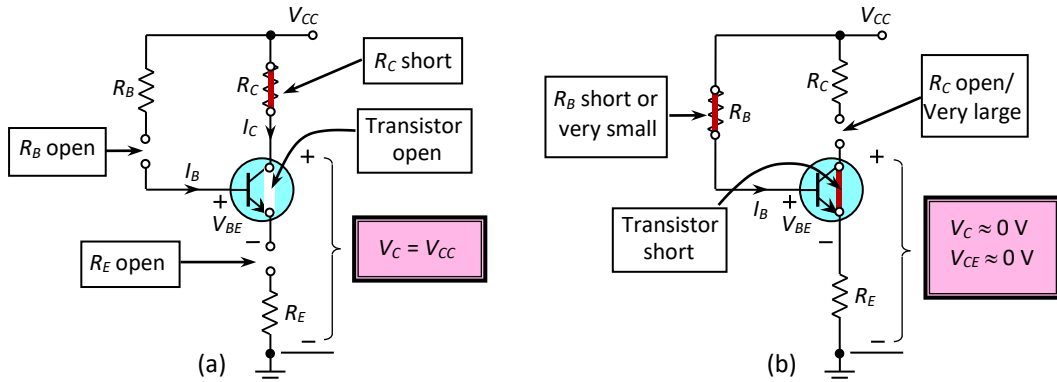


Fig.5-91: Inaccurate voltage levels of emitter bias circuit and their probable causes, (a) Probable causes for $V_C = V_{CC}$, and (b) Probable causes for $V_C = 0$ or $V_{CE} \approx 0$

5.35.3 Collector-base Bias

Some reasons for incorrect voltage levels across different terminals of a transistor in collector-to-bias are shown in Fig.5-92. If R_B is not connected properly, transistor will not conduct, so $V_C = V_{CC}$. Another reasons of $V_C = V_{CC}$ are emitter terminal is open and the transistor itself is open as shown in Fig.5-92(a). If $V_C \approx 0$, the transistor is short or saturated. The reasons behind this are: R_B may be short or very low, or the collector and emitter terminals of the transistor may be short as shown in Fig.5-92(b).

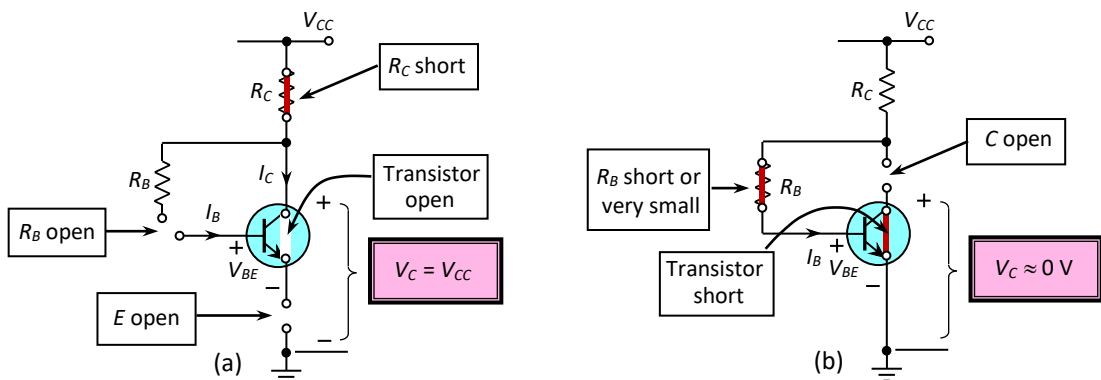


Fig.5-92: Incorrect voltage levels of collector-to-bias circuit and their probable reasons, (a) Probable reasons for $V_C = V_{CC}$, and (b) Probable reasons for $V_C \approx 0$

5.35.4 Voltage Divider Bias

Fig.5-93 shows some probable reasons for malfunctioning of a voltage divider bias circuit. If $V_C = V_{CC}$, either R_1 or R_E might be open circuited. Alternately, R_2 or R_C might be short circuited. Even the collector and emitter terminals of the transistor may be open circuited [Fig.5-93(a)]. If $V_C =$

0 or $V_{CE} \approx 0$, R_C and R_2 might be open or very larger than the required values or collector and emitter terminals of the transistor may be short circuited [Fig.5-93(b)].

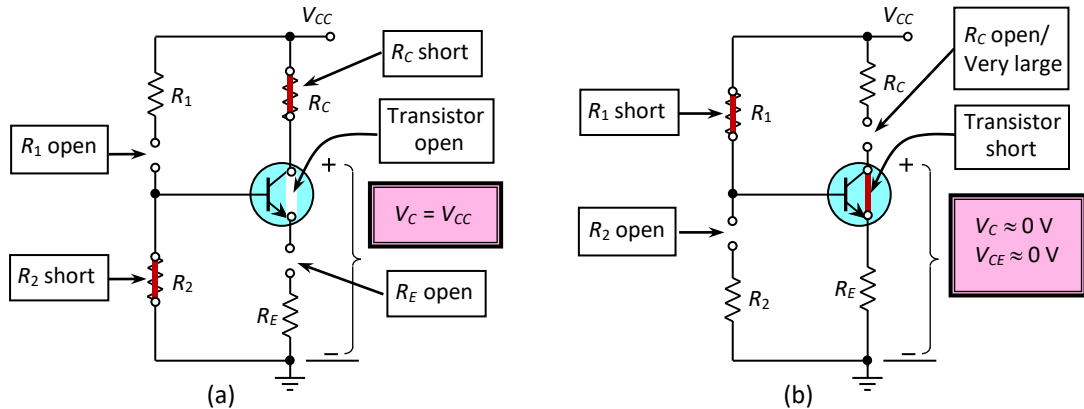


Fig.5-93: Inaccurate voltage levels of voltage divider bias circuit and their probable causes, (a) Probable causes for $V_C = V_{CC}$, and (b) Probable causes for $V_C = 0$ or $V_{CE} \approx 0$

5.36 Miscellaneous Biasing Circuits used with BJT

So far we have discussed four types of biasing circuits for different transistor configurations and different types of transistors (NPN or PNP). Besides these biasing circuits, some other variations can be used to improve some parameters of the circuits. Some of them are discussed here.

5.36.1 Voltage Feedback Biasing Circuit:

As shown in Fig.5-94, this is the combination of collector-to-base bias and emitter bias and called **collector-to-base bias with emitter feedback resistor**. In this circuit, both the collector resistor and the emitter resistor provide feedback for better stabilization. The equations to calculate its operating point will be the combination of those of collector-to-base bias and emitter bias. By applying KVL in the input circuit and output circuit the equations can easily be found. The equations are given below.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)(R_C + R_E)} \quad (5-113)$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (5-114)$$

If the total resistance of output circuit ($R_C + R_E$) of this circuit and the collector resistance (R_C) of collector-to-base bias is same, then the two biasing circuits provide the same stability against the β variation.

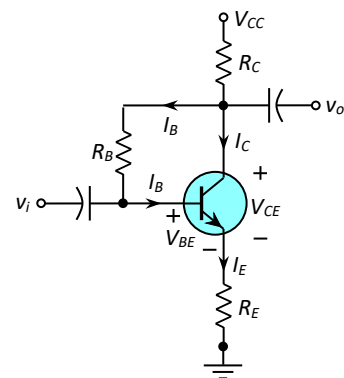


Fig.5-94: Voltage feedback bias

5.36.2 Emitter Bias with I_{CE0} Compensation using Diode

We know that with the increase in temperature, I_{CBO} increases and hence, I_{CEO} increases more rapidly. This problem can be overcome, up to a certain extent using a diode with different biasing circuits. Fig.5-95 shows an emitter biasing circuit with diode compensation. Look, the diode is connected in reverse bias polarity. So, normally it will draw a very small leakage current (or reverse saturation current) I_S . With the increase in temperature, the leakage current of the transistor (I_{CBO}) will increase and so does the reverse saturation current (I_S) of the diode. As I_S increases the voltage drop across R_B will increase and hence I_B decrease which is clear from Equ.5-115. If the temperature decreases, the base current adjustment will take place in the same method.

Using KVL in the base circuit, we can write,

$$V_{CC} - (I_B + I_S)R_B - V_{BE} - I_E R_E = 0$$

$$\therefore \boxed{I_B = \frac{V_{CC} - V_{BE} - I_E R_E - I_S R_B}{R_B}} \quad (5-115)$$

5.36.3 Voltage Divider Bias with V_{BE} Compensation using Diode

As discussed earlier, V_{BE} decreases with the increase in temperature ($2.5 \text{ mV}/^\circ\text{C}$). Thus the base current increases and shifts the operating point. This problem can be minimized using a diode in voltage divider bias circuit as shown in Fig.5-96. Note that the diode is conned as forward biased.

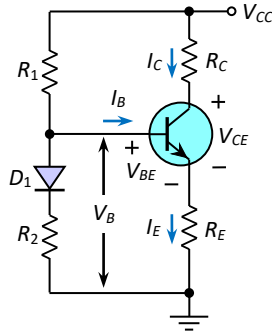


Fig.5-96: V_{BE} compensated voltage divider bias

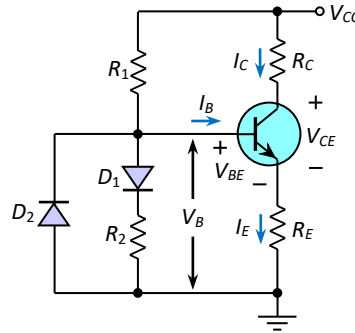


Fig.5-97: V_{BE} and I_{CBO} compensated voltage divider bias

The value of base voltage (V_B) for this circuit will be as given by Equ.5-116. If the base-emitter voltage of the transistor (V_{BE}) changes with temperature, the forward voltage drop of the diode (V_D) will also change at the same rate. Thus, if the required voltage of the base-emitter junction decreases due to temperature increase, according to Equ.5-116, the supply voltage will also decrease and hence, the base current I_B will remain constant.

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} + V_D \quad (5-116)$$

As shown in Fig.5-97, we can adopt both V_{BE} and I_{CBO} compensation using two diodes.

5.36.4 Voltage Divider Bias with Zener Diode:

The emitter voltage ($V_E = I_E R_E$) of a biasing circuit provides negative feedback to compensate any change of the collector current I_C . The compensation will work better if the base voltage V_B remains constant. In normal biasing circuit we try to do this, by flowing a very large current compared to I_{BQ} through the voltage divider network (R_1 and R_2). This can also be done by using a Zener diode. We have already known that a Zener diode can give almost constant voltage. As shown in Fig.5-98, the base voltage of the transistor will be equal to the Zener break down voltage, i.e., $V_B = V_Z$. In a Zener diode, for $V_Z < 5$ V, the breakdown mechanism is Zener breakdown, while for $V_Z > 5$ V, the breakdown process is avalanche breakdown. The Zener breakdown has the negative temperature coefficient, while the avalanche breakdown has positive temperature coefficient. Hence, if $V_B = V_Z \leq 5$ V, V_B will decrease with increase in temperature as does V_{BE} . So, the Zener diode, used in the biasing circuit of Fig.5-98, will provide thermal stabilization.

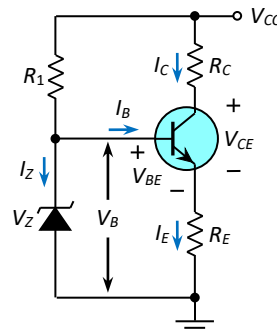


Fig.5-98: Voltage divider bias using Zener diode.

5.37 Multiple BJT Circuits

So far we have discussed different circuits with only a single transistor (BJT). In electronics, there are some circuits with more than one transistor. Some of the common circuits are multistage amplifier, cascode amplifier, Darlington pair, feedback pair, etc. Among these, multistage amplifier will be discussed later in another chapter and other amplifiers are discussed here.

5.37.1 Cascode Amplifier

Cascode amplifier is a two-stage amplifier that consists of a common-emitter amplifier and a common-base amplifier. As shown in Fig.5-99, the input signal is applied to the common-emitter amplifier that drives the common-base amplifier. The output is taken from the common-base amplifier. The main advantages of this amplifier are higher input impedance and **higher** output impedance. It also provides better isolation between the input and output. As there is no coupling between the input and output (by inter-electrode capacitance and resistance) it eliminates Miller effect (will be discussed later). Thus, the cascode amplifier provides higher bandwidth.

Operation

The input signal is applied to the base of CE amplifier through coupling capacitor C_1 and the output is taken from the collector of CB amplifier through another capacitor C_2 . The base of the CB amplifier is connected to ground by C_3 . As for the AC signal all the capacitors behave like a short circuit, the base of CB amplifier can be considered connected to ground. Now due to the input AC signal, the collector voltage of the CE amplifier will change which will also be applied to the emitter

of the CB amplifier. The common-base amplifier will again amplifies this signal to the collector and the load will get this amplified signal.

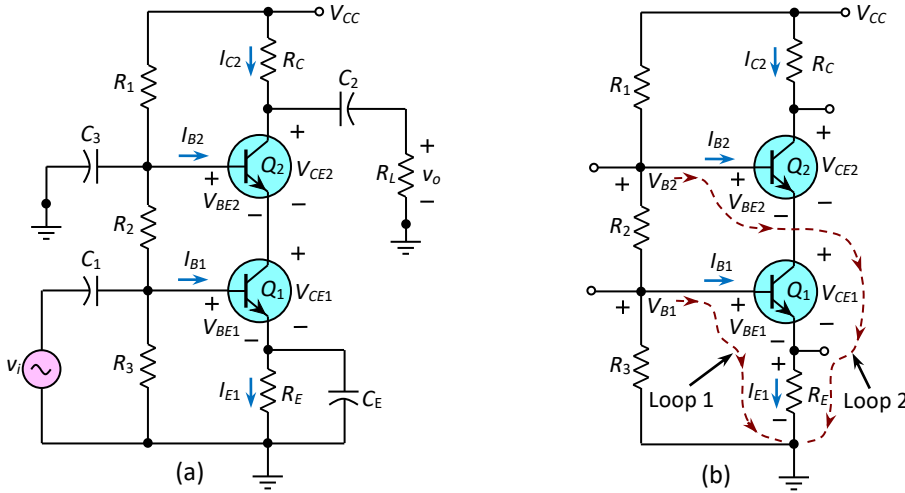


Fig.5-99 (a): Cascode amplifier, (b) DC equivalent circuit (with loops for KVL)

DC Analysis

The DC equivalent circuit of this amplifier can be drawn considering all the capacitors open which is shown in Fig.5-99(b). To simplify the analysis we will assume that the values of β of the transistors are very high. So I_{B1} and I_{B2} will be very small and will be neglected. Alternately, if the current flowing through the voltage divider network is quite larger than the transistors base currents, then also I_{B1} and I_{B2} can be neglected. Using voltage divider rule, the values of base voltages will be,

$$V_{B1} = \frac{V_{CC} R_3}{R_1 + R_2 + R_3} \quad (5-117)$$

and,

$$V_{B2} = \frac{V_{CC} (R_3 + R_2)}{R_1 + R_2 + R_3} \quad (5-118)$$

As I_{B1} and I_{B2} are neglected, $I_{C2} \approx I_{E2} \approx I_{C1} \approx I_{E1}$.

Now applying KVL to $B-E$ loop [loop 1 in Fig.5-99(b)] of transistor Q_1 ,

$$\begin{aligned} V_{B1} - V_{BE1} - I_{E1} R_E &= 0 \\ \therefore I_{E1} &= \frac{V_{B1} - V_{BE1}}{R_E} \end{aligned} \quad (5-119)$$

From the previous assumption, we can write

$$\therefore I_{C2} = I_{E2} = I_{C1} = \frac{V_{B1} - V_{BE1}}{R_E} \quad (5-120)$$

Again we can apply KVL starting from V_{B2} up to ground [loop 2 in Fig.5-99(b)] as,

$$V_{B2} - V_{BE2} - V_{CE1} - I_{E1} R_E = 0$$

$$\therefore V_{CE1} = V_{B2} - V_{BE2} - I_{E1}R_E \quad (5-121)$$

Once V_{CE1} is known, V_{CE2} can be calculated as,

$$\therefore V_{CE2} = V_{CC} - V_{CE1} - I_C(R_E + R_C) \quad (5-122)$$

If a split power supply is used, i.e., instead of ground if $-V_{EE}$ is connected to the emitter of Q_1 , we have to add V_{EE} with V_{CC} as,

$$V_{B1} = \frac{(V_{CC} + V_{EE})R_3}{R_1 + R_2 + R_3} \quad (5-123)$$

$$V_{B2} = \frac{(V_{CC} + V_{EE})(R_3 + R_2)}{R_1 + R_2 + R_3} \quad (5-124)$$

$$I_{E1} = \frac{V_{B1} + V_{EE} - V_{BE1}}{R_E} \quad (5-125)$$

$$\therefore V_{CE1} = V_{B2} - V_{BE2} - I_{E1}R_E \quad (5-126)$$

$$\therefore V_{CE2} = V_{CC} + V_{CE} - V_{CE1} - I_C(R_E + R_C) \quad (5-127)$$

Example 5-31

For the cascode amplifier of Fig.5-100, calculate the operating point of transistors. Assume $I_{B1} = I_{B2} \approx 0$.

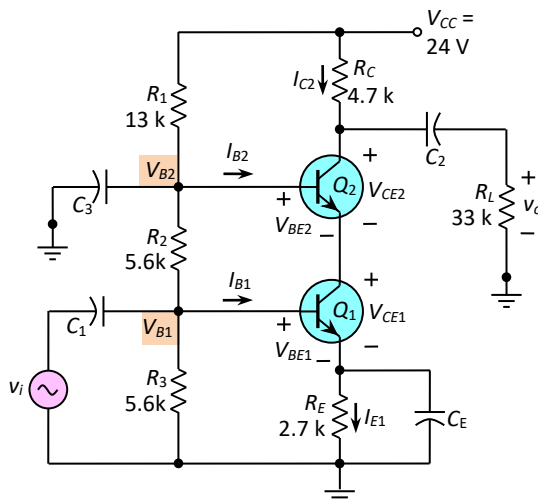


Fig.5-100: Cascode amplifier for Example 5-31

Solution:

Using Equ.(5-117),
$$V_{B1} = \frac{V_{CC}R_3}{R_1 + R_2 + R_3} = \frac{24 \text{ V} \times 5.6 \text{ k}}{13 \text{ k} + 5.6 \text{ k} + 5.6 \text{ k}} \approx 5.55 \text{ V}$$

Using Equ.(5-118),
$$V_{B2} = \frac{V_{CC}(R_3 + R_2)}{R_1 + R_2 + R_3} = \frac{24 \text{ V} \times (5.6 \text{ k} + 5.6 \text{ k})}{13 \text{ k} + 5.6 \text{ k} + 5.6 \text{ k}} \approx 11.1 \text{ V}$$

Using Equ.(5-119),
$$I_{E1} = \frac{V_{B1} - V_{BE1}}{R_E} = \frac{5.55 \text{ V} - 0.7 \text{ V}}{2.7 \text{ k}} \approx 1.796 \text{ mA} \approx I_{C1} \approx I_{C2}$$

Using Equ.(5-121),
$$V_{CE1} = 11.1 \text{ V} - 0.7 \text{ V} - 1.796 \text{ mA} \times 2.7 \text{ k}$$

or,
$$V_{CE1} \approx 5.55 \text{ V [Ans.]}$$

Using Equ.(5-122),
$$V_{CE2} = 24 \text{ V} - 5.55 \text{ V} - 1.796 \text{ mA} \times (4.7 \text{ k} + 2.7 \text{ k})$$

or,
$$V_{CE1} \approx 5.16 \text{ V [Ans.]}$$

Comments: As the supply voltage (V_{CC}) has to bias two amplifiers connected in series, so its value has to be higher compared to the supply voltage for a single stage amplifier.

5.37.2 Darlington Pair

To drive an output load, sometimes it is required to amplify an input current to a high output current. In that case, CC amplifiers are used with a transistor with higher current capacity, i.e., very high value of $I_{C(max)}$. But one problem is, a transistor with high current capacity has very low current gain (as described in section 5-22). This problem can be overcome by cascading a low power transistor (Q_1) with a high power transistor (Q_2) as shown in Fig.5-101 (sometimes identical transistors are also used). The arrangement is called **Darlington pair**. As shown in the figure, the collectors of two transistors are connected together, that work as the collector of the combination (C). The emitter of Q_1 is connected to the base of Q_2 . The base of the first transistor and the emitter of the second transistor are the base (B) and the emitter (E) of the Darlington pair, respectively. Darlington pair can be constructed by connecting two transistors or it is available as a single package. For example, TIP120 is a single package that contains a Darlington pair.

Now assuming proper biasing of the configuration, we will determine the overall current gain of the combination. Let us consider that a certain amount of current (I_B) is flowing through the base of Q_1 , i.e., through B . If the current gain of the first transistor is β_1 and that of the second transistor is β_2 , the value of I_{E1} and I_{C1} will be,

$$I_{E1} = (\beta_1 + 1)I_B \quad \text{and} \quad I_{C1} = \beta_1 I_B$$

From the circuit it is clear that $I_{E1} = I_{B2}$. The collector current of Q_2 will be

$$I_{C2} = \beta_2 I_{B2} = \beta_2 I_{E1} = \beta_2 (\beta_1 + 1) I_B$$

Now the total current flowing through C terminal is $I_{C1} + I_{C2}$. That is,

$$I_C = I_{C1} + I_{C2} = \beta_1 I_B + \beta_2 (\beta_1 + 1) I_B$$

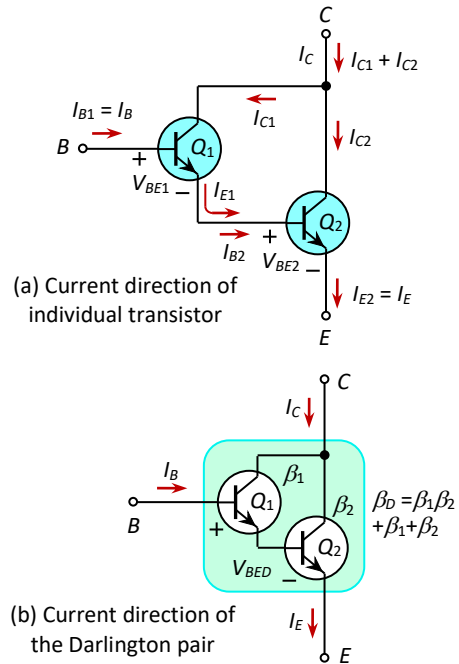


Fig.5-101: Darlington pair

$$\text{or,} \quad I_C = \beta_1 I_B + \beta_2 \beta_1 I_B + \beta_2 I_B$$

$$\text{or,} \quad I_C = (\beta_1 \beta_2 + \beta_1 + \beta_2) I_B$$

$$\therefore \quad \frac{I_C}{I_B} = (\beta_1 \beta_2 + \beta_1 + \beta_2)$$

Now if we consider the whole combination as a single transistor, its three terminal currents will be I_C , I_B and I_E as shown in Fig.5-101(b). Thus, the current gain of the combination (β_D) will be,

$$\boxed{\beta_D = \frac{I_C}{I_B} = (\beta_1 \beta_2 + \beta_1 + \beta_2)} \quad (5-128)$$

As $\beta_1 \beta_2 \gg (\beta_1 + \beta_2)$, generally we write $\beta_D \approx \beta_1 \beta_2$.

To calculate the emitter current of the combination (I_E), we can write

$$I_E = I_{E2} = I_{B2}(\beta_2 + 1) = I_{E1}(\beta_2 + 1) [\because I_{E1} = I_{B2}]$$

Putting the value of I_{E1} , we get

$$I_E = (\beta_1 + 1) I_B (\beta_2 + 1) = (\beta_1 \beta_2 + \beta_1 + \beta_2 + 1) I_B$$

$$I_E = (\beta_D + 1) I_B \text{ [same as a single transistor]}$$

Generally, Q_1 will be a low power transistor, but its maximum collector current capacity should be sufficient to drive Q_2 , i.e., $I_{C1(max)} \geq I_{C2(max)}/\beta_2$. The drawbacks of this configuration is the higher base-emitter voltage, that is now, $V_{BED} = V_{BE1} + V_{BE2}$. Another disadvantage is that the saturation voltage. If a single transistor has a saturation voltage of 0.2 V, the Darlington pair has a saturation voltage of 0.7V.

The AC resistance seen at the emitter terminal of this Darlington pair will,

$$r_{e(D)} = r_{e1} + r_{e2}$$

where, $r_{e1} \approx 26 \text{ mV}/I_{C1}$, and $r_{e2} \approx 26 \text{ mV}/I_{C2}$ (at room temperature). If this resistance is seen from base of the Darlington pair, its value will be $\beta_1(r_{e1} + \beta_2 r_{e2})$. As I_{C2} is very small, we can write $I_C \approx I_{C1}$. Then, $r_{e2} \approx 26 \text{ mV}/I_C$. It can be prove that the total emitter resistance of the Darlington pair will be, $r_{eD} = 2r_e = 2(26 \text{ mV}/I_C)$.

Although, Darlington pair can be used for any configuration, but it is most commonly used as a common collector amplifier as shown in Fig.5-102 for power amplification or impedance matching. This circuit is exactly same as that of Fig.4-67(d), except the Darlington pair. So the equations derived for that circuit can be used here by replacing β and V_{BE} by β_D and V_{BED} . The required expressions for DC analysis of this circuit will be as follows:

$$I_B = \frac{V_{CC} - V_{BED}}{R_B + (\beta_D + 1)R_E}$$

or,

$$\boxed{I_B = \frac{V_{CC} - V_{BE1} - V_{BE2}}{R_B + (\beta_D + 1)R_E}}$$

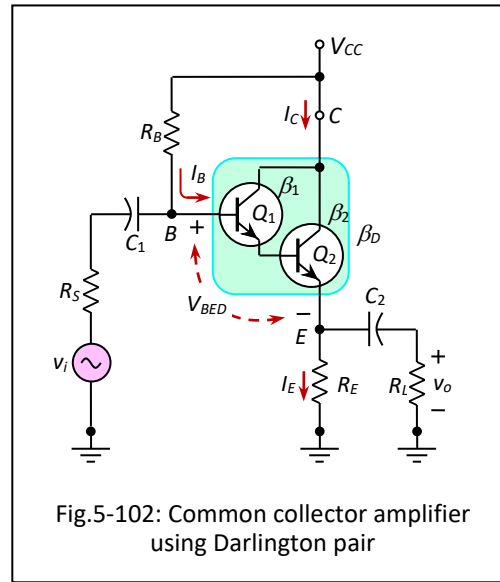


Fig.5-102: Common collector amplifier using Darlington pair

$$I_C = I_B \beta_D$$

$$V_{CE} = V_{CC} - I_E R_E = V_{CC} - (\beta_D + 1) I_B \quad (5-129)$$

Example 5-32

Design a common-collector amplifier using a Darlington pair for $I_{CQ} = 1.5 \text{ A}$ and $V_{CEQ} = 12 \text{ V}$. Assume the supply voltage is 24 V.

Solution:

As the quiescent current $I_{CQ} = 1.5 \text{ A}$, maximum current may be as high as 3 A. So Q_2 must be a power transistor with $I_{C2(max)} \geq 3 \text{ A}$. Let us use a 5 A transistor MJE105 with $\beta_{min} = 25$. The maximum base current of this transistor will be $I_{B2(max)} = 3 \text{ A} / 25 = 120 \text{ mA}$. Thus the maximum collector current of Q_1 should be $I_{C1(max)} \geq 120 \text{ mA}$. KSC1815YTA is 150 mA transistor with $\beta_{(min)} = 25$. Using these two transistors the current gain of the Darlington pair will be

$$\beta_D = \beta_1 \beta_2 + \beta_1 + \beta_2 = 25 \times 25 + 25 + 25 = 675 \text{ (minimum)}$$

$$V_{EQ} = V_{CC} - V_{CE} = 24 \text{ V} - 12 \text{ V} = 12 \text{ V}$$

$$\therefore R_E = \frac{V_{EQ}}{I_{EQ}} \approx \frac{V_{EQ}}{I_{CQ}} = \frac{12 \text{ V}}{1.5 \text{ A}} = 8 \Omega$$

$$I_{BQ} = \frac{I_{CQ}}{\beta_D} = \frac{1.5 \text{ A}}{675} \approx 2.2222 \text{ mA}$$

Using Equ.(5-105),

$$I_B = \frac{V_{CC} - V_{BE1} - V_{BE2}}{R_B + (\beta_D + 1) R_E}$$

or,

$$R_B = \frac{V_{CC} - V_{BE1} - V_{BE2}}{I_B} - (\beta_D + 1) R_E$$

or,

$$R_B = \frac{24 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V}}{2.2222 \text{ mA}} - (675 + 1) \times 8 \Omega \approx 4.76 \text{ k}\Omega$$

Results: $Q_1 = \text{KSC1815YTA}$, $Q_2 = \text{MJE105}$, $R_E = 8 \Omega$ and $R_B = 4.7 \text{ k}\Omega$ (standard)

Comments: This amplifier can be used as a power amplifier, e.g., to drive an 8Ω loud speaker.

5.37.3 Feedback Pair

Feedback pair is also known as a **Sziklai pair** according to George C. Sziklai who popularized the configuration. It is also called the **complementary feedback pair** or **complementary Darlington pair**. The arrangement is same as the Darlington pair, but here an NPN transistor and a PNP transistor are used as shown in Fig.5-103. Interchanging the position of the PNP and the NPN transistor, feedback pair can be constructed in two configurations as shown in Fig.5-103(a) and Fig.5-103(b). Here, the collector of the first transistor drives the base of the second transistor. Though Darlington pair is available in a single package, feedback pair is not. This arrangement has the same advantage of high current gain. For the circuit of Fig.5-103(a), $\beta_S = \beta_1 \beta_2 + \beta_1$ and for that of Fig.5-103(b) $\beta_S = \beta_1 \beta_2 + \beta_1 + 1$. It has one important difference from the Darlington pair.

The value of V_{BE} of a feedback pair is just half of that Darlington pair. For Si transistor Darlington pair $V_{BED} \approx 1.4$ V, but for a feedback pair its value (V_{BES}) is 0.7 V, same as a single transistor. The saturation voltage is also higher than a single transistor (≈ 0.7 V). Biasing of the circuit of Fig.5-103(a) will be same as the Darlington pair. But the input side of the feedback pair

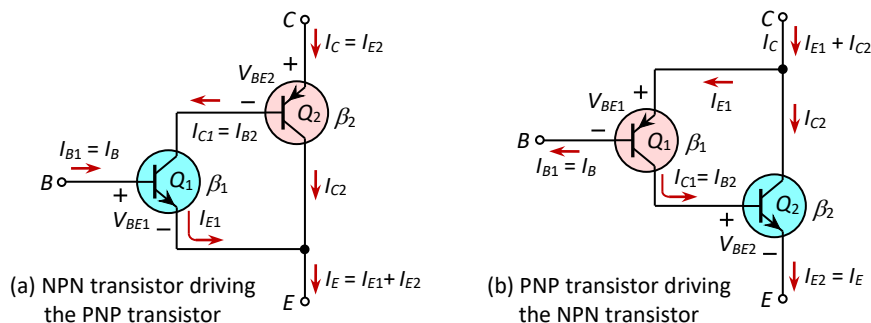


Fig.5-103: Feedback pair or Sziklai pair

of Fig.5-103(b) has to be biased as a PNP transistor, i.e., low input voltage to turn the pair ON. The biasing circuit of feedback pair of Fig.5-103(b) is shown in Fig.5-104. The base resistor R_B is connected to ground instead of V_{CC} . The analysis procedure is same as Darlington pair, except $V_{BES} = 0.7$ V.

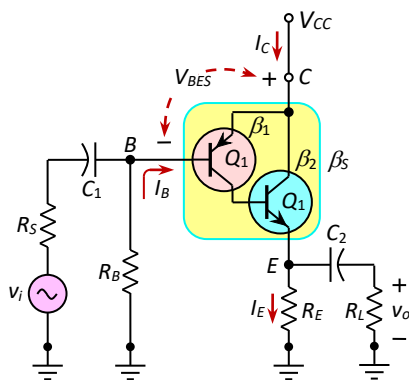


Fig.5-104: Common collector amplifier using Sziklai pair

5.38 Sample Questions

1. What is load line and operating point? Explain how the load line and operating point of a transistor circuit can be determined.
2. Explain how the position of load line changes with the load resistance.
3. Explain how the position of load line changes with the change in supply voltage, V_{CC} .
4. Explain how the position of load line changes with the change in base circuit, I_{BQ} .

5. Explain how the operating point swings when an AC signal is applied to the input of a properly biased transistor circuit.
6. Discuss how the cutoff and saturation region limit the swing of the operating point.
7. Why we try to set the DC operating point at the middle of the load line?
8. Explain the parameters of a transistor that are sensitive to temperature.
9. What is thermal runaway? Explain how thermal runaway may occur in transistor circuits.
10. What is stability factor? Define various types of stability factors.
11. Why biasing is necessary for transistors? Write the names of different types of transistor biasing circuits.
12. Draw the schematic diagram of a fixed bias circuit and explain how we can determine its operating point.
13. Derive the expression for the stability factor, $S(I_{CBO})$, for the fixed bias circuit.
14. Draw the schematic diagram of an emitter bias circuit and explain how we can determine its operating point.
15. Derive the expression for the stability factor, $S(I_{CBO})$, for the emitter bias circuit.
16. Draw the schematic diagram of a collector-to-base bias circuit and explain how we can determine its operating point.
17. Derive the expression for the stability factor, $S(I_{CBO})$, for the collector-to-base bias circuit.
18. Draw the schematic diagram of a voltage divider bias circuit and explain how we can determine its operating point.
19. Derive the expression for the stability factor, $S(I_{CBO})$, for the voltage divider bias circuit.
20. Why there has less possibility for a transistor to operate in cutoff region in a biasing circuit, due to wrong parameters and components selection.
21. Why there has more possibility for a transistor to operate in saturation region than cutoff region in a biasing circuit, due to wrong parameters and components selection.
22. Explain how a resistance is feedback from emitter to base in a transistor circuit.
23. Explain how a resistance is feedback from base to emitter in a transistor circuit.
24. Draw the base bias circuit with a PNP transistor and explain how its operating point can be determined.
25. Draw the collector feedback bias circuit with a PNP transistor and explain how its operating point can be determined.
26. Draw the collector feedback bias circuit with a PNP transistor and explain how its operating point can be determined using exact analysis and approximate analysis.
27. What is single polarity power supply and dual polarity power supply?
28. Draw the base bias circuit with split power supply and explain.
29. Draw the schematic diagram of emitter to base bias with split power supply and describe.
30. Draw the schematic diagram of collector to base bias with split power supply and describe.
31. Draw the schematic diagram of voltage divider bias circuit with split power supply and determine its operating point using both exact method and approximate method.
32. Draw the schematic diagram of emitter current bias circuit and explain.
33. Describe the design procedure of i) base bias, ii) emitter bias, and iii) voltage divider bias circuits.
34. Draw the biasing circuit of an NPN transistor in Common Base Configuration (CBC) and describe.

35. Draw the biasing circuit of a PNP transistor in Common Base Configuration (CBC) and describe.
36. Draw the biasing circuit of an NPN transistor in Common Collector Configuration (CCC) and describe.
37. Draw the biasing circuit of a PNP transistor in Common Collector Configuration (CCC) and describe.
38. Explain how a transistor can be used as a power switch.
39. Explain how a transistor can be used as an inverter (A low power switch).
40. Explain the design procedure of a transistor switch.
41. When should we use a PNP transistor in a switching circuit? – Explain.
42. Explain the limitations of transistor switch for high power controlling.
43. Draw the circuit diagram of a constant current source using BJT and explain.
44. In trouble shooting a fixed biased BJT circuit, the transistor is found open ($V_{CE} = V_{CC}$). What are the probable regions – explain.
45. In trouble shooting a fixed biased BJT circuit, the transistor is found short ($V_{CE} = 0$). What are the probable regions – explain.
46. In trouble shooting a self biased BJT circuit, the transistor is found open ($V_{CE} = V_{CC}$). What are the probable regions – explain.
47. In trouble shooting a self biased BJT circuit, the transistor is found short ($V_{CE} = 0$). What are the probable regions – explain.
48. In trouble shooting a self biased BJT circuit, the transistor is found open ($V_{CE} = V_{CC}$). What are the probable regions – explain.
49. In trouble shooting a voltage divider biased circuit for BJT, the transistor is found short ($V_{CE} = 0$). What are the probable regions – explain.
50. In trouble shooting a voltage divider biased circuit for BJT, the transistor is found open. What are the probable regions – explain.
51. Draw the voltage feedback biasing circuit and explain.
52. How I_{CEO} variation in a BJT circuit can be compensated using diodes – explain.
53. Draw the voltage divider bias circuit with V_{BE} compensation using diode and explain this compensation is achieved.
54. Explain how Zener diode can be used to design transistor biasing circuit.
55. What is cascade amplifier and what are the advantages of cascade amplifiers?
56. Draw the biasing circuit diagram of a cascade amplifier and describe how to determine its operating point.
57. What is Darlington pair? Why we need a Darlington pair?
58. Draw the schematic diagram of Darlington pair and determine its current gain (β_D) and base emitter voltage (V_{BED}).
59. Draw a CCC circuit using Darlington pair and explain how to determine its operating point.
60. Draw the schematic diagram of a feedback pair and determine its current gain (β_S) and base emitter voltage (V_{BES}).
61. Draw a CCC circuit using feedback pair and explain how to determine its operating point.

5.39 Exercise:

1. Calculate the terminal currents (I_B , I_C , and I_E) and the terminal voltages (V_{BE} , V_{CE} , and V_{CB}) for the circuits given in Fig.5-1. Assume Si transistor and $\beta = 120$.

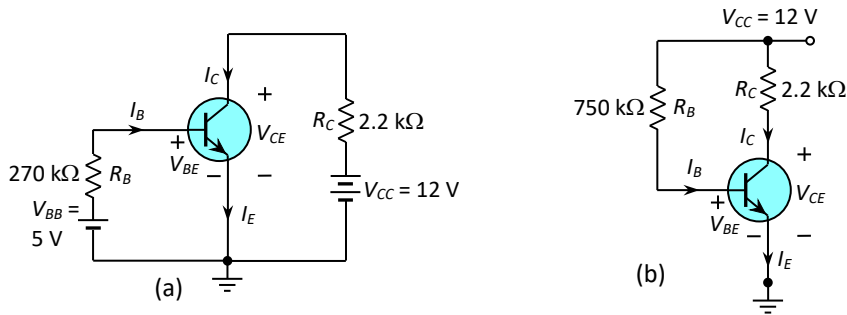


Fig.5-1: Biasing circuit for Exercise no. 1

2. Draw the DC load line and operating point for the BJT circuits of Fig.5-1.
3. Draw the DC load line and operating point for the BJT circuits of Fig.5-1 considering $V_{CC} = 6\text{ V}$, 12 V , 18 V , and 24 V on the same coordinate system.
4. Draw the DC load line and operating point for the BJT circuits of Fig.5-1 considering $R_C = 1\text{ k}$, 2 k , and 3 k on the same coordinate system.
5. The values of I_{CEO} and $V_{CE(sat)}$ of a power transistor are 0.5 mA and 1.2 V , respectively. Considering $V_{CC} = 20\text{ V}$ and $R_L = 10\text{ }\Omega$, (i) Draw the load line, (ii) Maximum swing span of the operating point and (iii) Position of the operating point. Show them in a graph.
6. For the biasing circuit of Fig.5-2, determine the operating point and load line. Also calculate the value of I_C and I_B for which the transistor saturates that is $V_{CE} = V_{CE(sat)} = 0.5\text{ V}$.

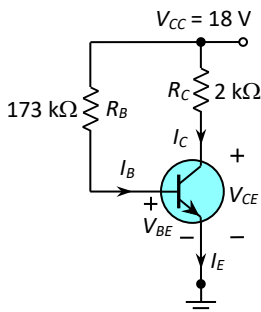


Fig.5-2: Biasing circuit of a transistor for Exercise 6

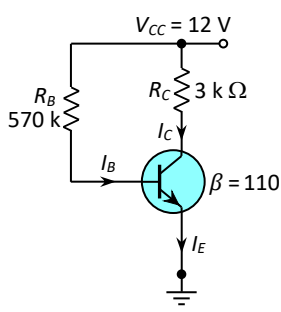


Fig.5-3: Biasing circuit of a transistor for Exercise 6

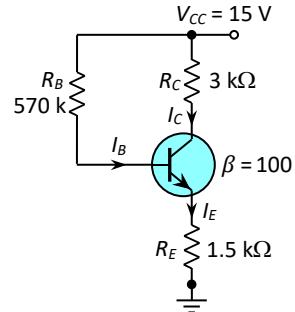


Fig.5-4: Emitter bias for Ex 10

7. Assuming $V_{CC} = 15\text{ V}$, $R_B = 160\text{ k}\Omega$, $R_C = 1.2\text{ k}\Omega$ and a silicon transistor with $\beta = 120$ for the circuit of Fig.5-2, draw the load line and show the position of the operating point.
8. A fixed bias circuit is designed with $V_{CC} = 12\text{ V}$ and a transistor with $\beta = 140$ which is shown in Fig.5-3. Calculate the operating point. To check how higher values of β and R_C and lower value of R_B saturate the transistor, calculate the operating points considering $\beta = 260$, $R_B = 250\text{ k}\Omega$ and $R_C = 8.3\text{ k}\Omega$, one by one.

9. The following table shows the parameter values for a typical transistor at temperature 10°C and 50°C. Find the total change in DC collector current over the temperature range for a base bias circuit. Assume the values of $V_{CC} = 12\text{ V}$ and $R_B = 220\text{ k}\Omega$.

	10°C	50°C	Change
I_{CBO}	0.02 μA	0.17 μA	$\Delta I_{CBO} = 0.15\text{ } \mu\text{A}$
V_{BE}	0.74 V	0.59 V	$\Delta V_{BE} = -0.15\text{ V}$
β	80	100	$\Delta\beta = 20$

10. Fig.5-4 shows a fixed bias circuit. Calculate the operating point for the given values of components in the figure. To get idea of saturation, calculate the operating points considering $\beta = 280$, $R_B = 120\text{ k}\Omega$ and $R_C = 10\text{ k}\Omega$, one by one. Also to prove that R_E cannot saturate the transistor repeat the problem for $R_E = 100\text{ }\Omega$ and $R_E = 520\text{ k}\Omega$.
11. Fig.5-23(a) shows a fixed bias circuit with emitter resistor. The circuit has been designed to set the operating point at the middle of load line considering a typical β value of 150. Calculate the load line and operating point of this circuit. If the transistor has $\beta_{min} = 50$ and $\beta_{max} = 250$, calculate the lowest and the highest position of operating point.

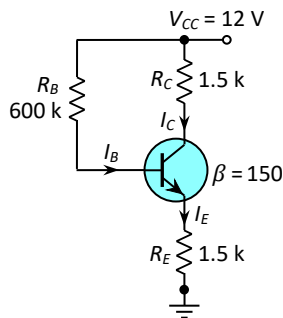


Fig.5-5: Circuit diagram and graph for Example 11

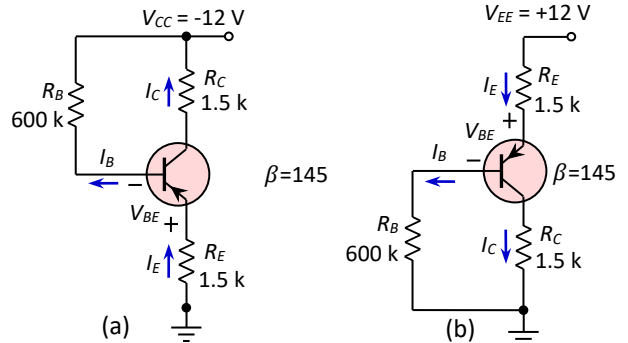


Fig.5-6: Emitter feedback bias with PNP for Ex. 12

12. Calculate the values of I_{BQ} , I_{CQ} , and V_{CEQ} for the emitter bias circuits of PNP transistor given in Fig.5-26.
13. Calculate the operating point of the circuit of Fig.5-7. Verify the collector current using Equ.(5-37).

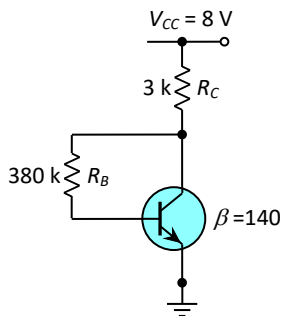


Fig.5-7: Circuit for Example 5-10

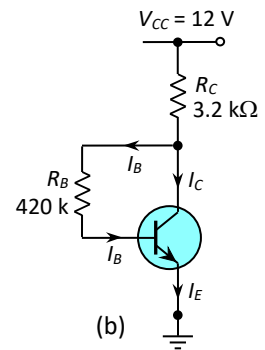
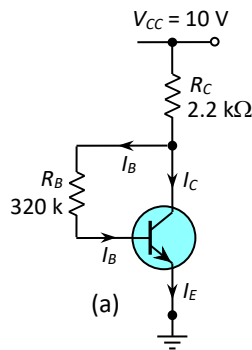


Fig.5-8: Circuit for Example 15-

14. If the value of β of the transistor of Fig.5-7 increases from 140 to 200, calculate the change in collector current using normal procedure and using Equ.(5-53).
15. Fig.5.8 shows a collector-to-base bias circuit, designed to set the operating point at the middle of load line considering a typical β value of 150. Calculate the load line and operating point of this circuit. If the transistor has $\beta_{min} = 50$ and $\beta_{max} = 240$, calculate also the lowest and the highest operating point.

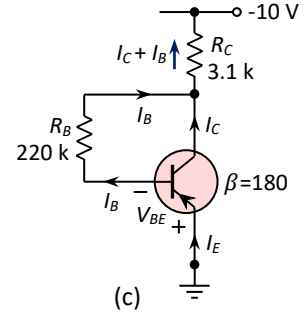
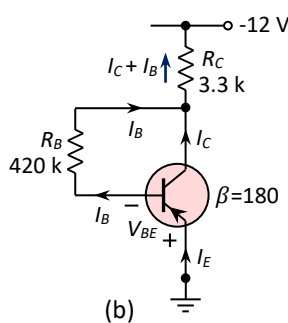
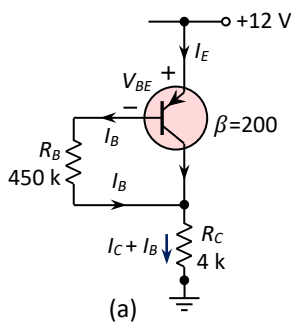


Fig.5-9: Collector feedback bias for PNP transistor

16. For the voltage divider biasing circuit of Fig.5-9, calculate (i) I_{BQ} , (ii) I_{CQ} , (iii) V_{CEQ} , (iv) V_E , and (v) V_{RC} using exact analysis and approximate analysis and compare the results.

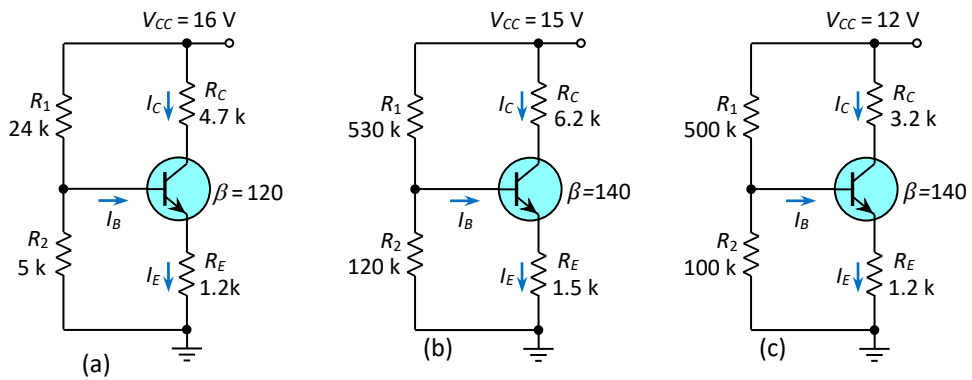


Fig.5-10: Voltage divider bias circuit for Ex.17

17. Even the condition for approximate analysis is not true for the voltage divider bias circuit of Fig.5-10, determine the operating point using exact and approximate method and show the differences in the results.
18. Fig.5.39(a) shows a voltage divider bias circuit considering a typical β value of 145. Calculate the load line and operating point of this circuit. If we consider BD135 transistor, that has $\beta_{min} = 50$ and $\beta_{max} = 250$, calculate the lowest and the highest operating point corresponding to these β values. **Compare the results with the results of Fig.5-19, and Fig.5-23.**

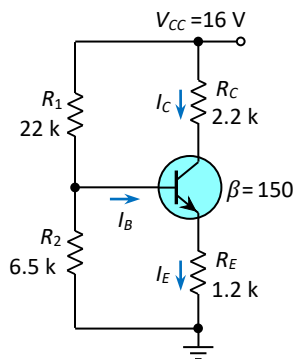


Fig.5-11: Voltage divider bias circuit for Example 4.14

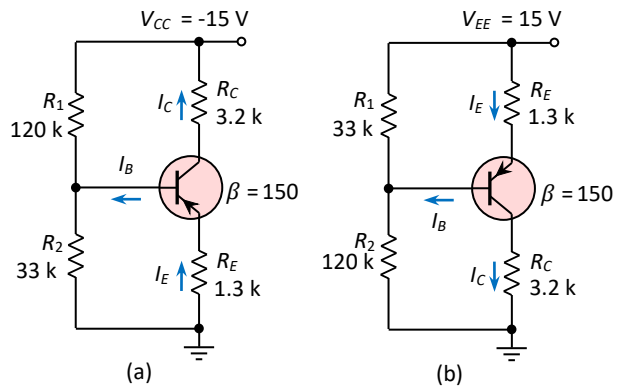


Fig.5-12: PNP transistor biasing circuits for Ex. 19

19. For the voltage divider circuits of Fig.5-12, calculate the operating points.
20. Calculate the operating point of the base bias circuit of Fig.5-13.

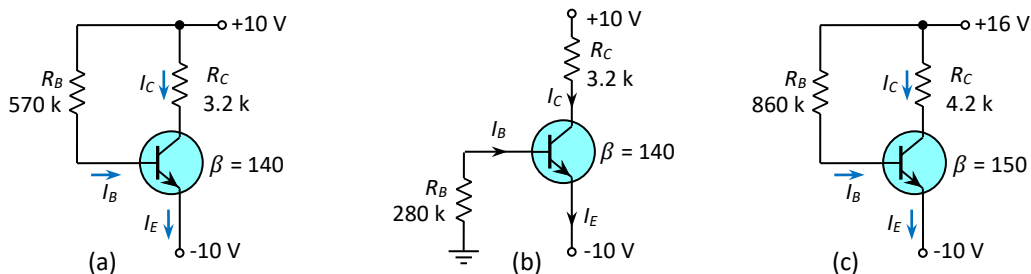


Fig.5-13: Circuit for Example 5-20

21. For the voltage divider biasing circuit of Fig.5-14, determine the operating point, draw the load line and place the operating point on the load line.

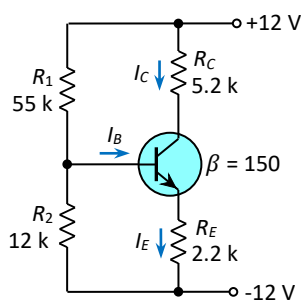


Fig.5-14: Voltage divider bias circuit for Example 5-21

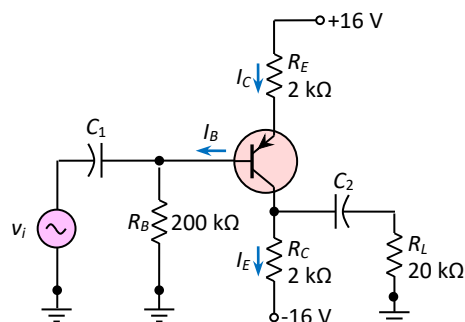


Fig.5-15: PNP transistor biasing circuits with

22. For the voltage divider bias circuit of Fig.5-15, draw the load-line and calculate the operating point.
23. Design a bias circuit that will be operated by two batteries - $V_{CC} = 14\text{ V}$ and $V_{BB} = 5\text{ V}$. Assume the current gain of the transistor is $h_{FE} = 150$.
24. Design a base bias circuit that will be operated by a supply voltage of $V_{CC} = 20\text{ V}$. Assume the current gain of the transistor is $h_{FE} = 120$.
25. Design an emitter bias circuit that will be operated by a supply voltage of $V_{CC} = 16\text{ V}$. Assume that the transistor has the current gain of 150.
26. Design a collector-to-base bias circuit to have $V_{CE} = 16\text{ V}$ and $I_C = 4\text{ mA}$. Assume that the supply voltage is 12 V and the transistor has $\beta = 130$.
27. Design a voltage divider bias circuit considering the supply voltage of 18 V and the transistor has $\beta = 120$.
28. For the biasing circuit of common-base configuration in Fig.5-16, determine the load line and the operating point.

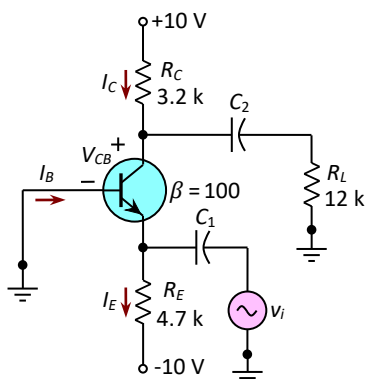
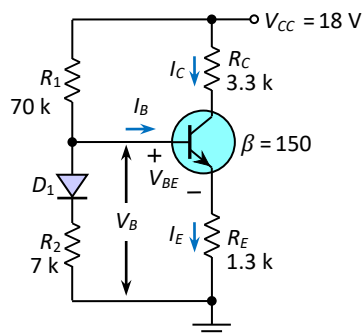


Fig.5-16: CB circuit for Example 28

Fig.5-96: V_{BE} compensated voltage divider bias

29. Design a common-base amplifier with emitter bias using a PNP transistor of $h_{FE} = 140$. Assume the supply voltage is ± 15 V.
30. For the common-collector biasing circuit in Fig.5-76(a) determine the load line and operating point.

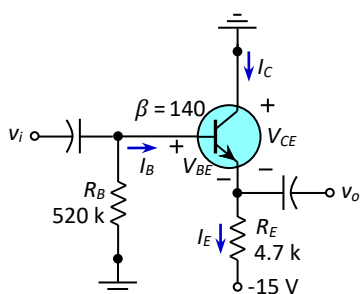


Fig.5-17: CC circuit for Example 5-30

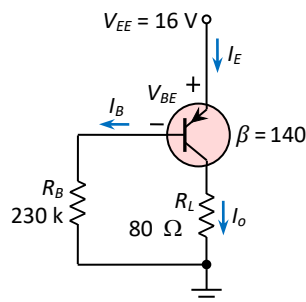


Fig.5-18: Constant current source for Example 5-30

31. Design a common-collector amplifier with voltage divider bias using a PNP transistor of $h_{FE} = 120$. Assume the supply voltage is 16 V.
32. A controlling circuit requires to ON/OFF a 24 V, 500 W heating coil by a 12 V relay. If the output of the controlling circuit is TTL standard, design a transistor switching circuit to drive the relay and draw the schematic diagram of the full system.
33. Design an inverter circuit to drive a 2.2 kΩ load that requires TTL logic levels. Assume $h_{FE(min)} = 80$.
34. Design an inverter circuit to drive a 2.2 kΩ load that requires TTL logic levels. Assume $h_{FE(min)} = 80$ using PNP transistor.
35. For the constant current source of Fig.5-18, calculate the output current and output voltage. Determine the maximum value of R_L that can be used with this current source.

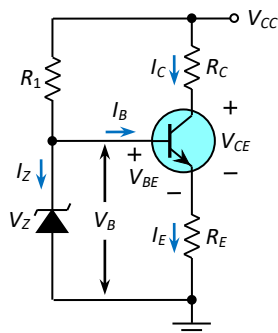


Fig.5-19: Voltage divider bias using Zener diode.

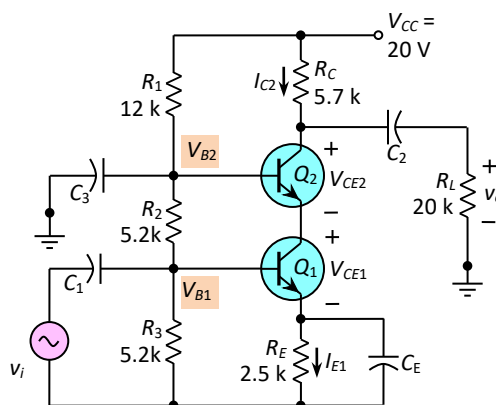


Fig.5-21: Cascode amplifier for Exercise

36. For the cascode amplifier of Fig.5-19, calculate the operating point of transistors. Assume $I_{B1} = I_{B2} \approx 0$.
37. The current gain of the transistors used in a Darlington pair are: $\beta_1 = 120$ and $\beta_2 = 40$. Determine the current gain of the Darlington pair.
38. The current gain of the transistors used in a Sziklai pair are: $\beta_1 = 110$ and $\beta_2 = 50$. Determine the current gain of the Sziklai pair for the configurations shown in Fig.5-.
39. Design a common-collector amplifier using a Darlington pair for $I_{CQ} = 2.5$ A and $V_{CEQ} = 10$ V. Assume the supply voltage is 30 V.
40. Design a common-collector amplifier using a Sziklai pair for $I_{CQ} = 3.5$ A and $V_{CEQ} = 12$ V. Assume the supply voltage is 35 V.
41. Fig.5- 106 shows a fixed bias circuit for an NPN transistor. The transistor has the following maximum ratings: $P_{D(max)} = 500$ mW, $V_{CE(max)} = 16$ V, and $I_{C(max)} = 100$ mA. Determine the maximum value of the supply voltage (V_{CC}) that can be applied to this biasing circuit. Which rating will be exceeded first?
42. A BD145 transistor is used in the circuit of Fig.5-106. Determine the maximum value to which V_{CC} can be increased without exceeding its maximum ratings. Refer to the datasheet of Fig.4-39, given in Chapter 4.

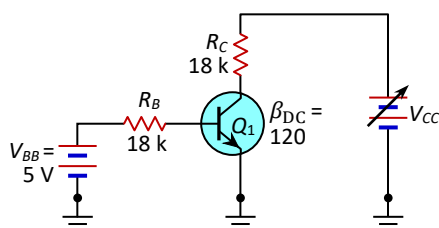


Fig.5-20: BJT biasing circuit for Exercise 41 and 42

43. Fig5-21 shows a fixed bias circuit with emitter beedbac resistor. For some reason, the transistor has been open. Now determine the value of R_C for which the transistor will go to the cutoff condition.

44. As shown in Fig. 22 the emitter terminal of the transistor is open. For what value of R_C the entire supply voltage will be dropped across the transistor, i.e, $V_{CE} = V_{CC} = 12\text{ V}$.

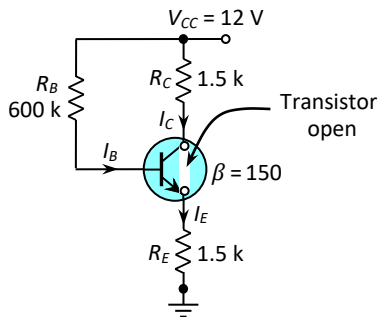


Fig.5-21: Circuit diagram and graph for Example 43

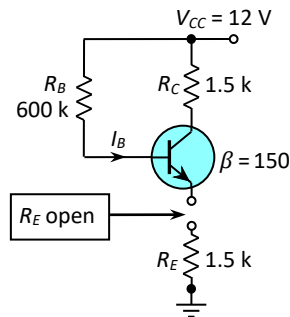


Fig.5-22: Circuit diagram and graph for Example 44

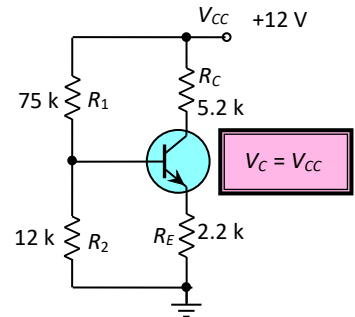


Fig.5-23: Circuit diagram and graph for Example 45

45. For what value of R_1 , R_2 , and R_E the transistor's operating point in circuit of Fig. will shift to the cutoff position, that is, $V_{CE} = V_{CC} = 12\text{ V}$.

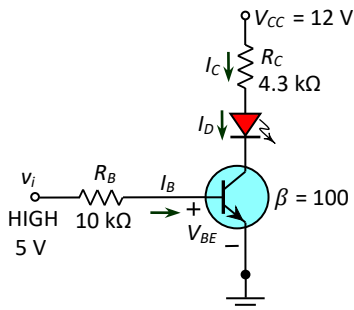


Fig.5-: Switching circuit to turn ON/OFF an LED

46. A switching circuit is designed to turn ON/OFF a red LED as shown in Fig.5-. Determine the voltage dropped across the diode and the current flowing through the diode. The forward voltage drop of a red LED is 1.8 V . [Ans. : $V_D \approx 1.8\text{ V}$ and $I_D = 2.37\text{ mA}$]